

META - Overview

Ticker: META

Rows in minute df: 405010

Intraday abnormal sequences: 58864

META - Trigger Counts - Intraday trigger counts

pattern	seq_window_mins	q_label	n_sequences
trade_volume_above_mid	5	90	4376
trade_volume_above_mid	10	90	2621
trade_volume_above_mid	20	90	1601
trade_volume_above_mid	30	90	1214
trade_volume_above_mid	60	90	750
trade_volume_above_mid	5	95	2711
trade_volume_above_mid	10	95	1665
trade_volume_above_mid	20	95	990
trade_volume_above_mid	30	95	715
trade_volume_above_mid	60	95	464
trade_volume_above_mid	5	99	774
trade_volume_above_mid	10	99	452
trade_volume_above_mid	20	99	284
trade_volume_above_mid	30	99	214
trade_volume_above_mid	60	99	134

META - Trigger Counts - Intraday trigger counts

pattern	seq_window_mins	q_label	n_sequences
trade_volume_at_mid	5	90	5147
trade_volume_at_mid	10	90	2987
trade_volume_at_mid	20	90	1785
trade_volume_at_mid	30	90	1280
trade_volume_at_mid	60	90	781
trade_volume_at_mid	5	95	2933
trade_volume_at_mid	10	95	1812
trade_volume_at_mid	20	95	1065
trade_volume_at_mid	30	95	794
trade_volume_at_mid	60	95	469
trade_volume_at_mid	5	99	811
trade_volume_at_mid	10	99	490
trade_volume_at_mid	20	99	318
trade_volume_at_mid	30	99	240
trade_volume_at_mid	60	99	161

META - Trigger Counts - Intraday trigger counts

pattern	seq_window_mins	q_label	n_sequences
trade_volume_below_mid	5	90	4287
trade_volume_below_mid	10	90	2569
trade_volume_below_mid	20	90	1628
trade_volume_below_mid	30	90	1218
trade_volume_below_mid	60	90	750
trade_volume_below_mid	5	95	2659
trade_volume_below_mid	10	95	1620
trade_volume_below_mid	20	95	982
trade_volume_below_mid	30	95	737
trade_volume_below_mid	60	95	460
trade_volume_below_mid	5	99	803
trade_volume_below_mid	10	99	468
trade_volume_below_mid	20	99	284
trade_volume_below_mid	30	99	205
trade_volume_below_mid	60	99	156

META - Trigger Counts - Daily trigger counts

pattern	q_label	n_abnormal_days
trade_volume_above_mid	90	150
trade_volume_above_mid	95	105
trade_volume_above_mid	99	68

META - Trigger Counts - Daily trigger counts

pattern	q_label	n_abnormal_days
trade_volume_at_mid	90	155
trade_volume_at_mid	95	107
trade_volume_at_mid	99	61

META - Trigger Counts - Daily trigger counts

pattern	q_label	n_abnormal_days
trade_volume_below_mid	90	157
trade_volume_below_mid	95	116
trade_volume_below_mid	99	79

META - Trigger Counts - Weekly trigger counts

pattern	q_label	n_abnormal_weeks
trade_volume_above_mid	90	40
trade_volume_above_mid	95	29
trade_volume_above_mid	99	18

META - Trigger Counts - Weekly trigger counts

pattern	q_label	n_abnormal_weeks
trade_volume_at_mid	90	36
trade_volume_at_mid	95	28
trade_volume_at_mid	99	14

META - Trigger Counts - Weekly trigger counts

pattern	q_label	n_abnormal_weeks
trade_volume_below_mid	90	35
trade_volume_below_mid	95	24
trade_volume_below_mid	99	16

META - Trigger Counts - Intraday imbalance trigger counts

pattern	seq_window_mins	q_label	n_sequences
imbalance_above_mid	5	90	7325
imbalance_above_mid	10	90	4633
imbalance_above_mid	20	90	2850
imbalance_above_mid	30	90	2121
imbalance_above_mid	60	90	1329
imbalance_above_mid	5	95	4023
imbalance_above_mid	10	95	2540
imbalance_above_mid	20	95	1510
imbalance_above_mid	30	95	1138
imbalance_above_mid	60	95	775

META - Trigger Counts - Intraday imbalance trigger counts

pattern	seq_window_mins	q_label	n_sequences
imbalance_above_mid	5	99	922
imbalance_above_mid	10	99	548
imbalance_above_mid	20	99	350
imbalance_above_mid	30	99	269
imbalance_above_mid	60	99	204
imbalance_below_mid	5	90	7244
imbalance_below_mid	10	90	4637
imbalance_below_mid	20	90	2785
imbalance_below_mid	30	90	2123
imbalance_below_mid	60	90	1324

META - Trigger Counts - Intraday imbalance trigger counts

pattern	seq_window_mins	q_label	n_sequences
imbalance_below_mid	5	95	3978
imbalance_below_mid	10	95	2530
imbalance_below_mid	20	95	1466
imbalance_below_mid	30	95	1145
imbalance_below_mid	60	95	722
imbalance_below_mid	5	99	887
imbalance_below_mid	10	99	556
imbalance_below_mid	20	99	353
imbalance_below_mid	30	99	282
imbalance_below_mid	60	99	211

META - Trigger Counts - Daily imbalance trigger counts

pattern	q_label	n_abnormal_days
imbalance_above_mid	90	154
imbalance_above_mid	95	95

META - Trigger Counts - Daily imbalance trigger counts

pattern	q_label	n_abnormal_days
imbalance_above_mid	99	56
imbalance_below_mid	90	144

META - Trigger Counts - Daily imbalance trigger counts

pattern	q_label	n_abnormal_days
imbalance_below_mid	95	95
imbalance_below_mid	99	57

META - Trigger Counts - Weekly imbalance trigger counts

pattern	q_label	n_abnormal_weeks
imbalance_above_mid	90	37
imbalance_above_mid	95	26

META - Trigger Counts - Weekly imbalance trigger counts

pattern	q_label	n_abnormal_weeks
imbalance_above_mid	99	16
imbalance_below_mid	90	35

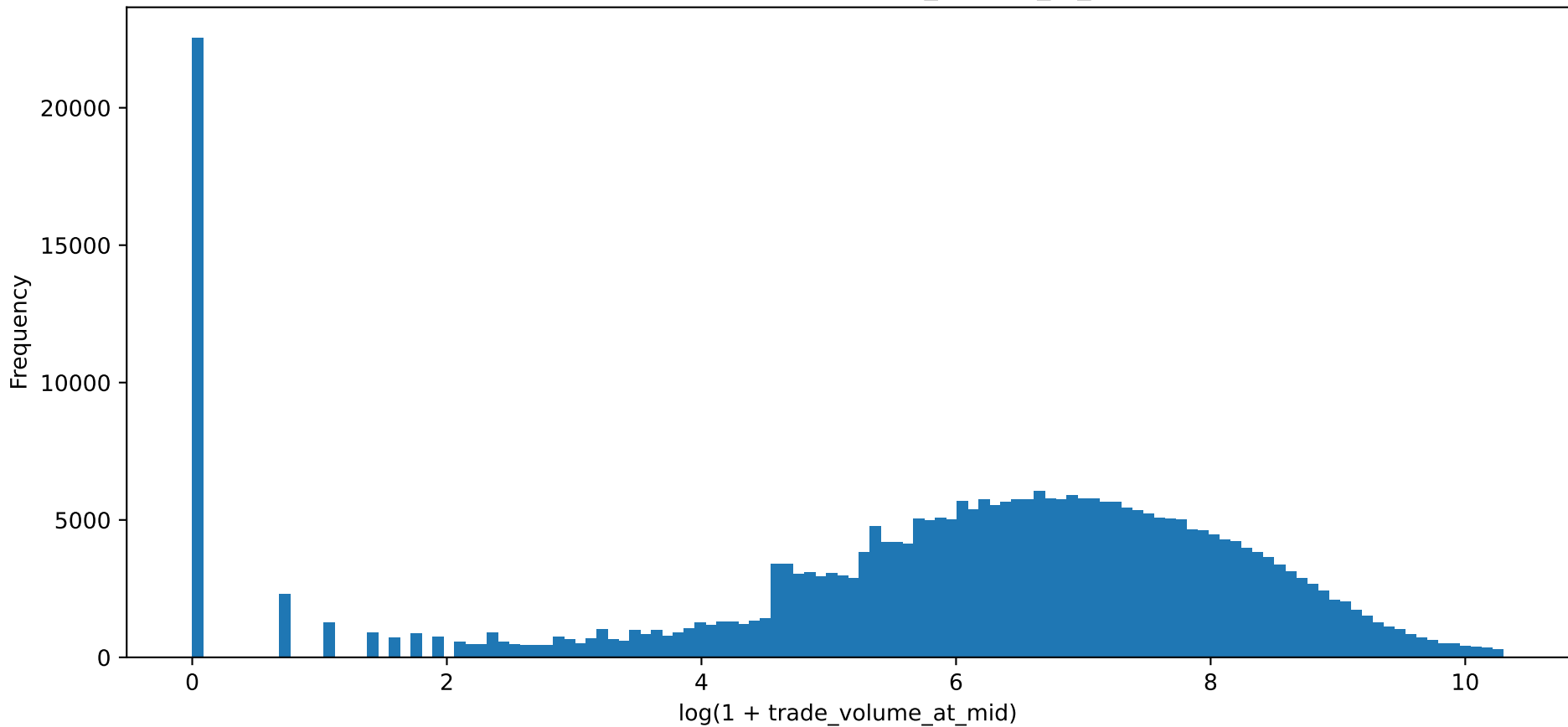
META - Trigger Counts - Weekly imbalance trigger counts

pattern	q_label	n_abnormal_weeks
imbalance_below_mid	95	22
imbalance_below_mid	99	12

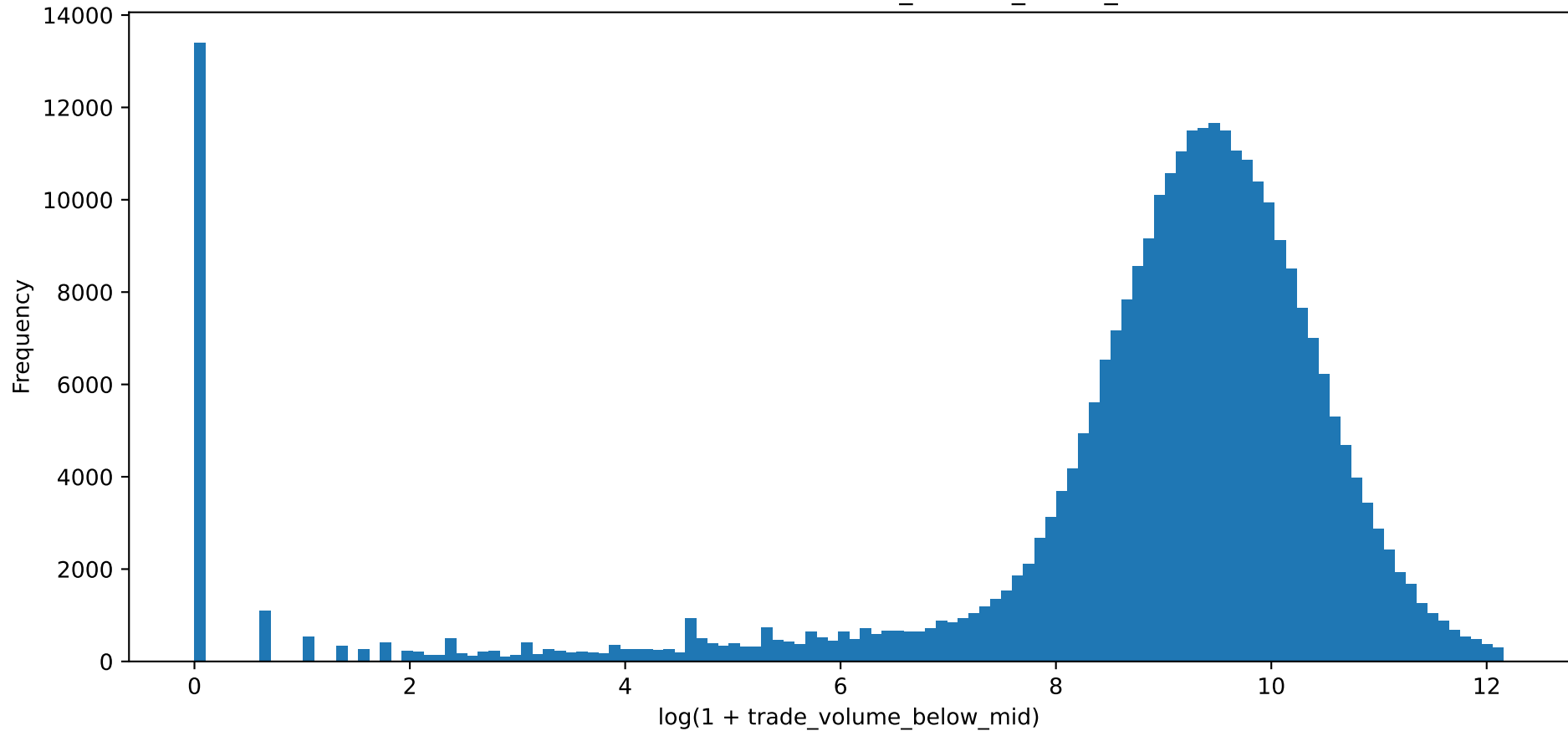
META - Microstructure

This section contains distribution and monthly plots for merged minute data.

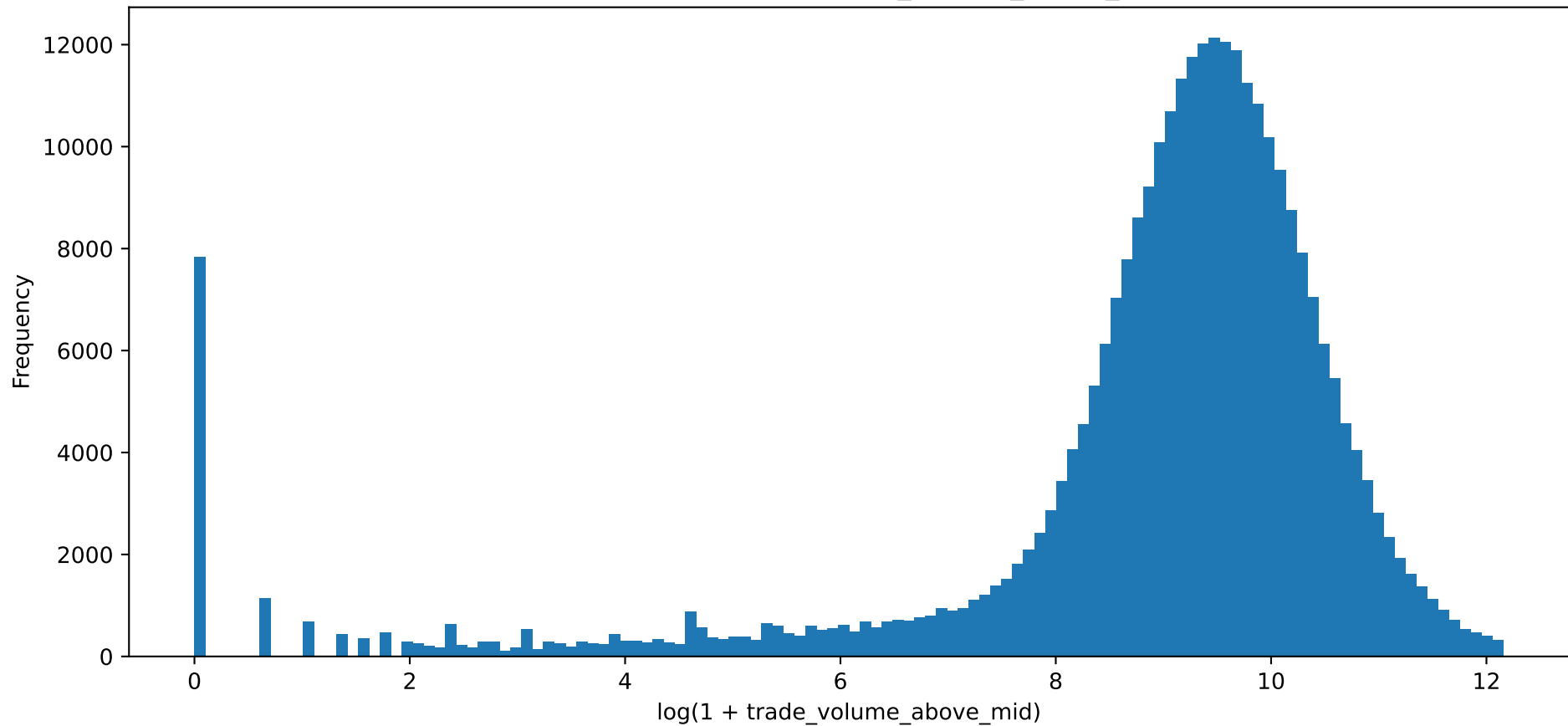
META - Distribution of trade_volume_at_mid



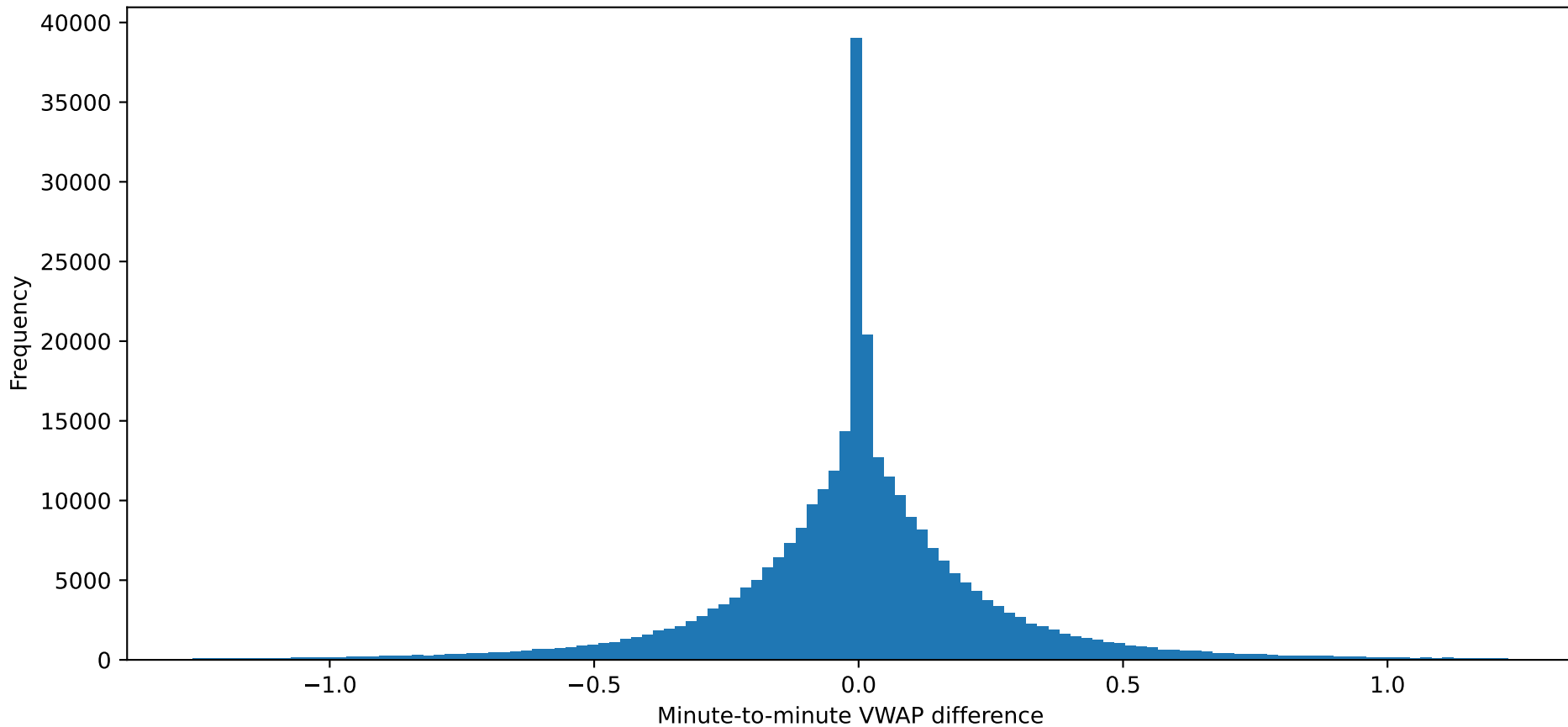
META - Distribution of trade_volume_below_mid



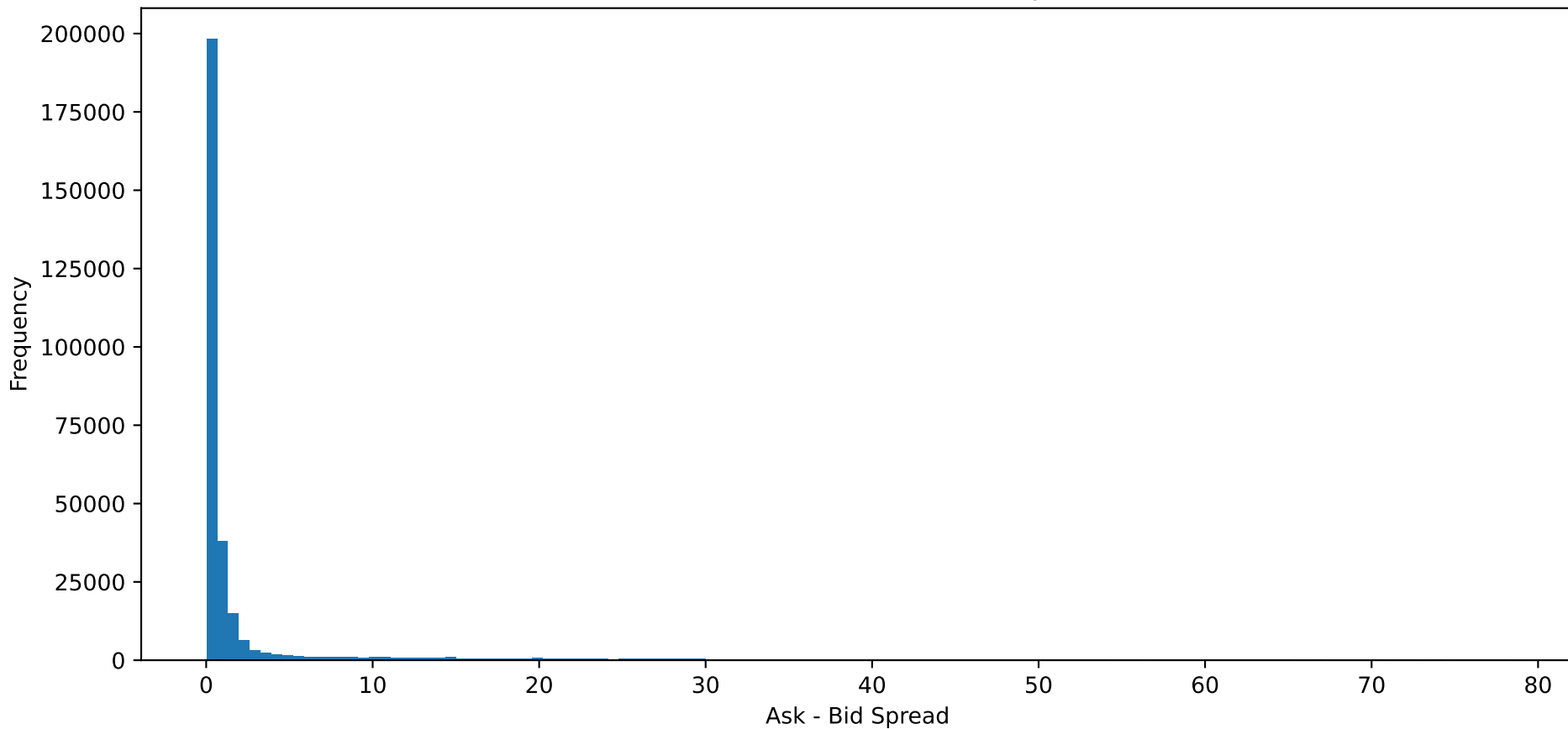
META - Distribution of trade_volume_above_mid



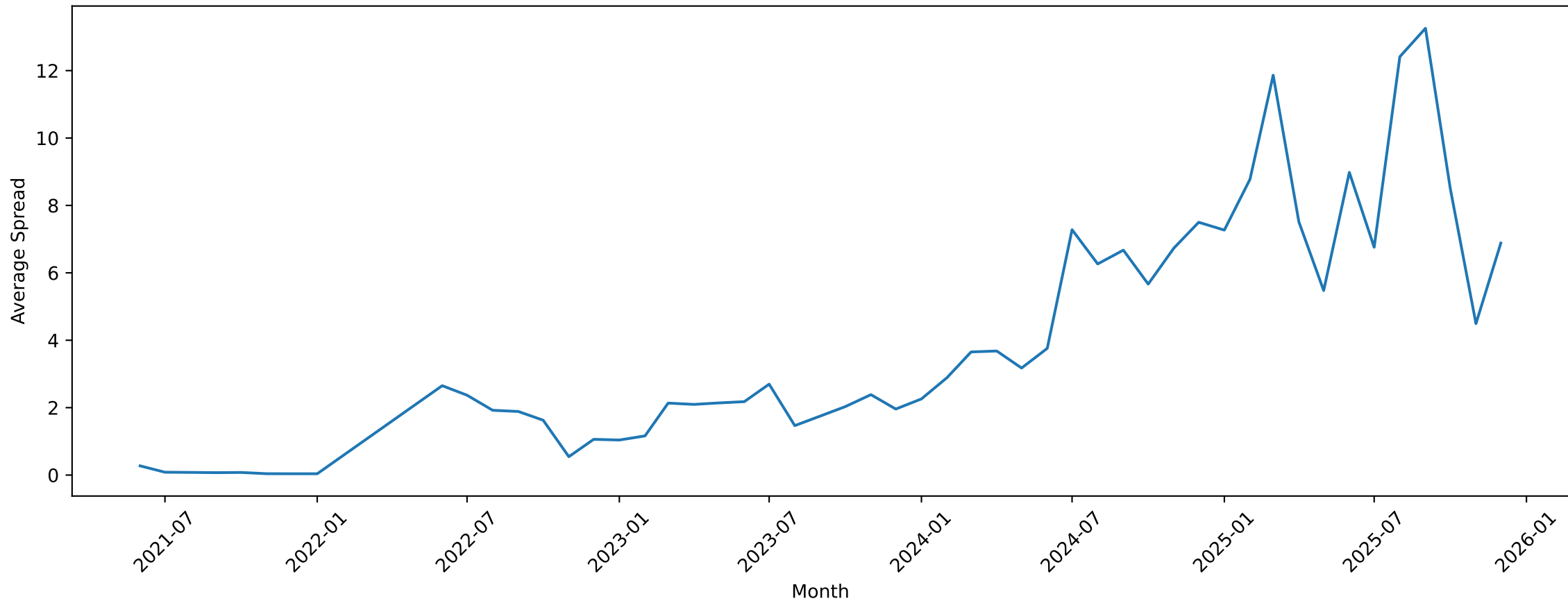
META - Distribution of VWAP Differences



META - Distribution of Bid-Ask Spread



META - Monthly Average Bid-Ask Spread



META - Intraday Multi-Quantile Results

q90 / q95 / q99 intraday excessive-sequence results versus baseline.

Post-abnormal future drift means the future Up / Down / Abs excursion after the abnormal sequence has been

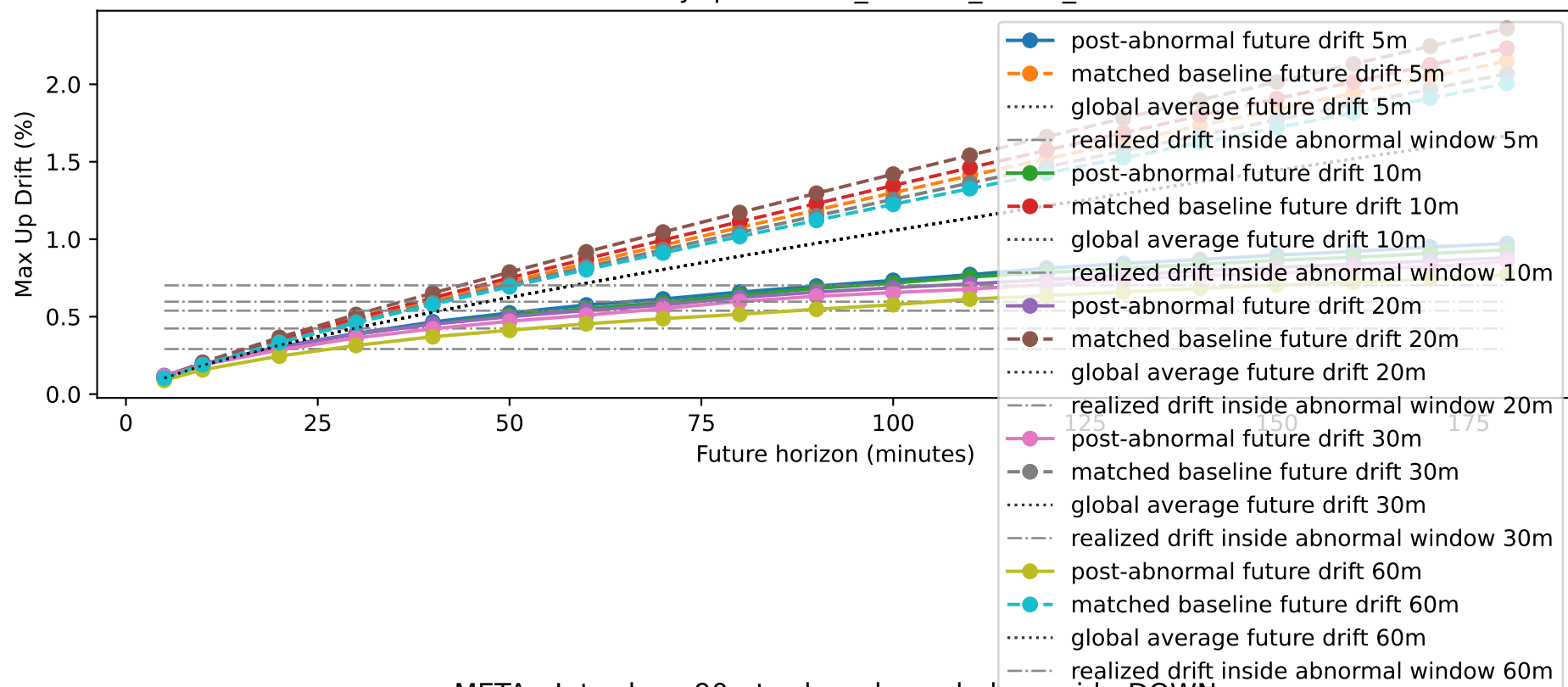
Matched baseline future drift is the corresponding non-abnormal baseline reference.

Dotted lines show global average future drift; dash-dot lines show realized drift already inside the abnormal

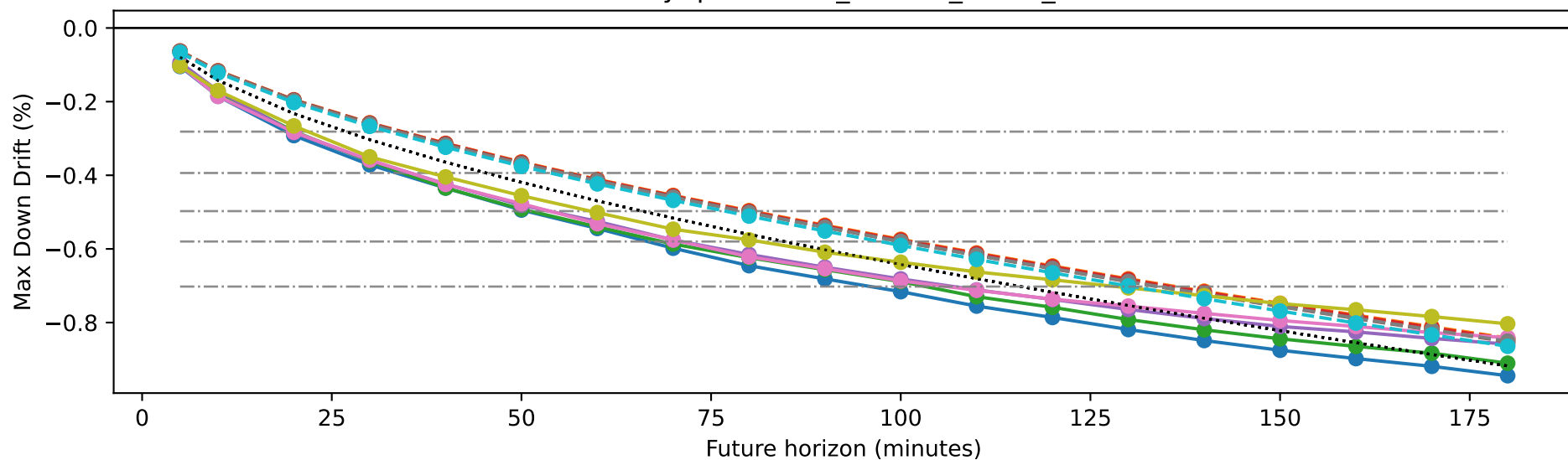
Down-side values are signed and are typically negative.

P-values are from t-tests on event-level excess versus rolling normal baseline excursion.

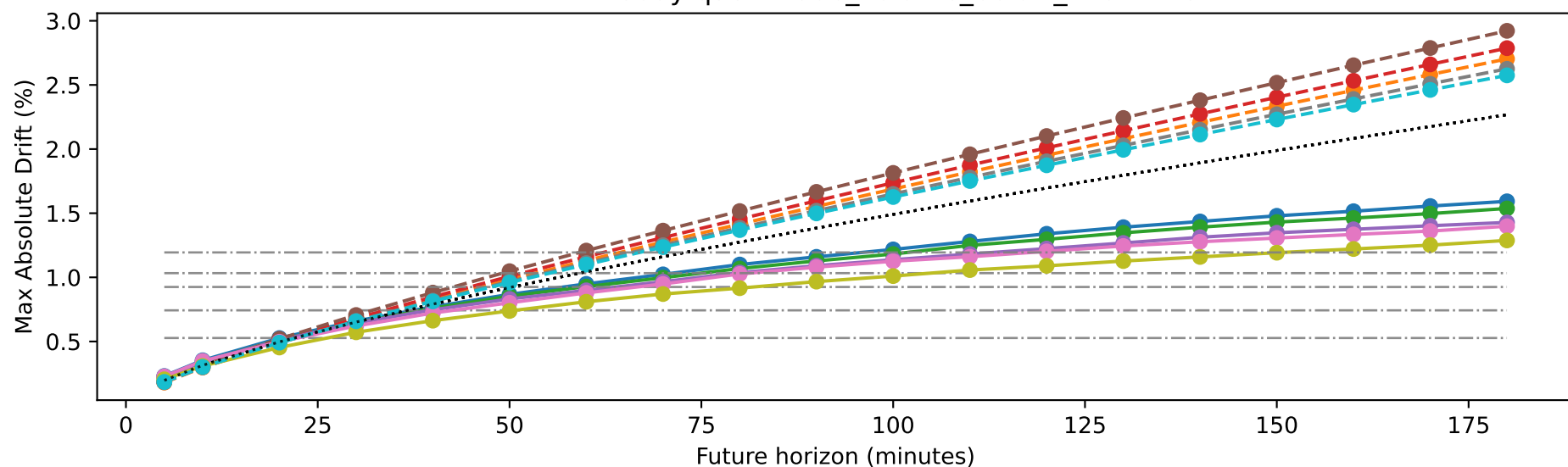
META - Intraday q90 - trade_volume_below_mid - UP



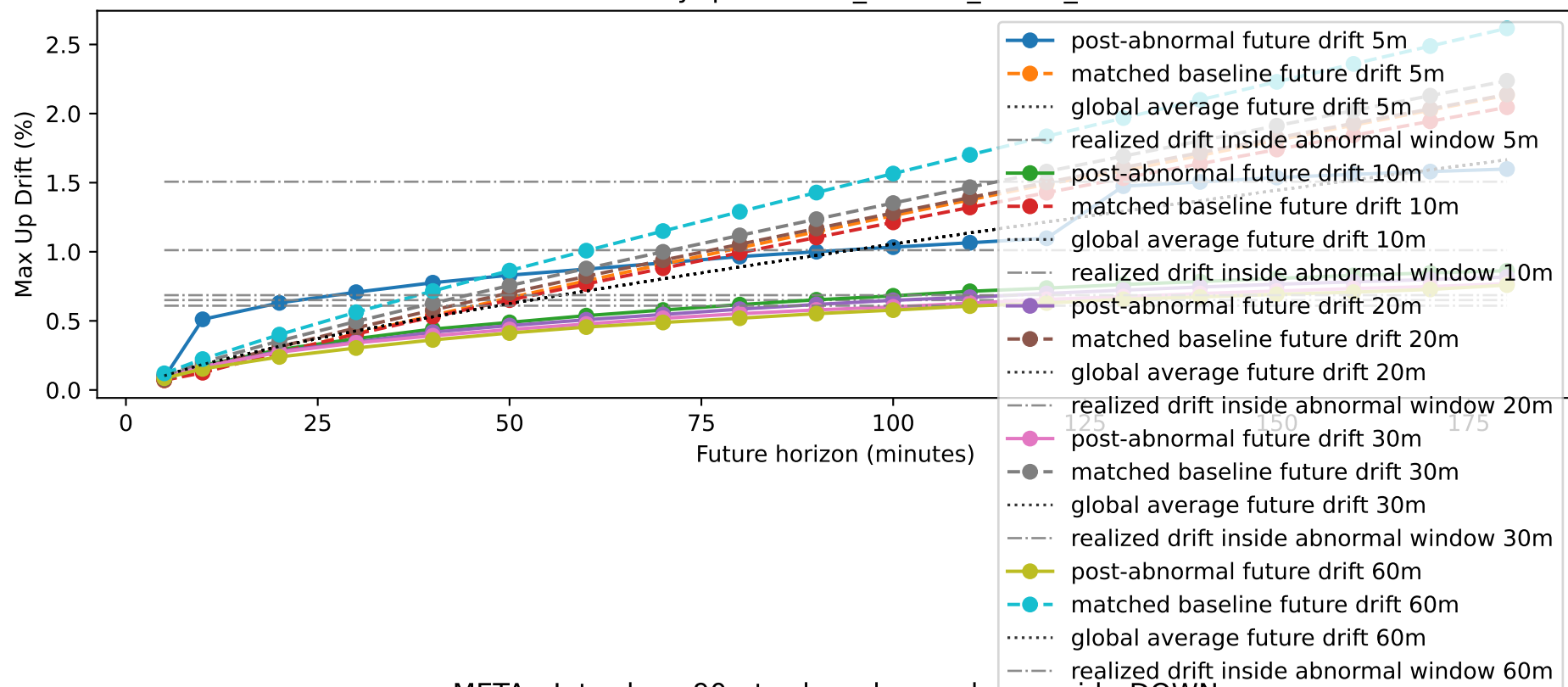
META - Intraday q90 - trade_volume_below_mid - DOWN



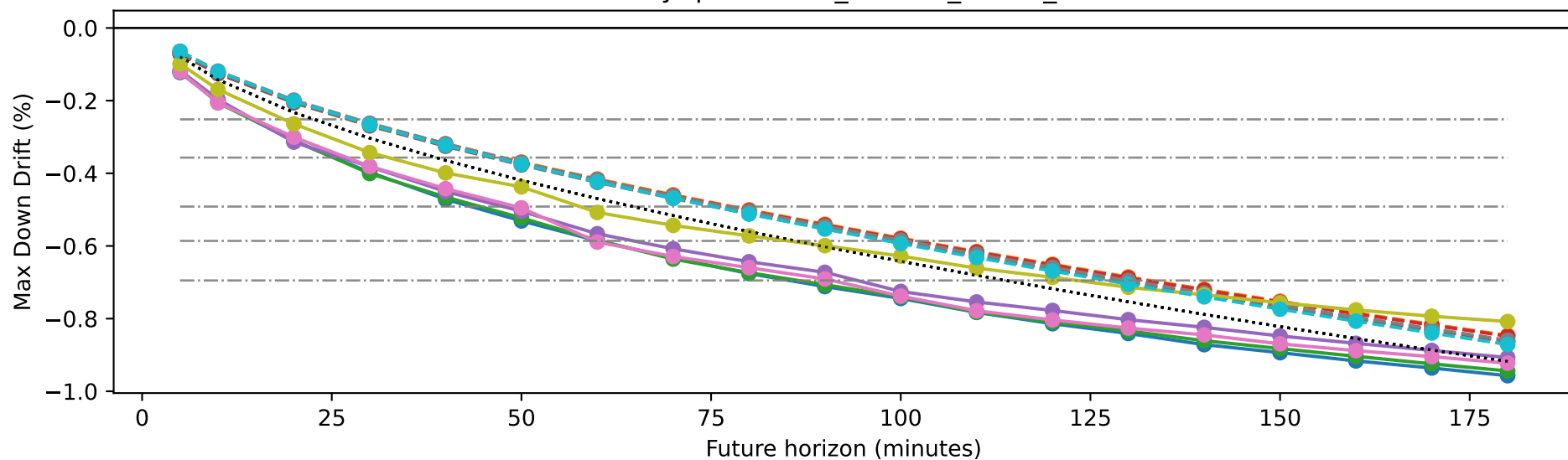
META - Intraday q90 - trade_volume_below_mid - ABS



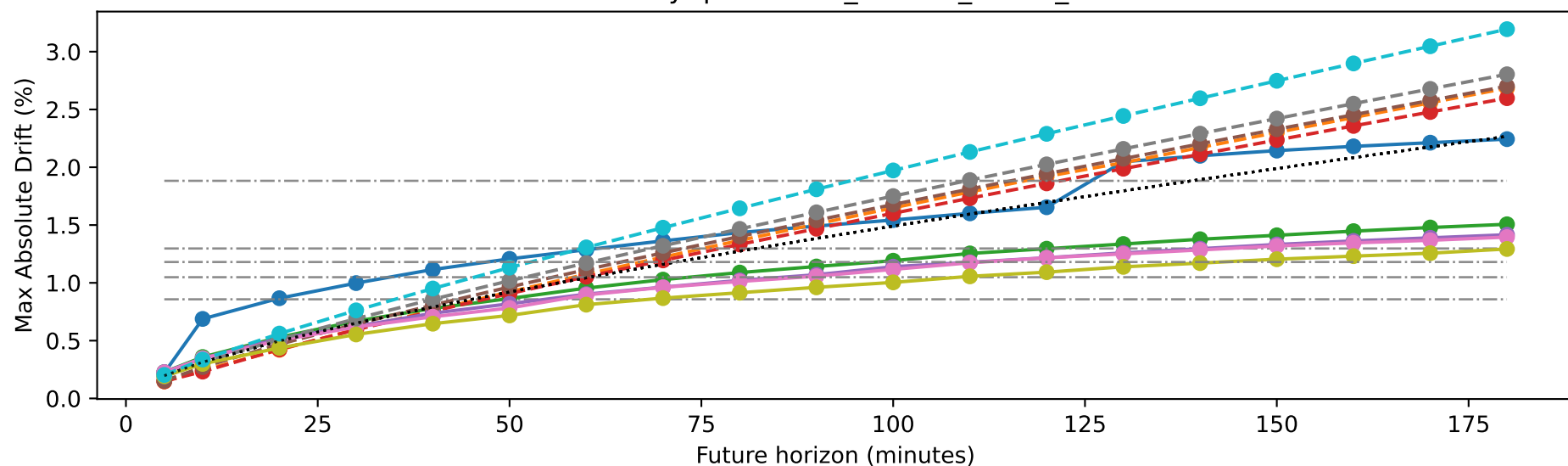
META - Intraday q90 - trade_volume_above_mid - UP



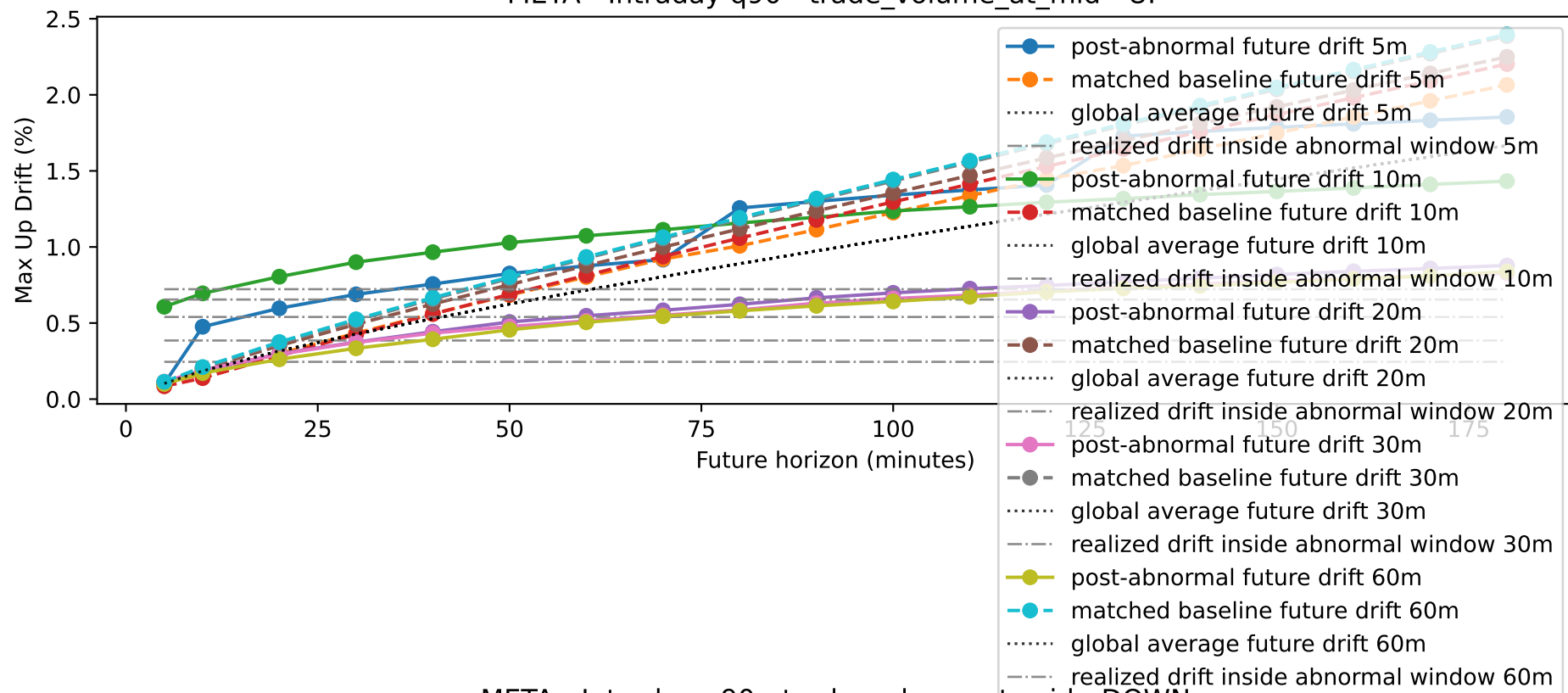
META - Intraday q90 - trade_volume_above_mid - DOWN



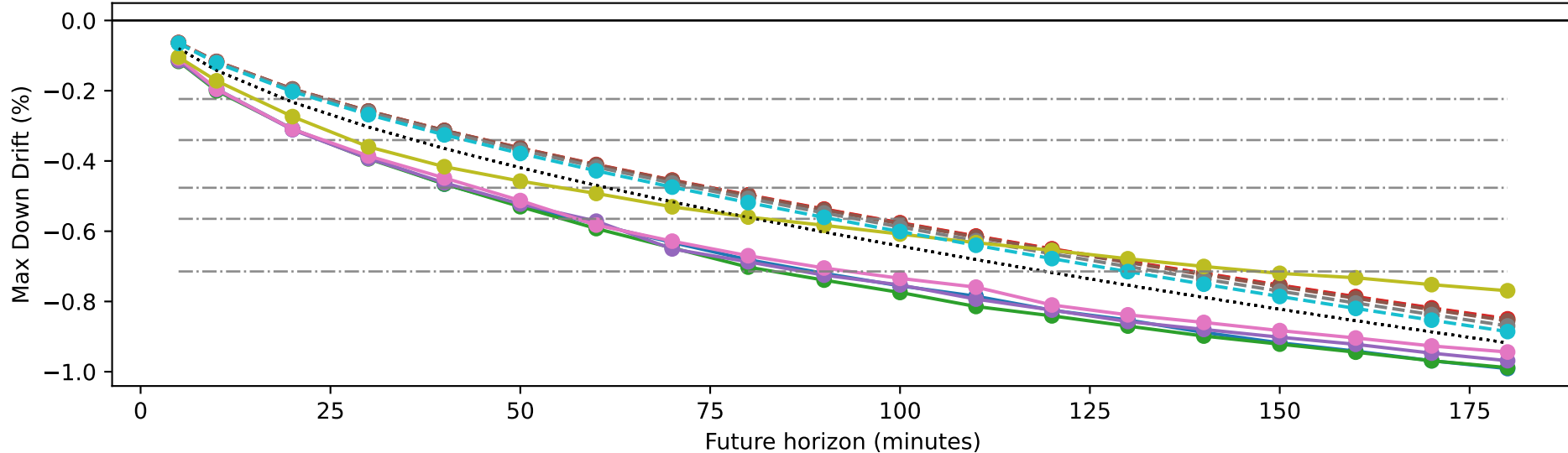
META - Intraday q90 - trade_volume_above_mid - ABS



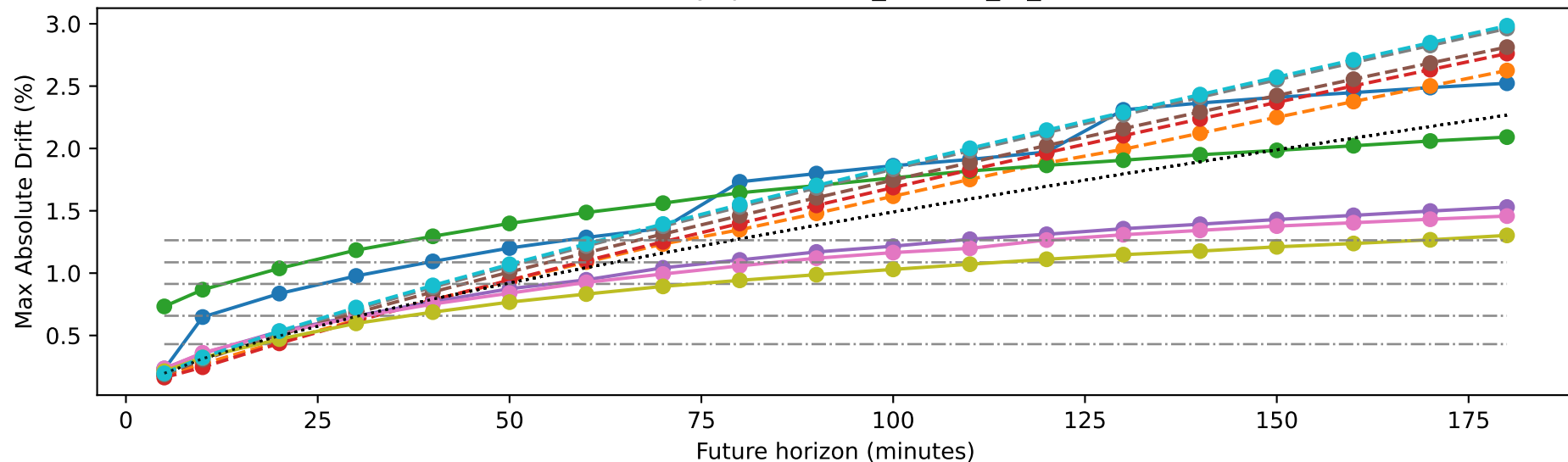
META - Intraday q90 - trade_volume_at_mid - UP



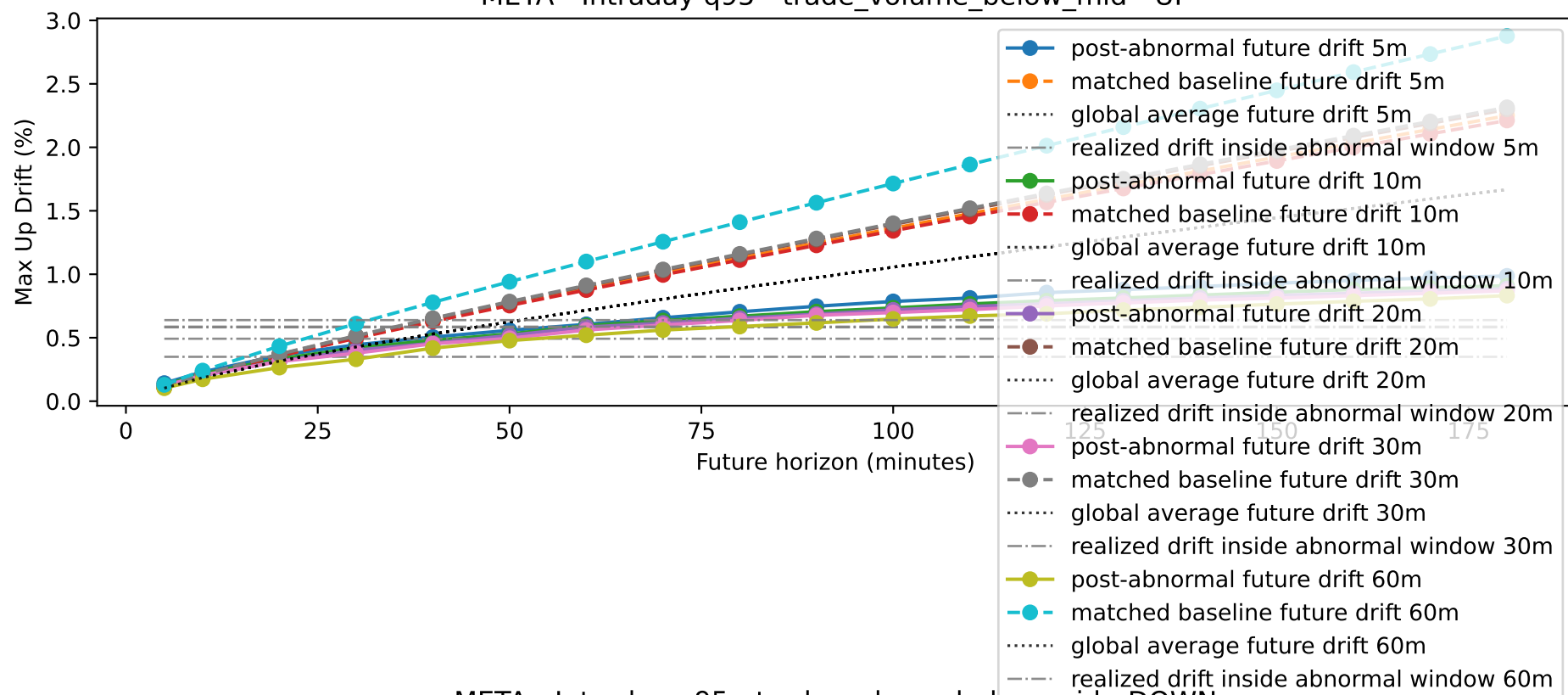
META - Intraday q90 - trade_volume_at_mid - DOWN



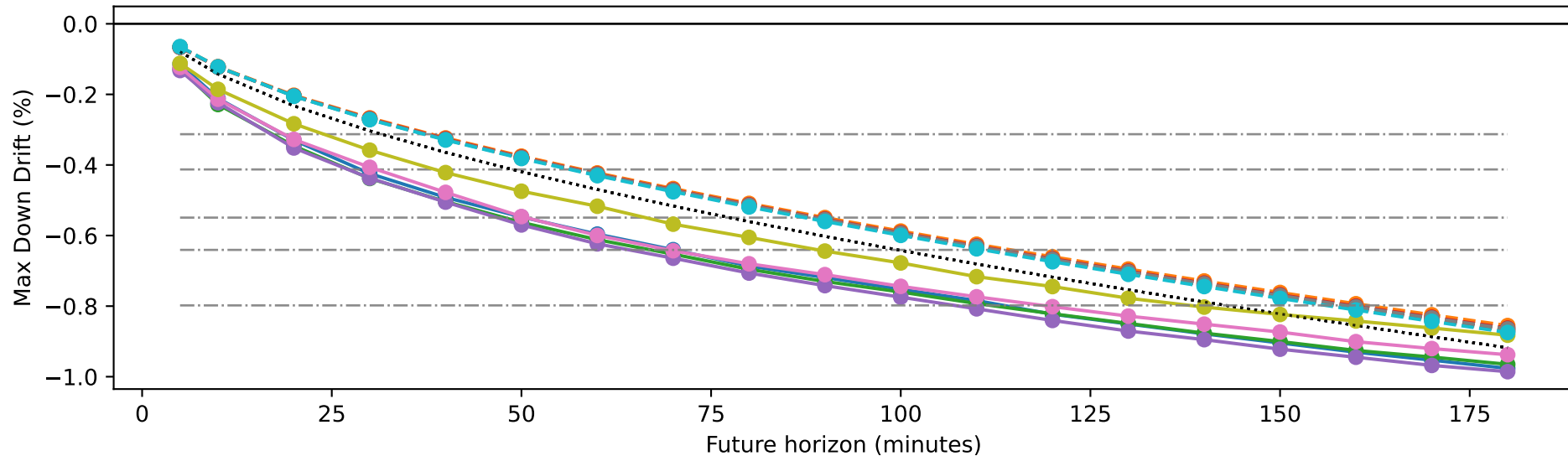
META - Intraday q90 - trade_volume_at_mid - ABS



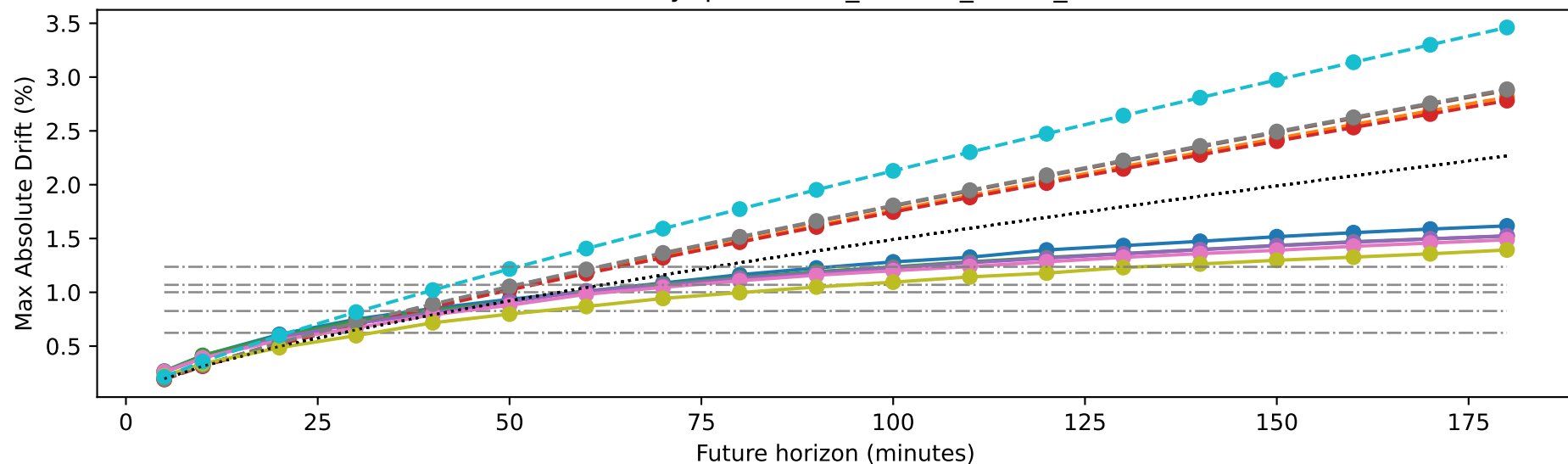
META - Intraday q95 - trade_volume_below_mid - UP



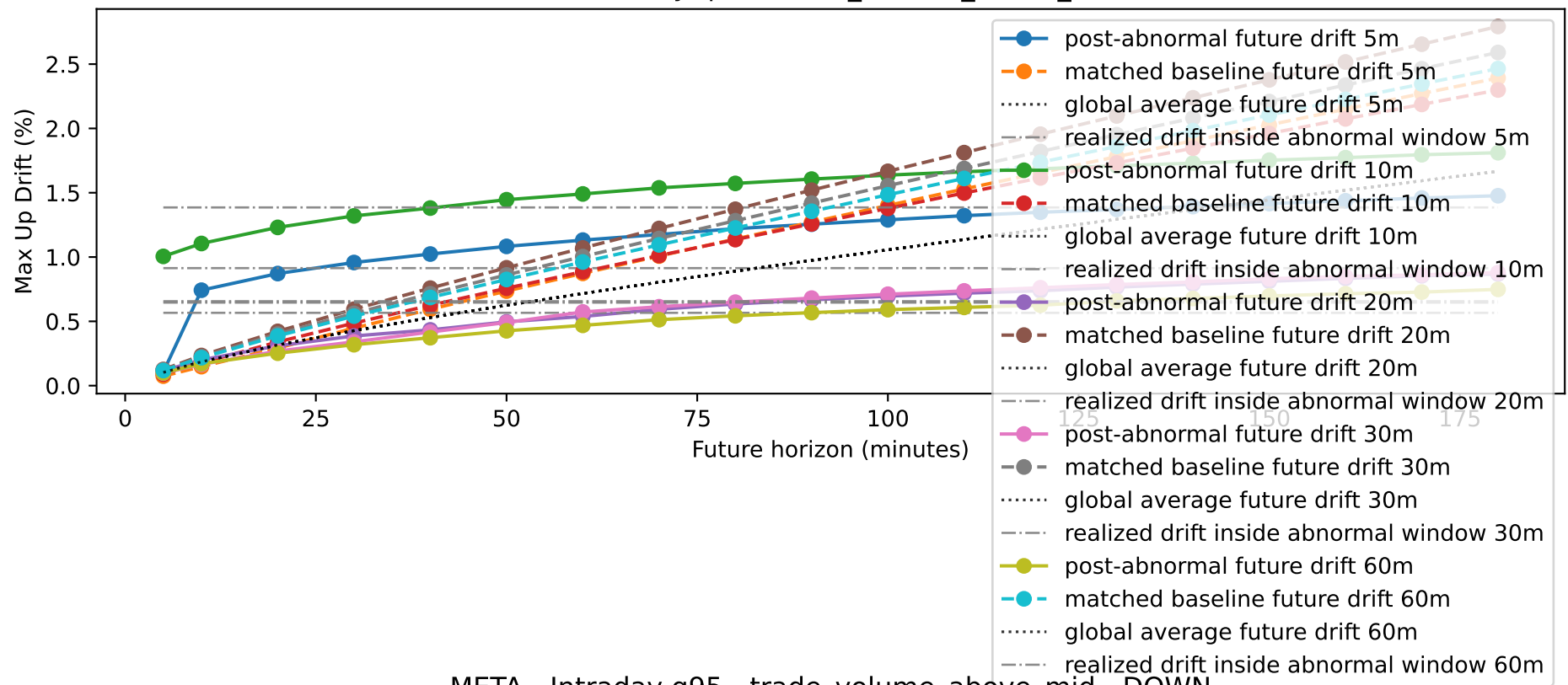
META - Intraday q95 - trade_volume_below_mid - DOWN



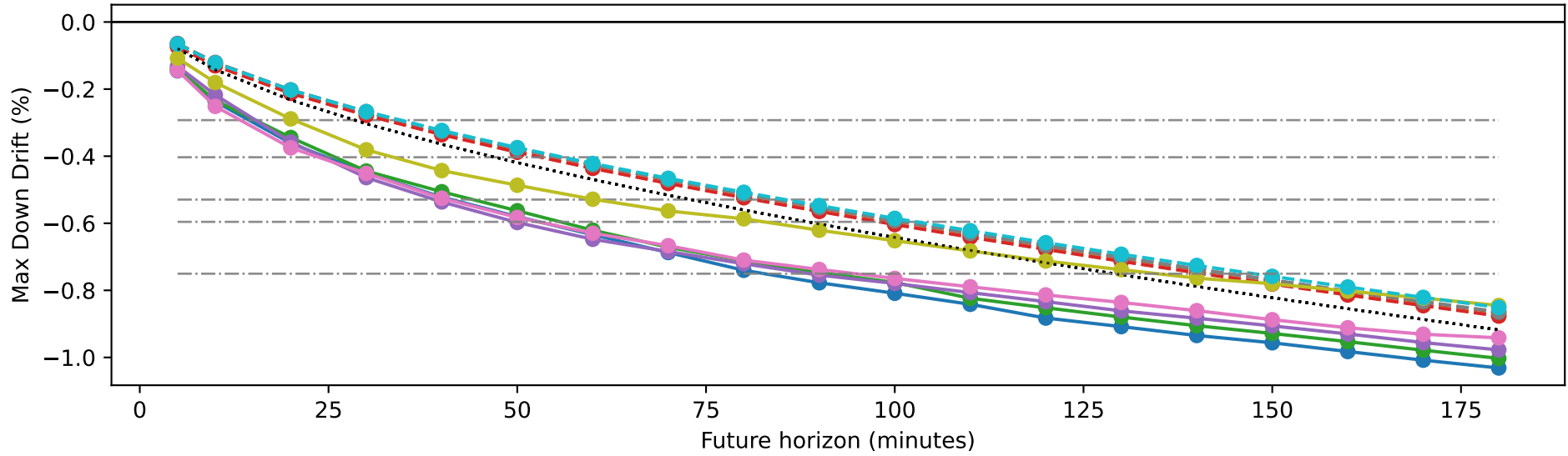
META - Intraday q95 - trade_volume_below_mid - ABS



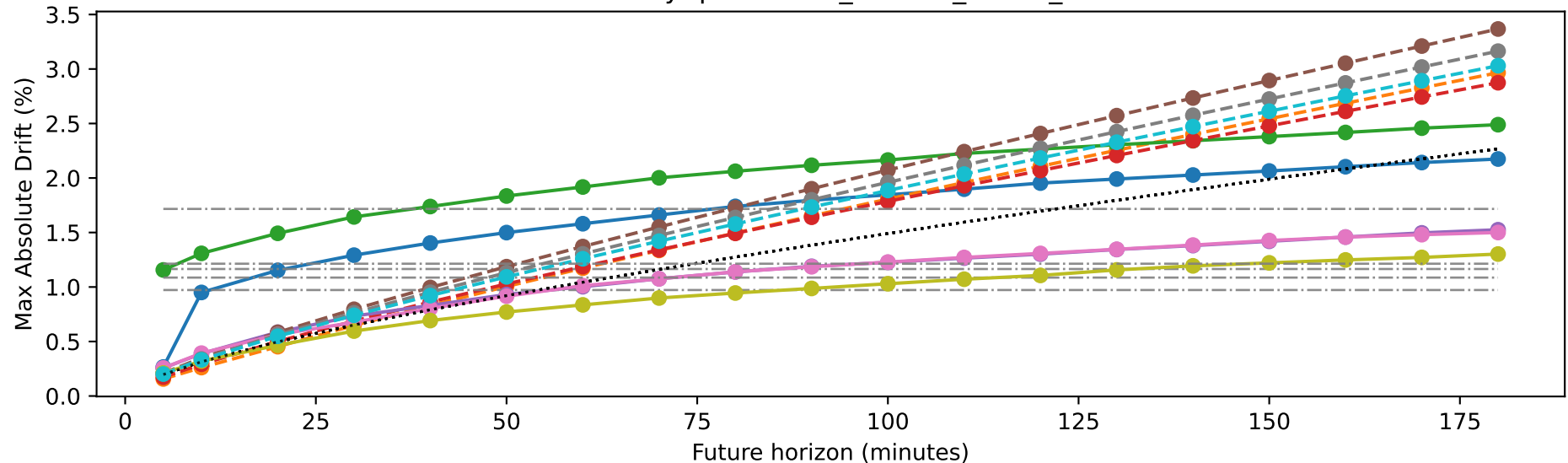
META - Intraday q95 - trade_volume_above_mid - UP



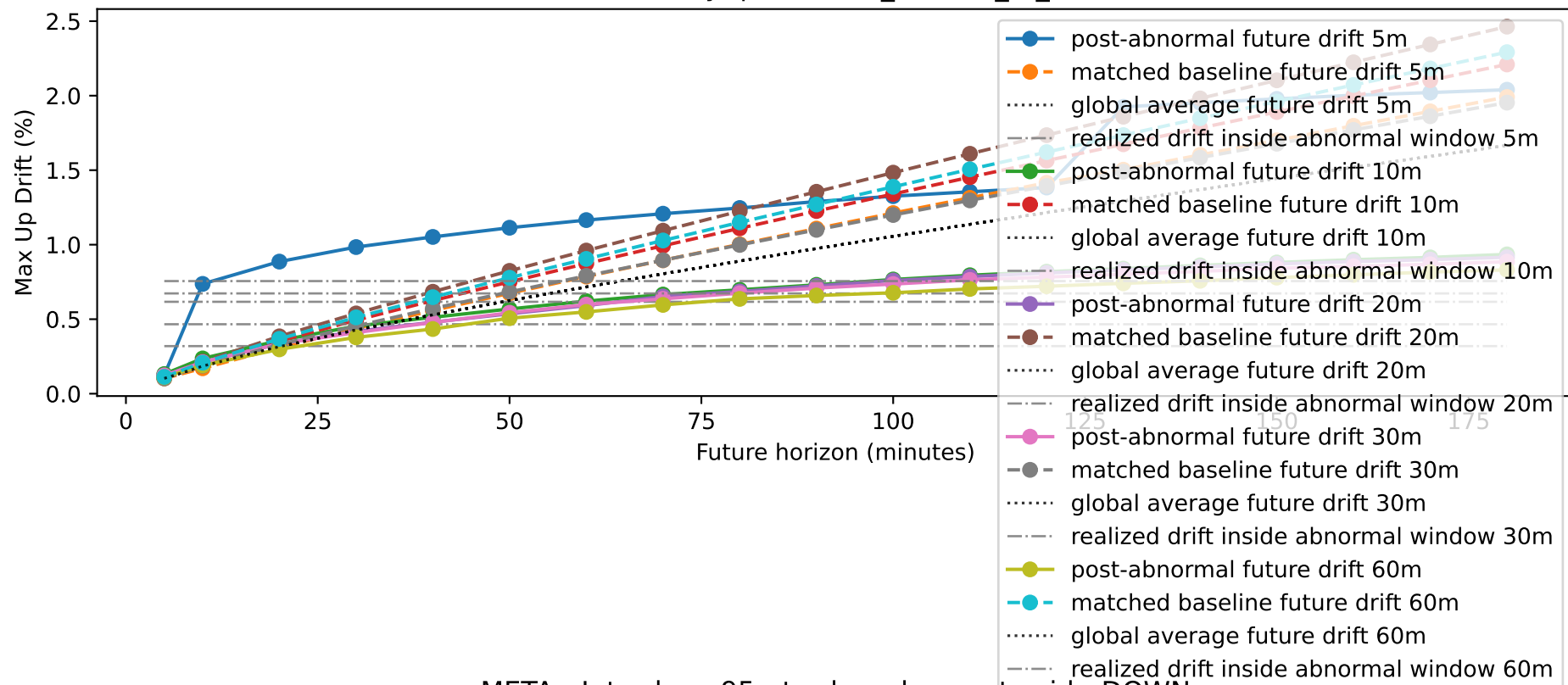
META - Intraday q95 - trade_volume_above_mid - DOWN



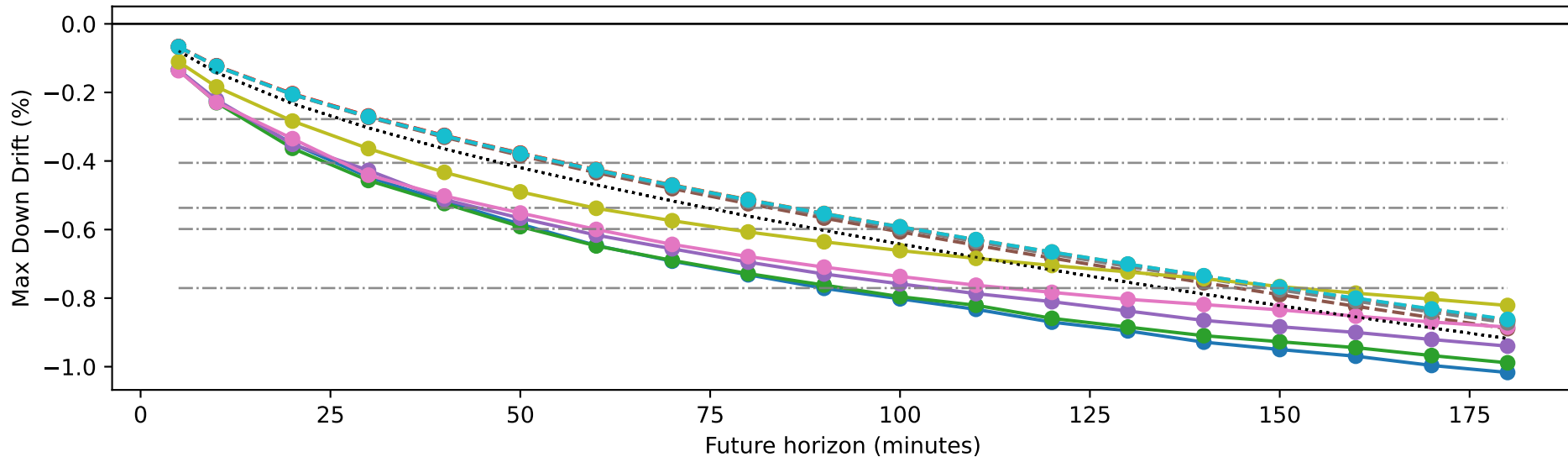
META - Intraday q95 - trade_volume_above_mid - ABS



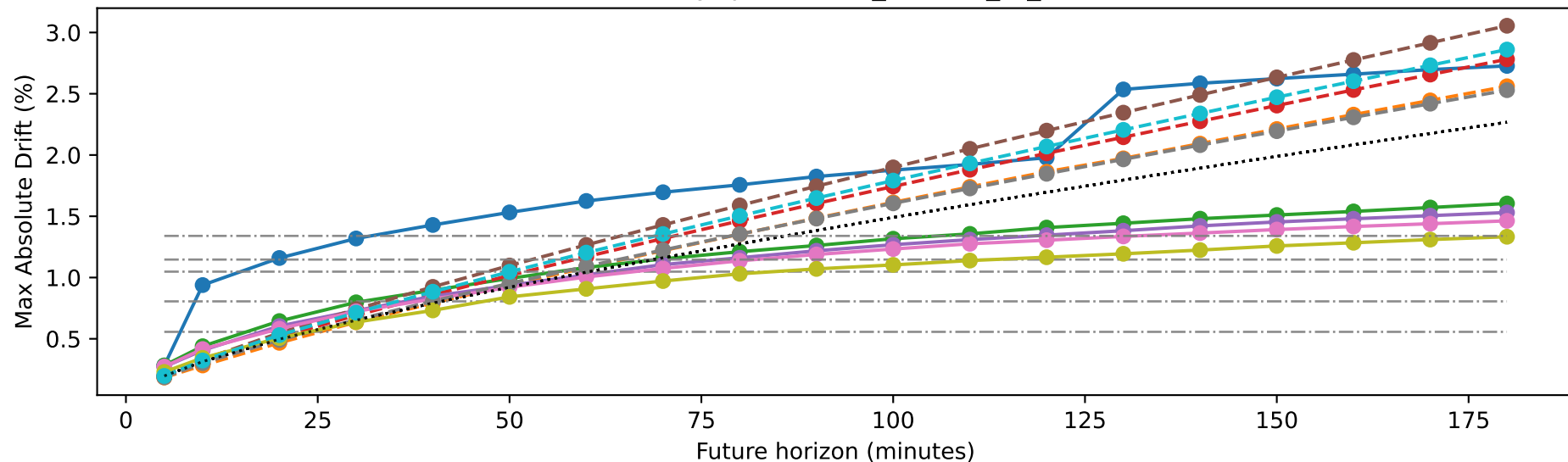
META - Intraday q95 - trade_volume_at_mid - UP



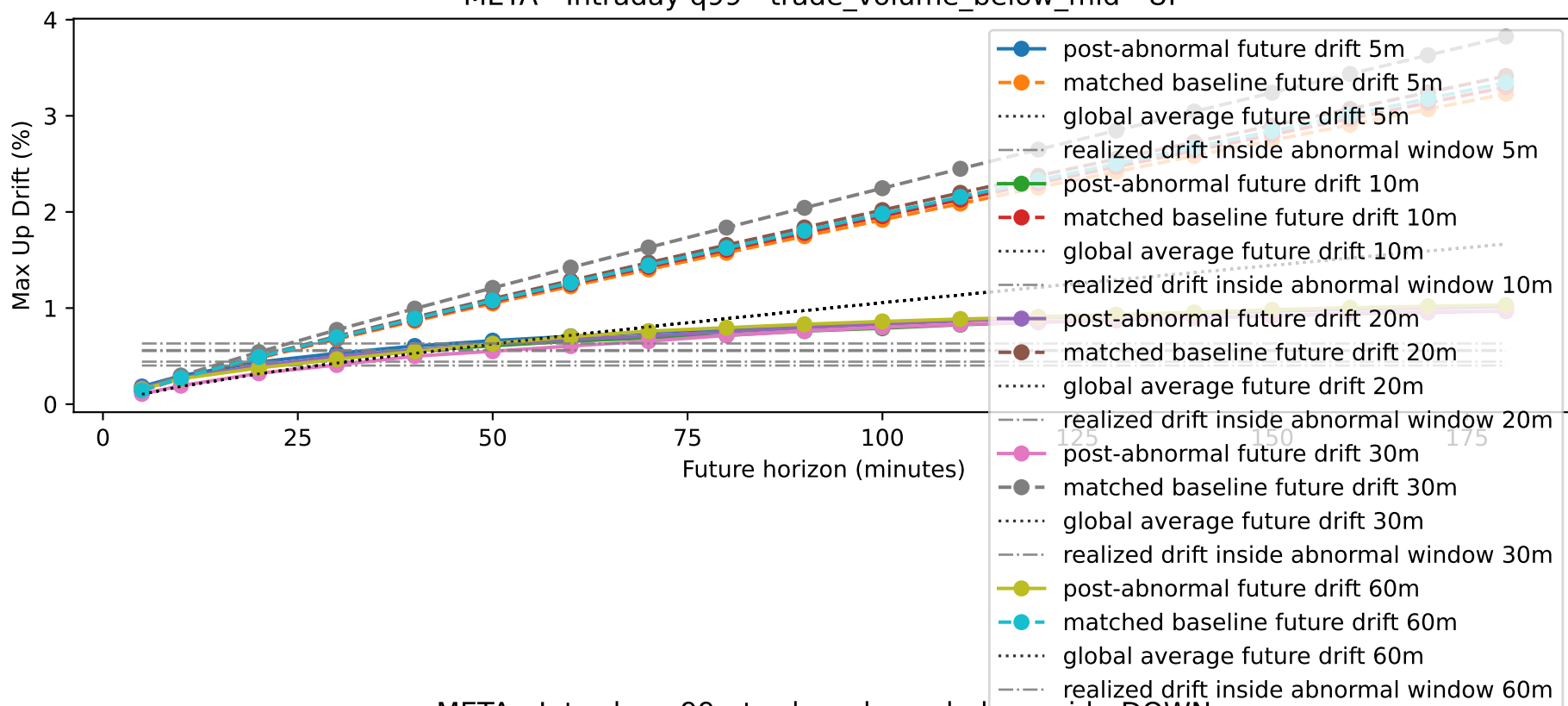
META - Intraday q95 - trade_volume_at_mid - DOWN



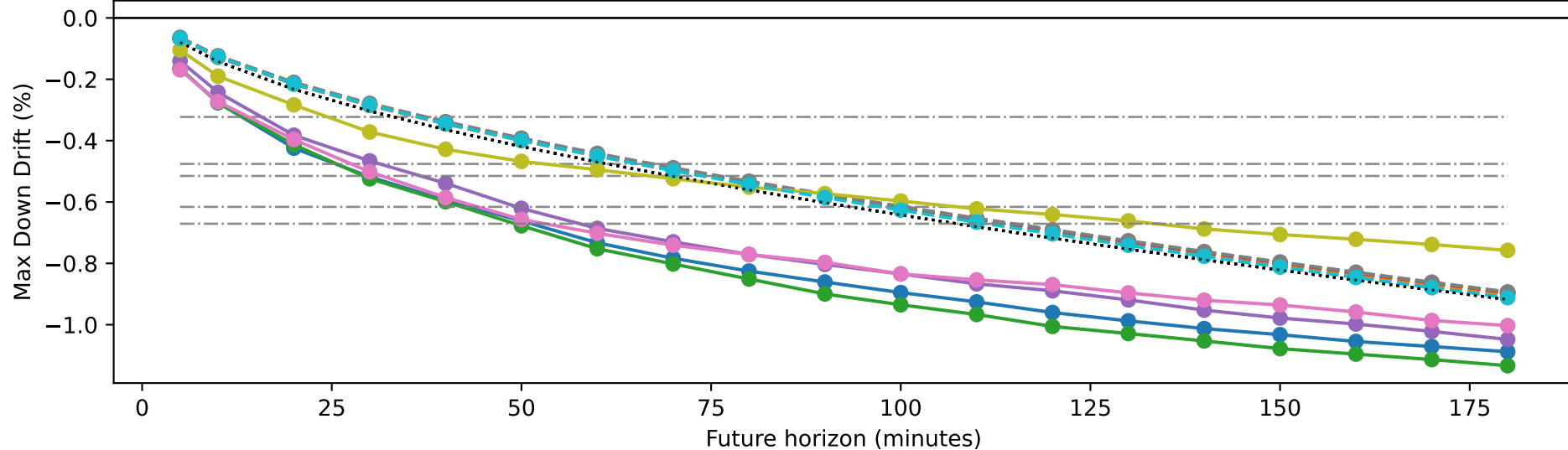
META - Intraday q95 - trade_volume_at_mid - ABS



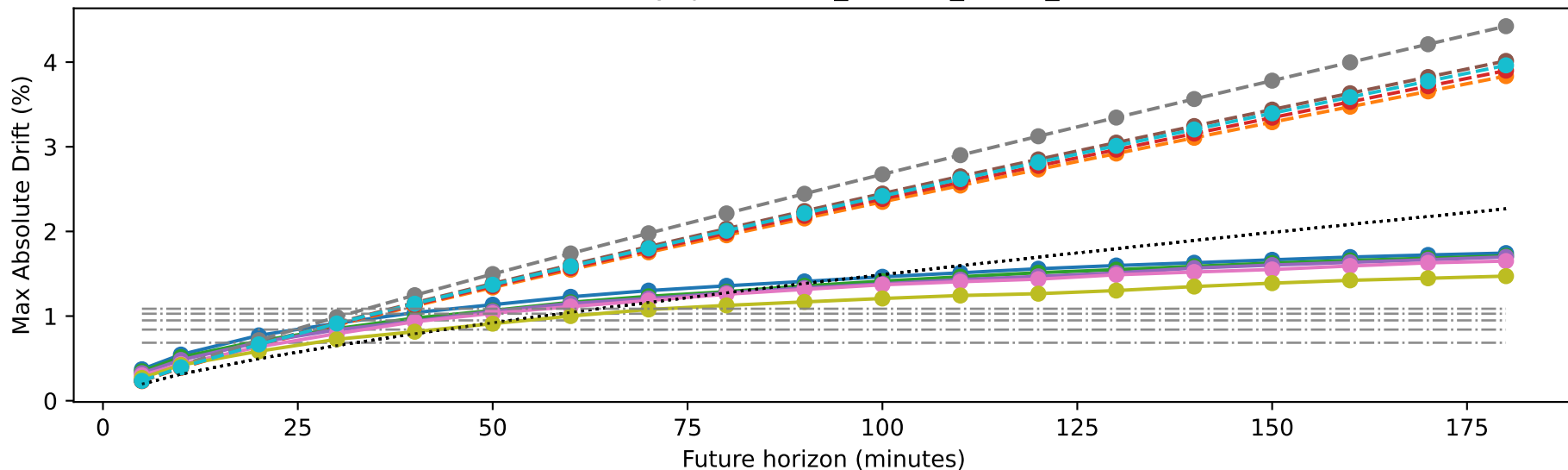
META - Intraday q99 - trade_volume_below_mid - UP



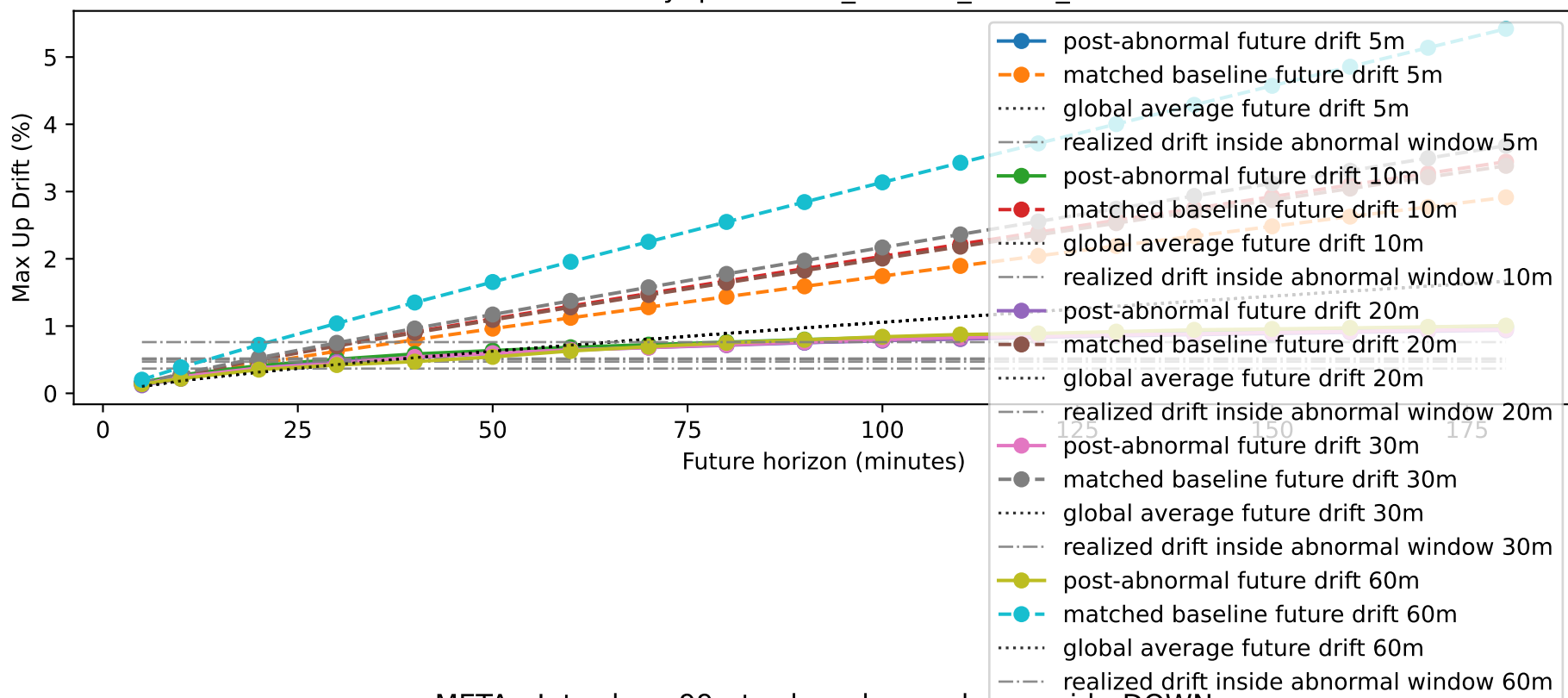
META - Intraday q99 - trade_volume_below_mid - DOWN



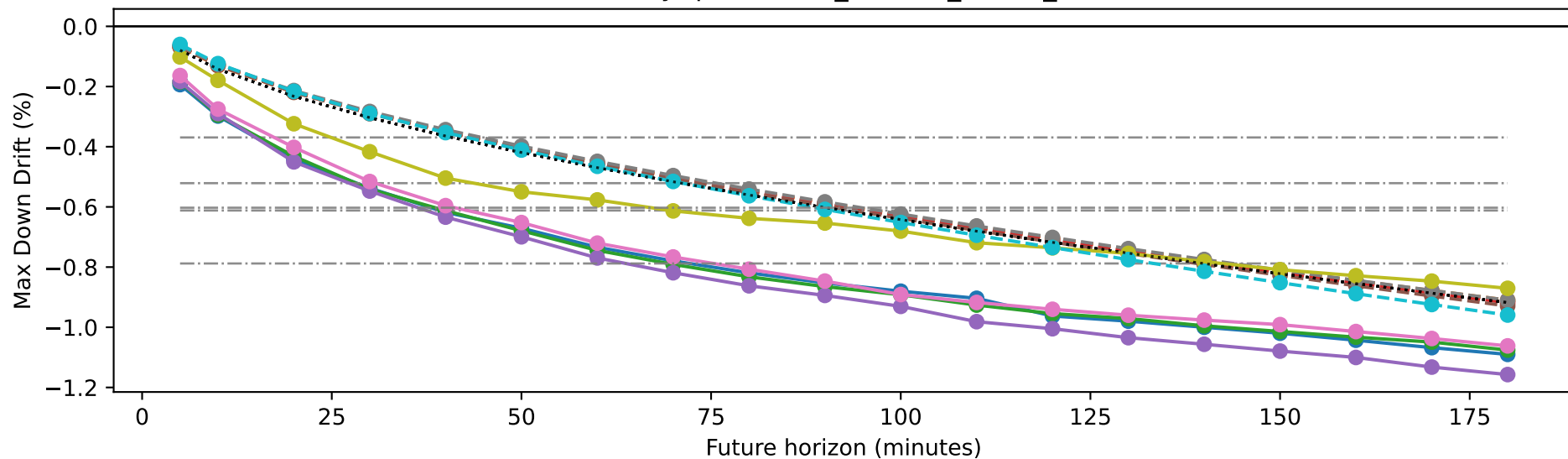
META - Intraday q99 - trade_volume_below_mid - ABS



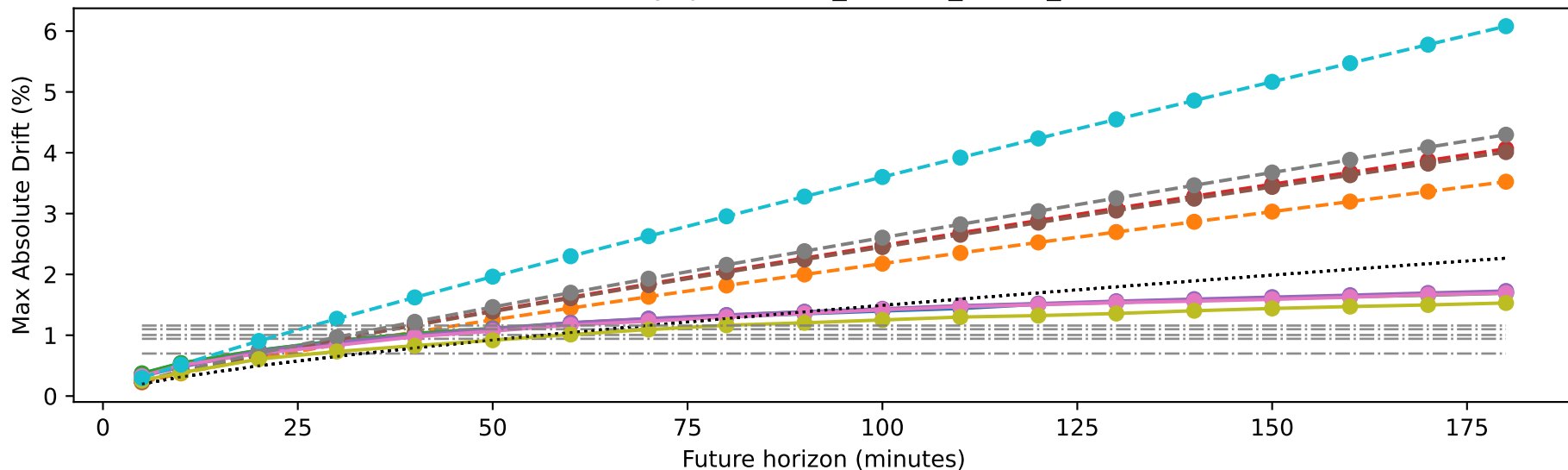
META - Intraday q99 - trade_volume_above_mid - UP



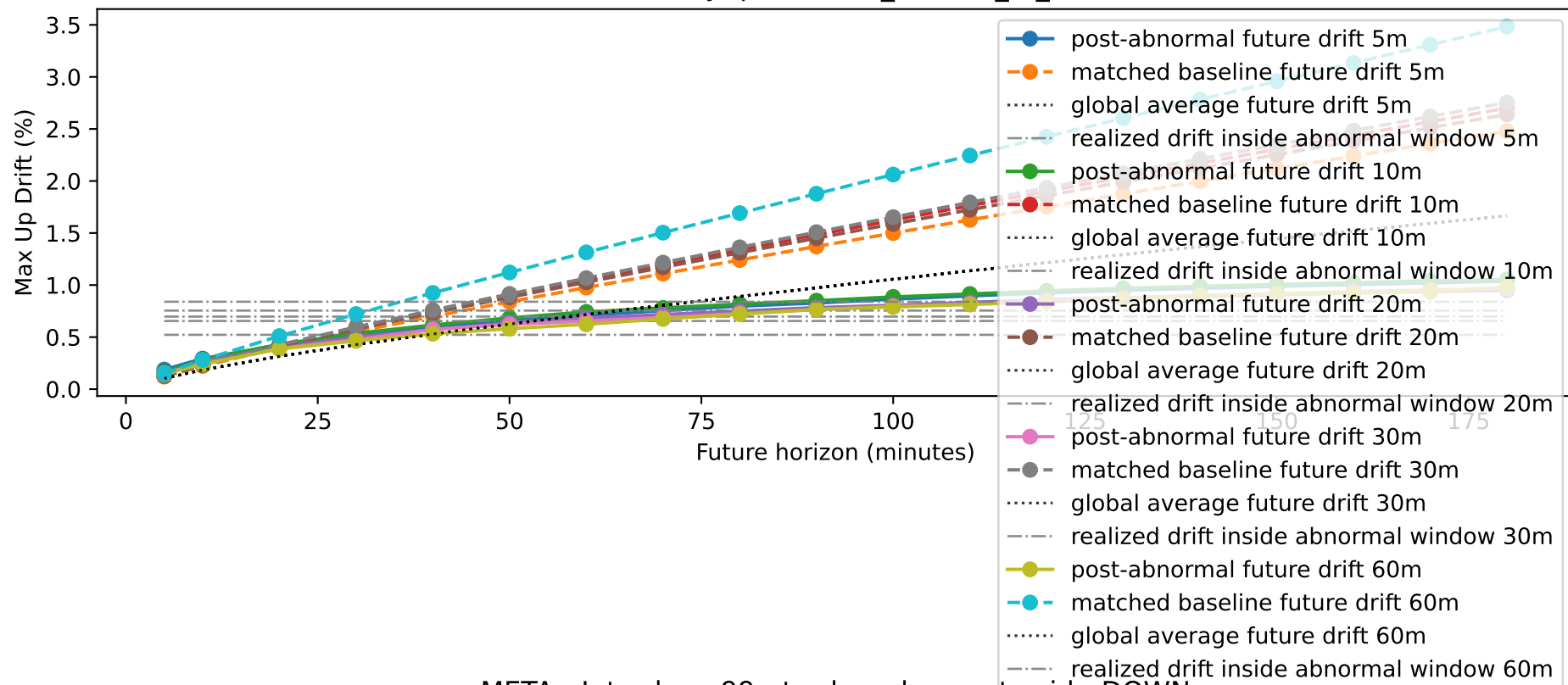
META - Intraday q99 - trade_volume_above_mid - DOWN



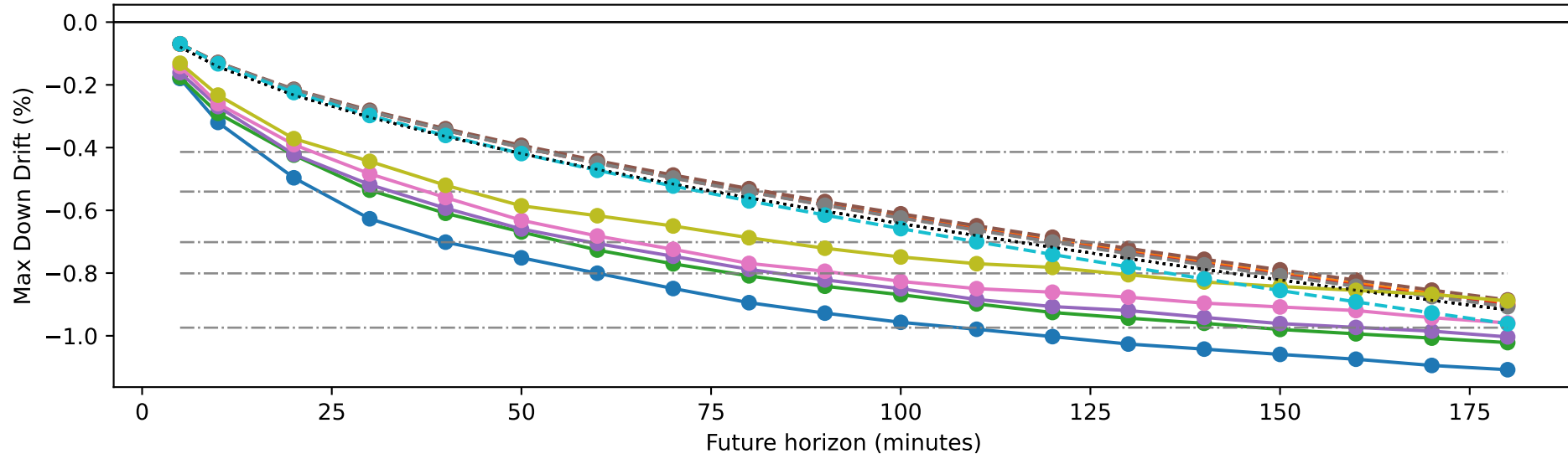
META - Intraday q99 - trade_volume_above_mid - ABS



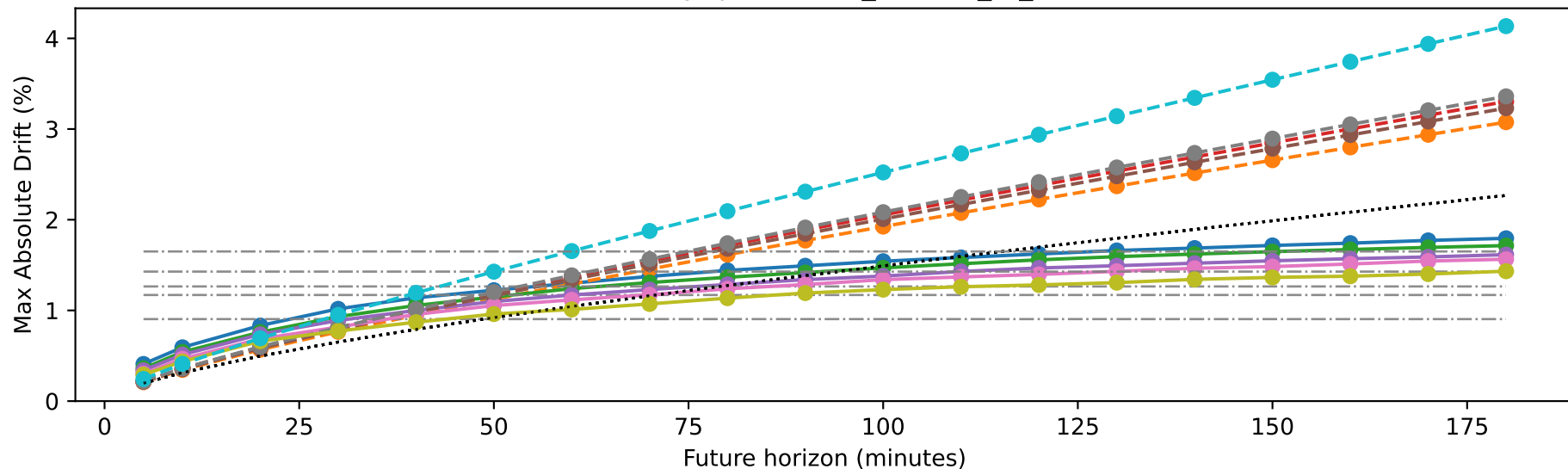
META - Intraday q99 - trade_volume_at_mid - UP



META - Intraday q99 - trade_volume_at_mid - DOWN



META - Intraday q99 - trade_volume_at_mid - ABS



META - Daily Multi-Quantile Results

q90 / q95 / q99 daily excessive-volume results versus baseline.

Post-abnormal future drift means next-day Up / Down / Abs excursion after the abnormal day has been identified.

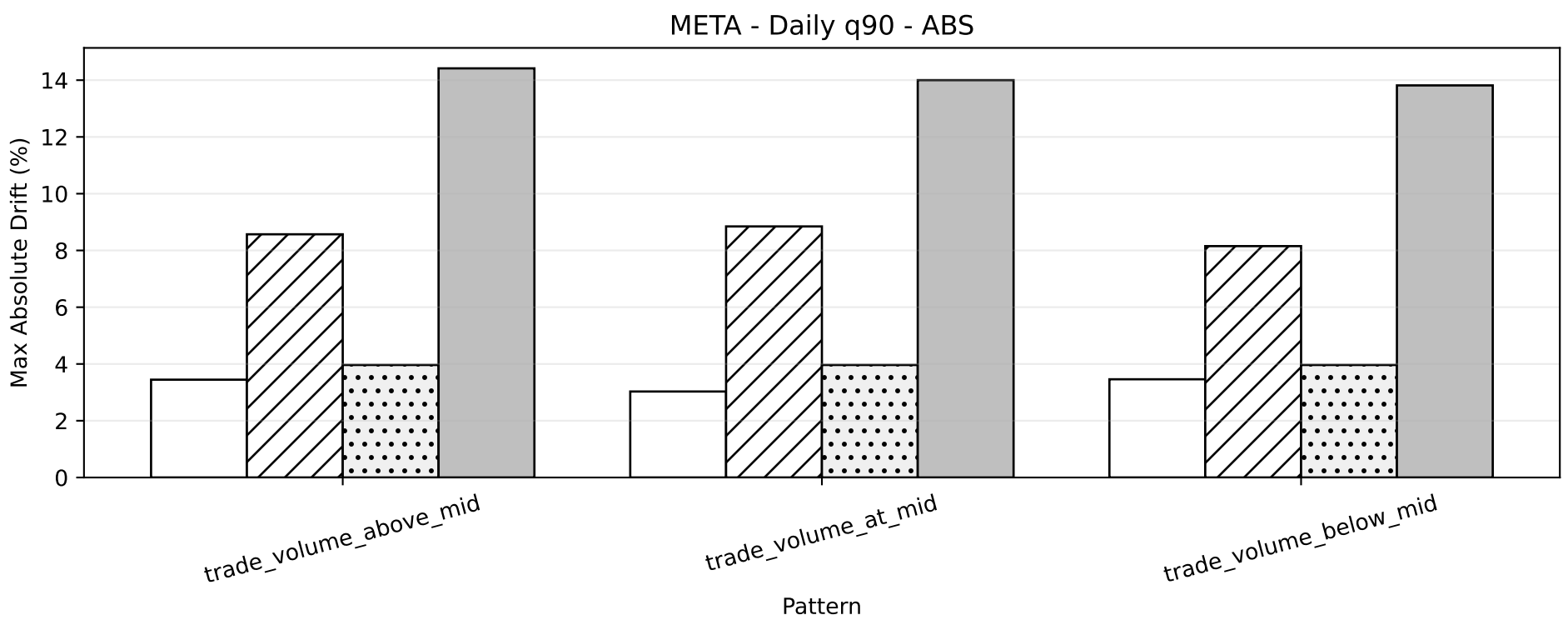
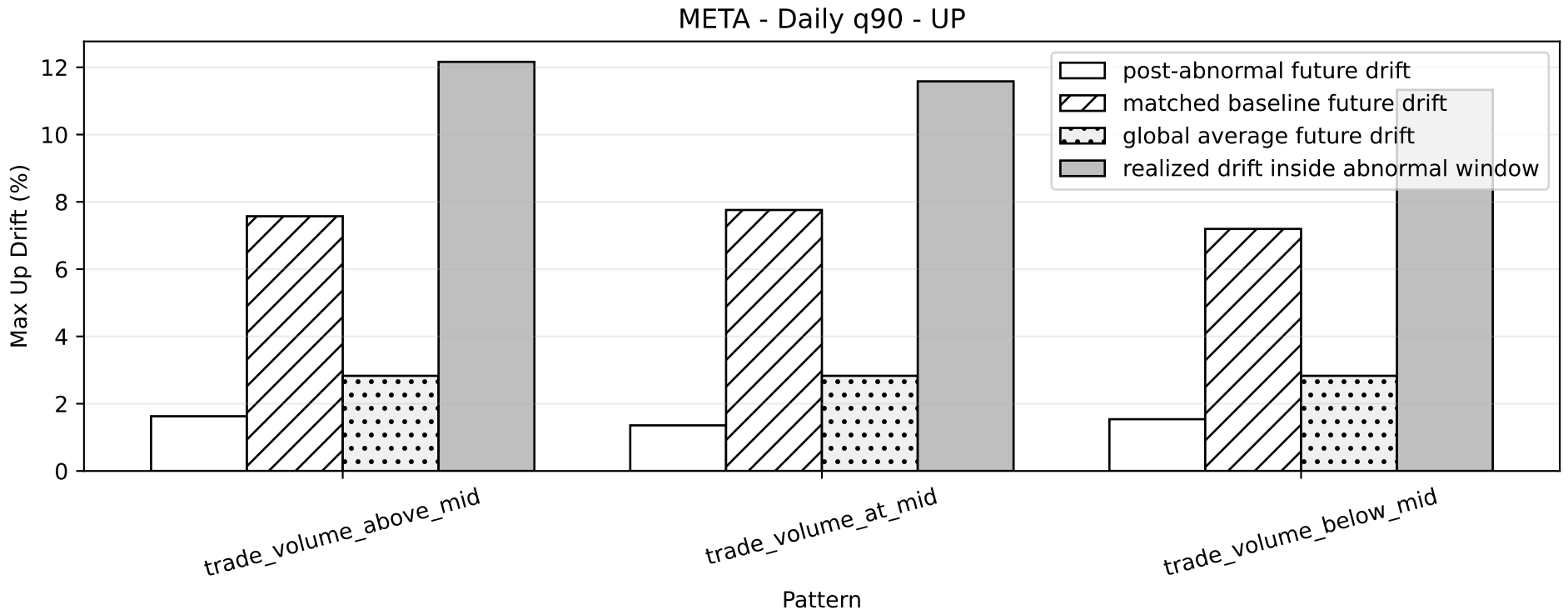
Matched baseline future drift is the corresponding non-abnormal baseline reference.

Additional bars show global average next-day drift and realized drift already inside the abnormal day itself.

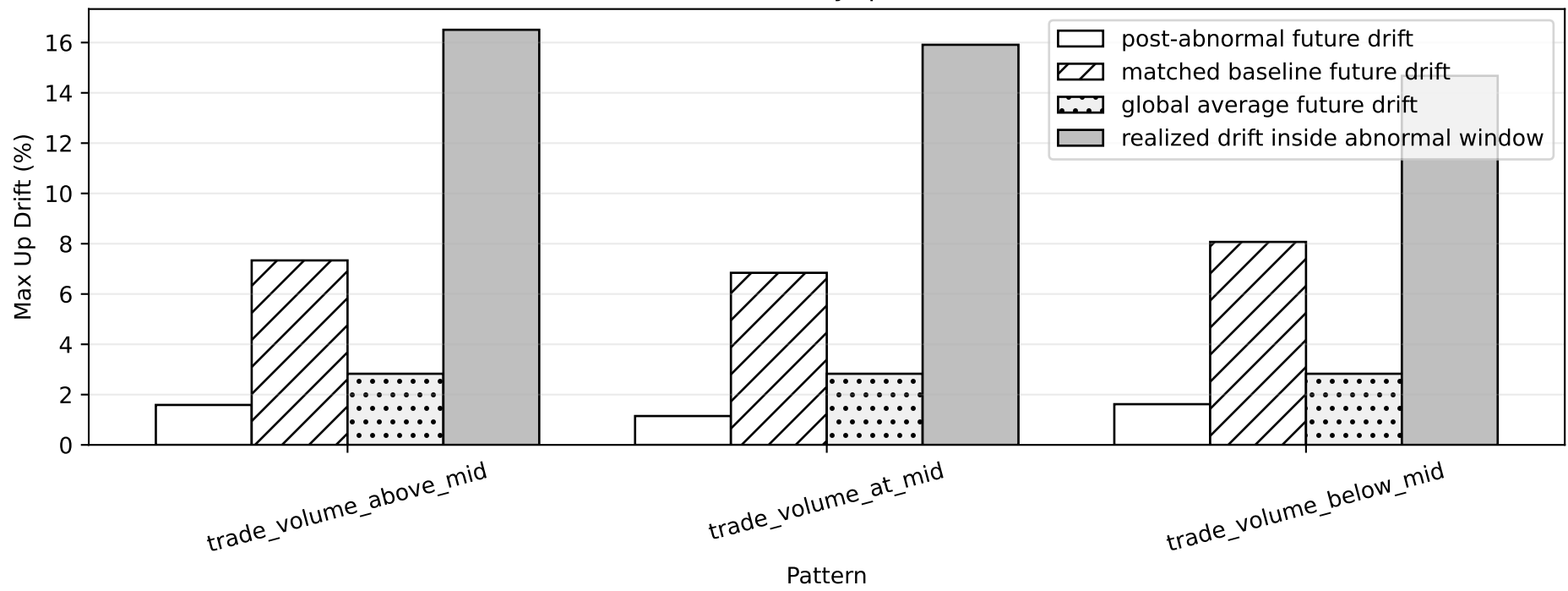
Independent line-view charts are also included, using quantile on the x-axis.

Down-side values are signed and are typically negative.

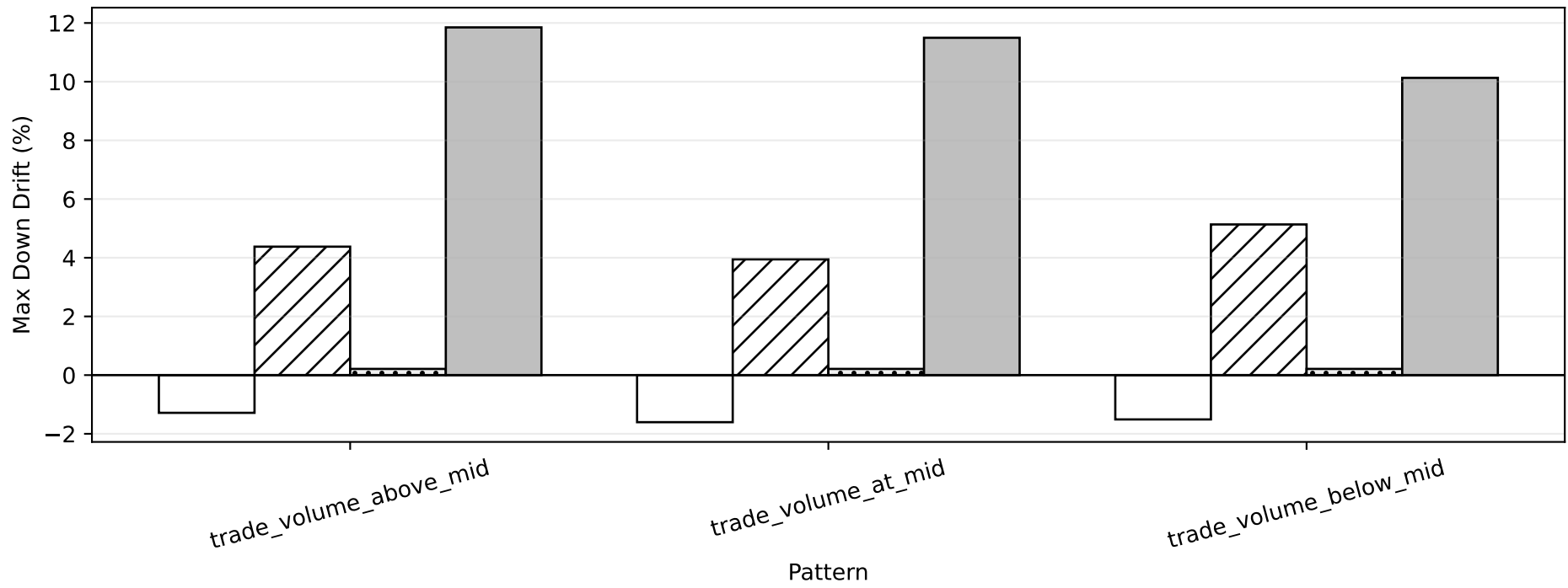
P-values are from excessive-day excess over rolling normal-day baseline.



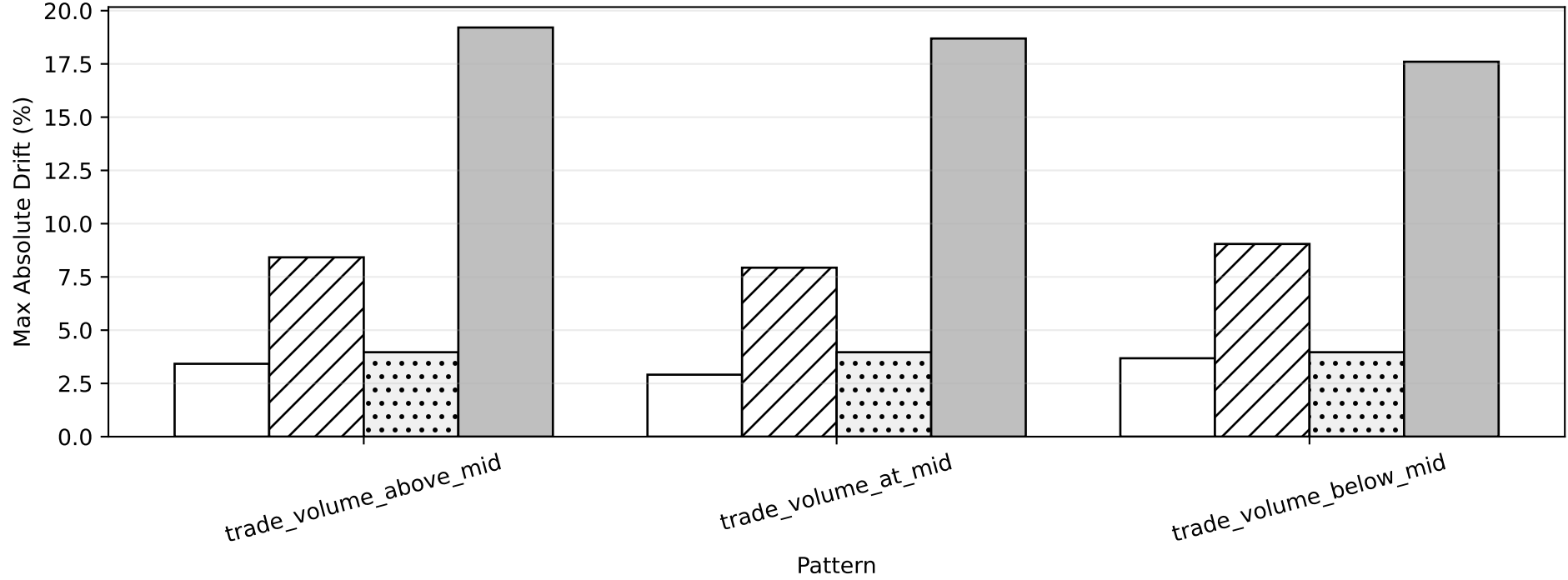
META - Daily q95 - UP



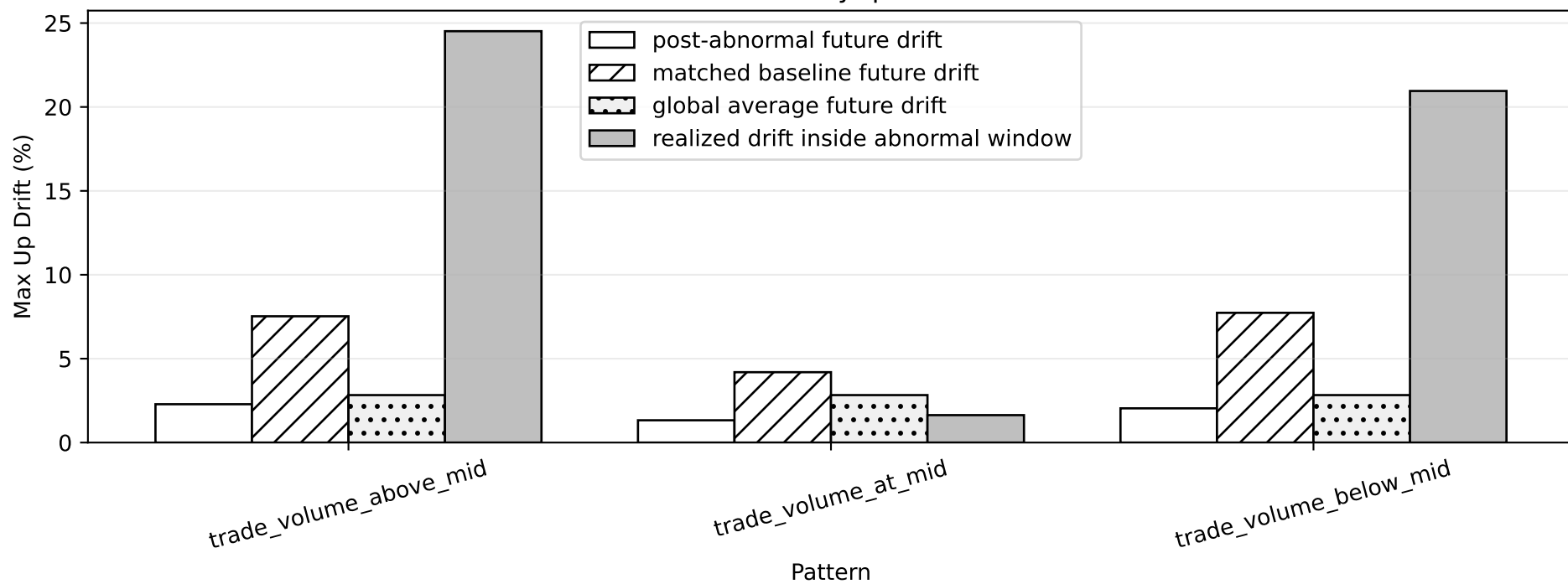
META - Daily q95 - DOWN



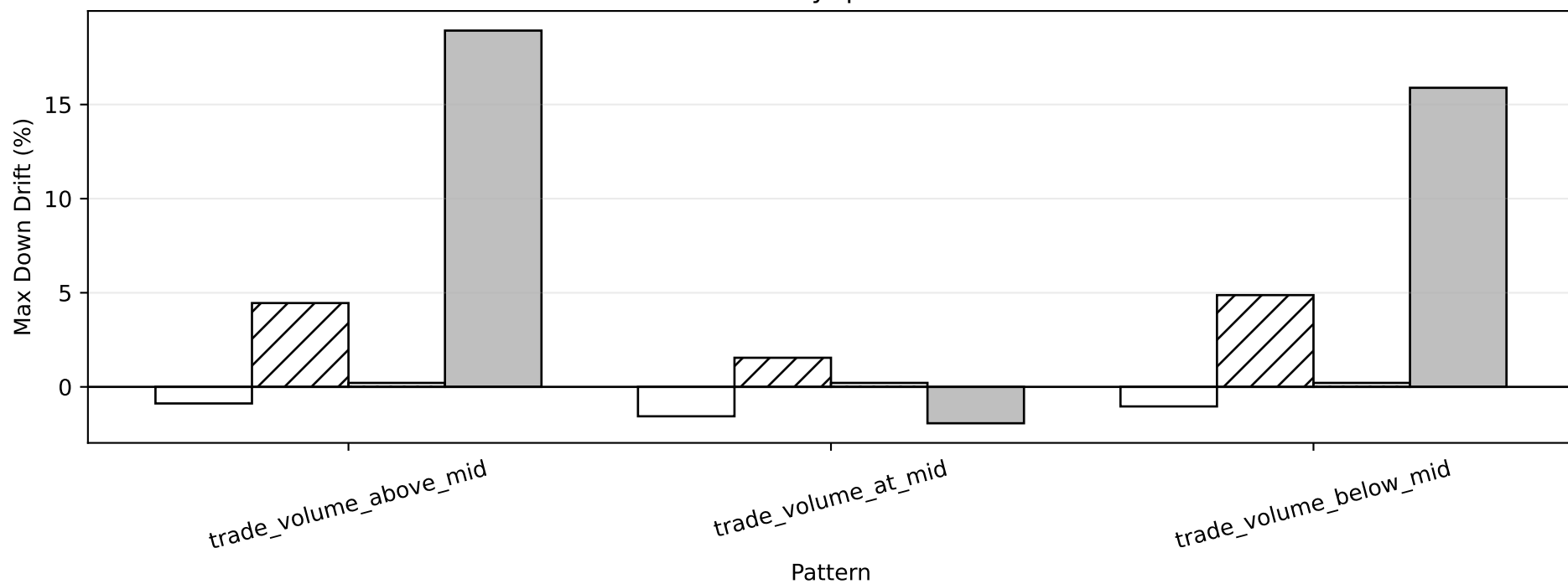
META - Daily q95 - ABS



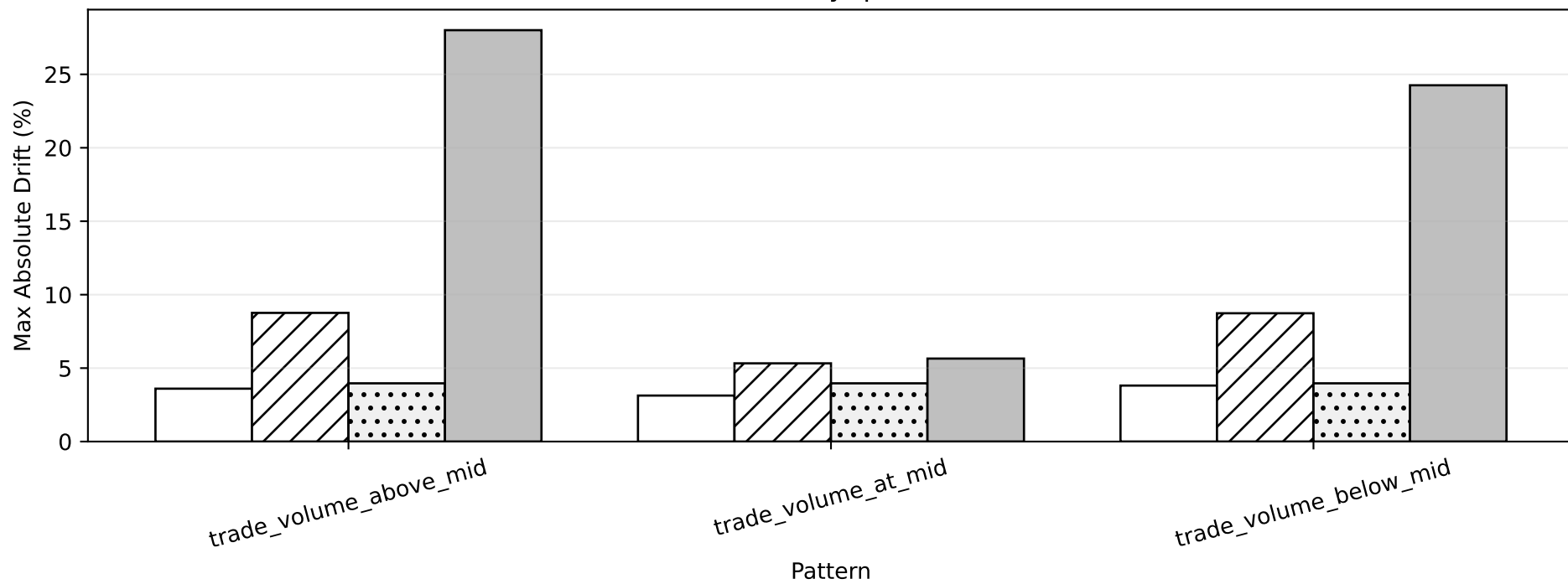
META - Daily q99 - UP



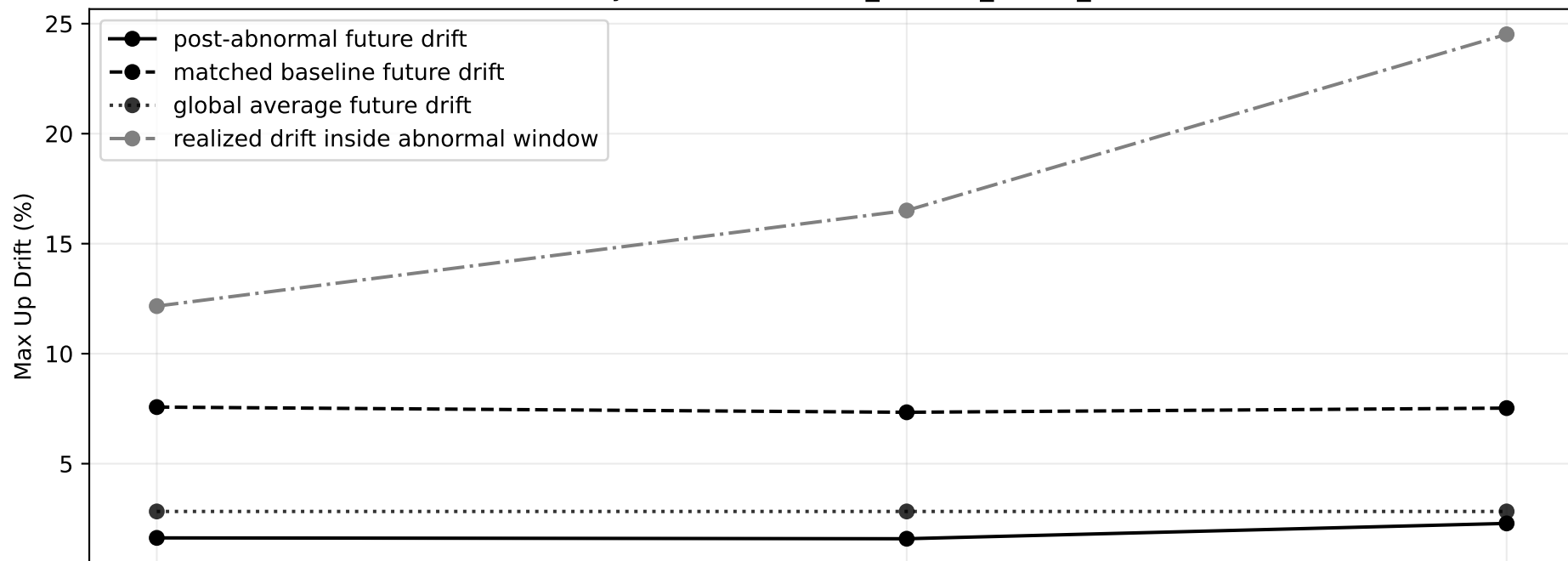
META - Daily q99 - DOWN



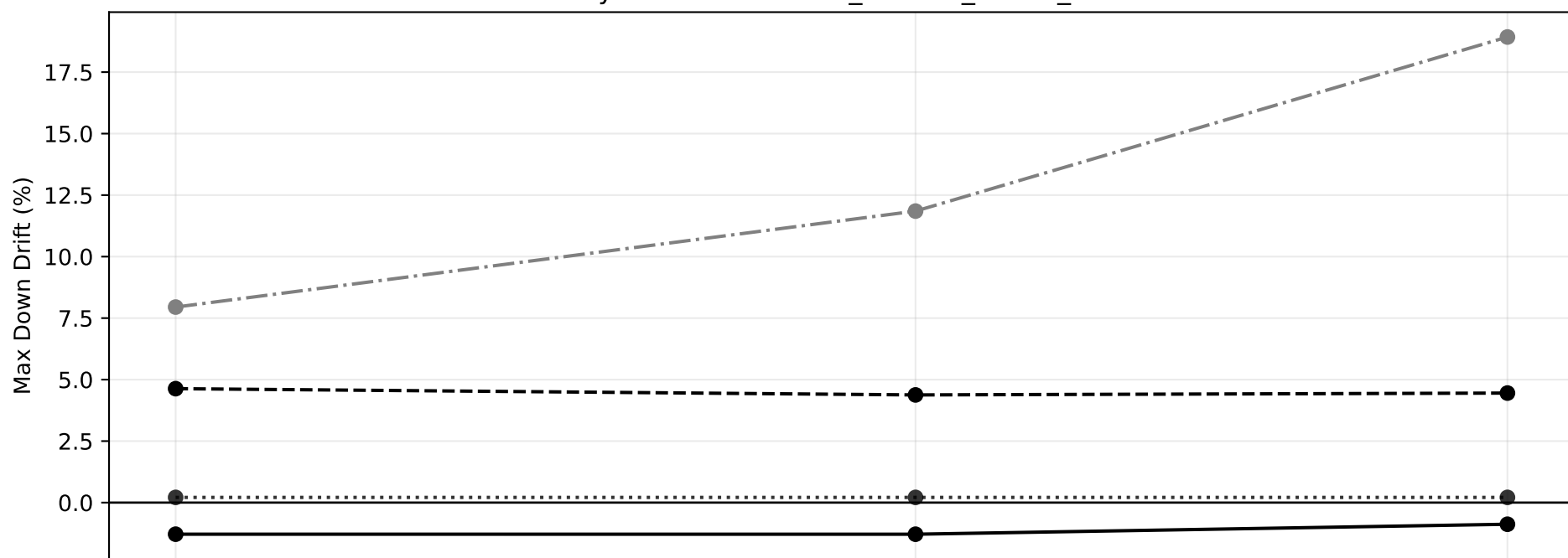
META - Daily q99 - ABS



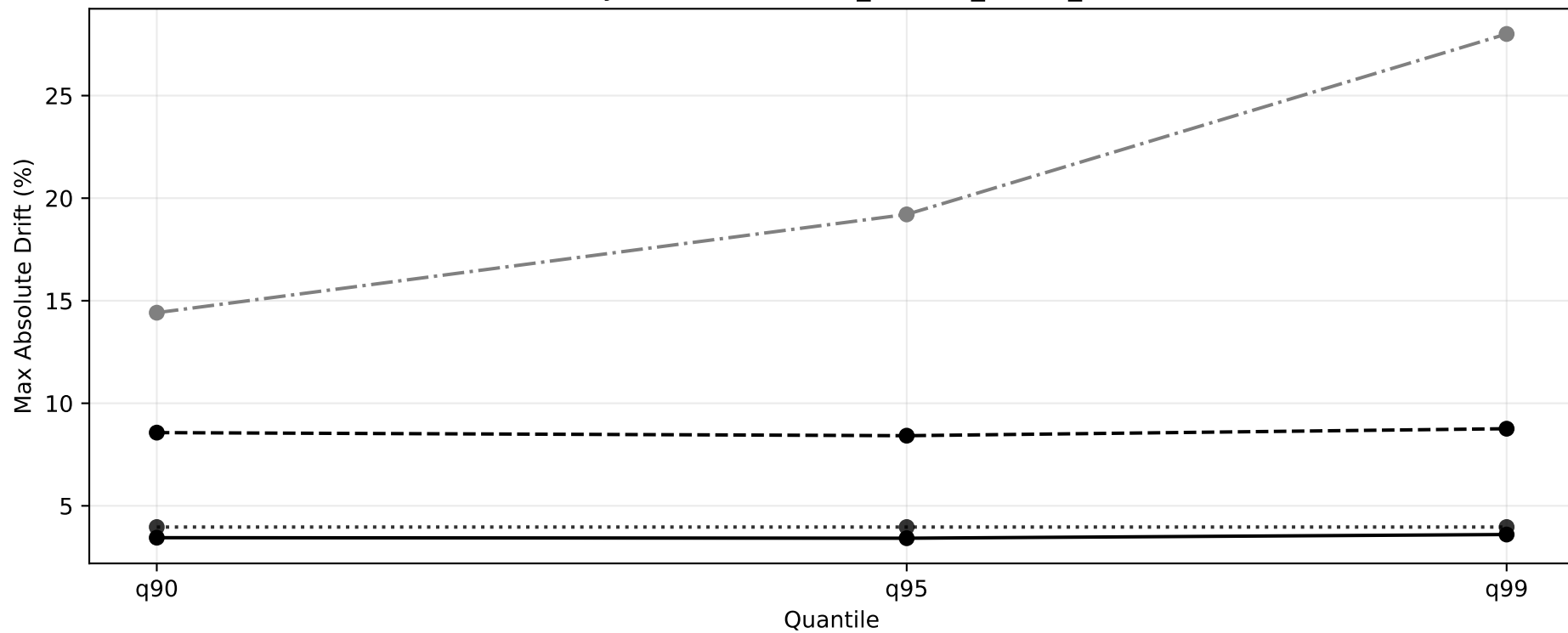
META - Daily Line View - trade_volume_above_mid - UP



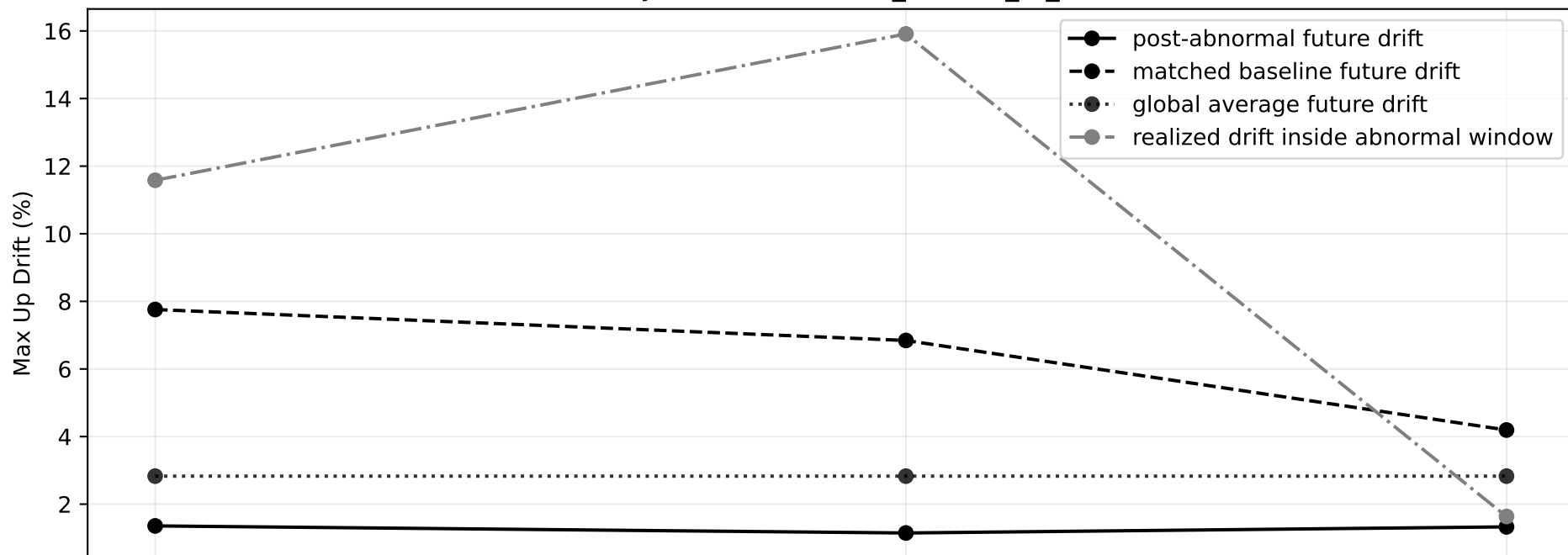
META - Daily Line View - trade_volume_above_mid - DOWN



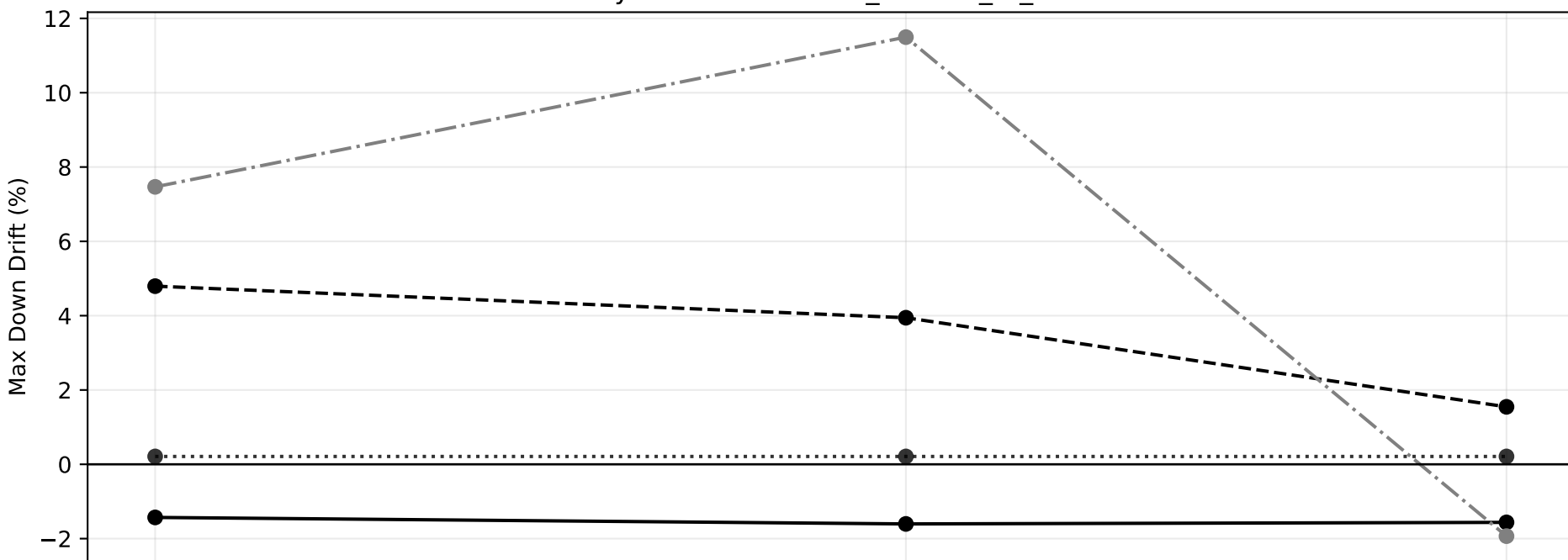
META - Daily Line View - trade_volume_above_mid - ABS



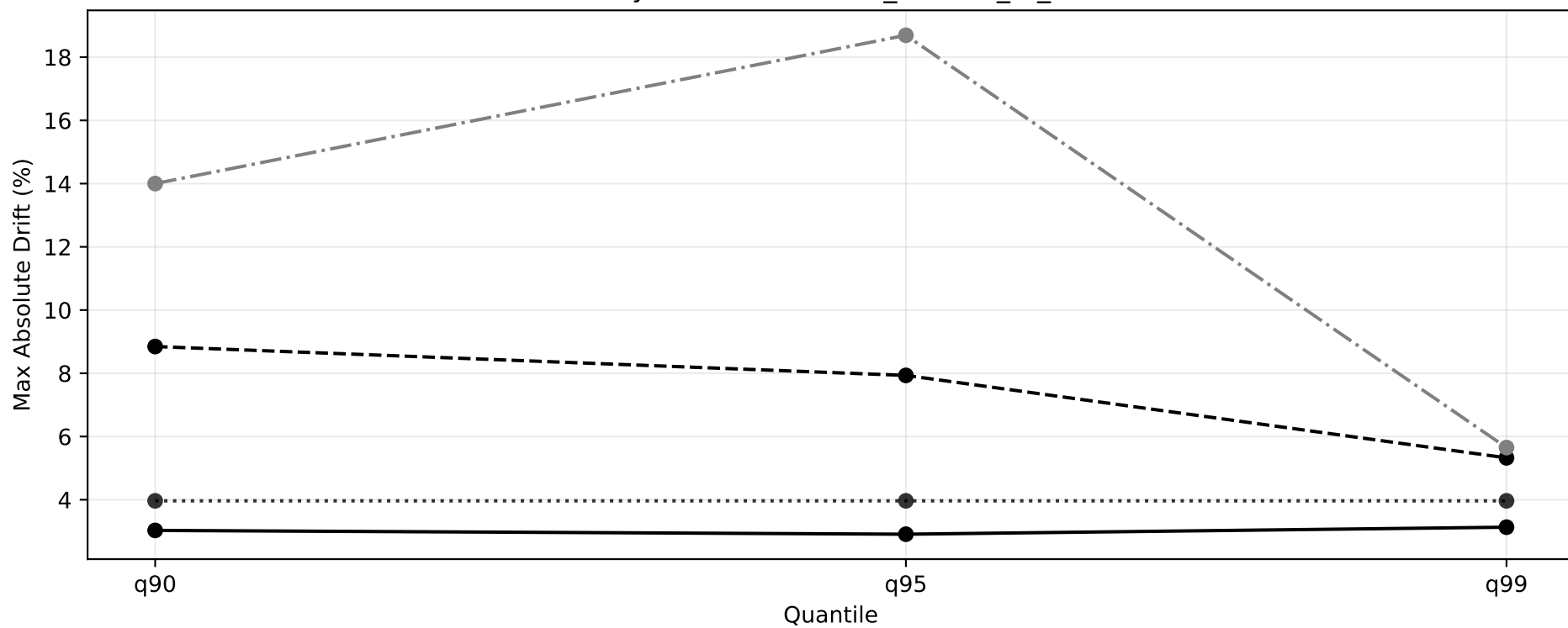
META - Daily Line View - trade_volume_at_mid - UP



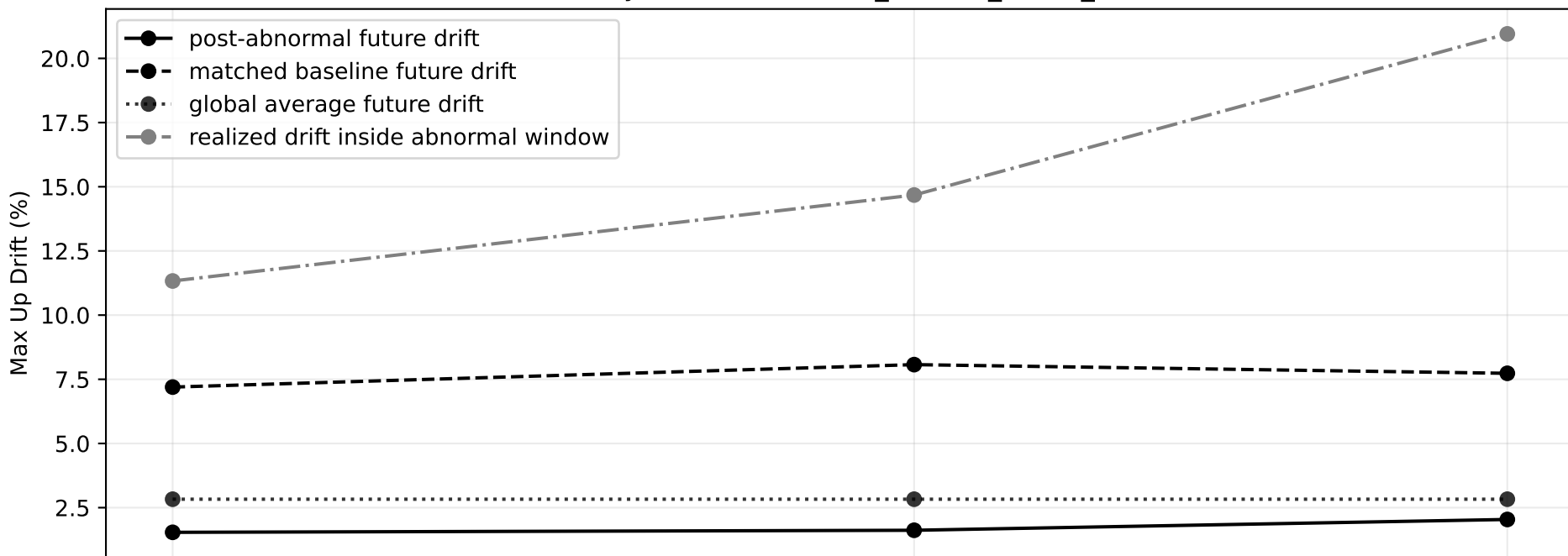
META - Daily Line View - trade_volume_at_mid - DOWN



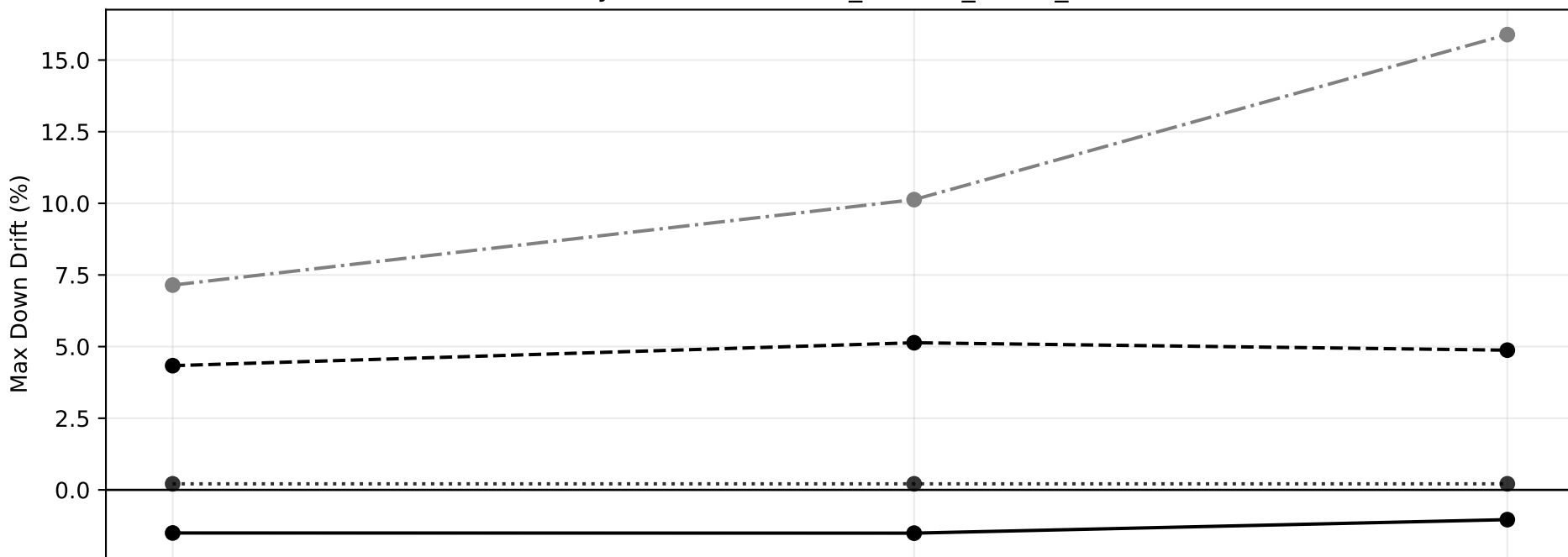
META - Daily Line View - trade_volume_at_mid - ABS



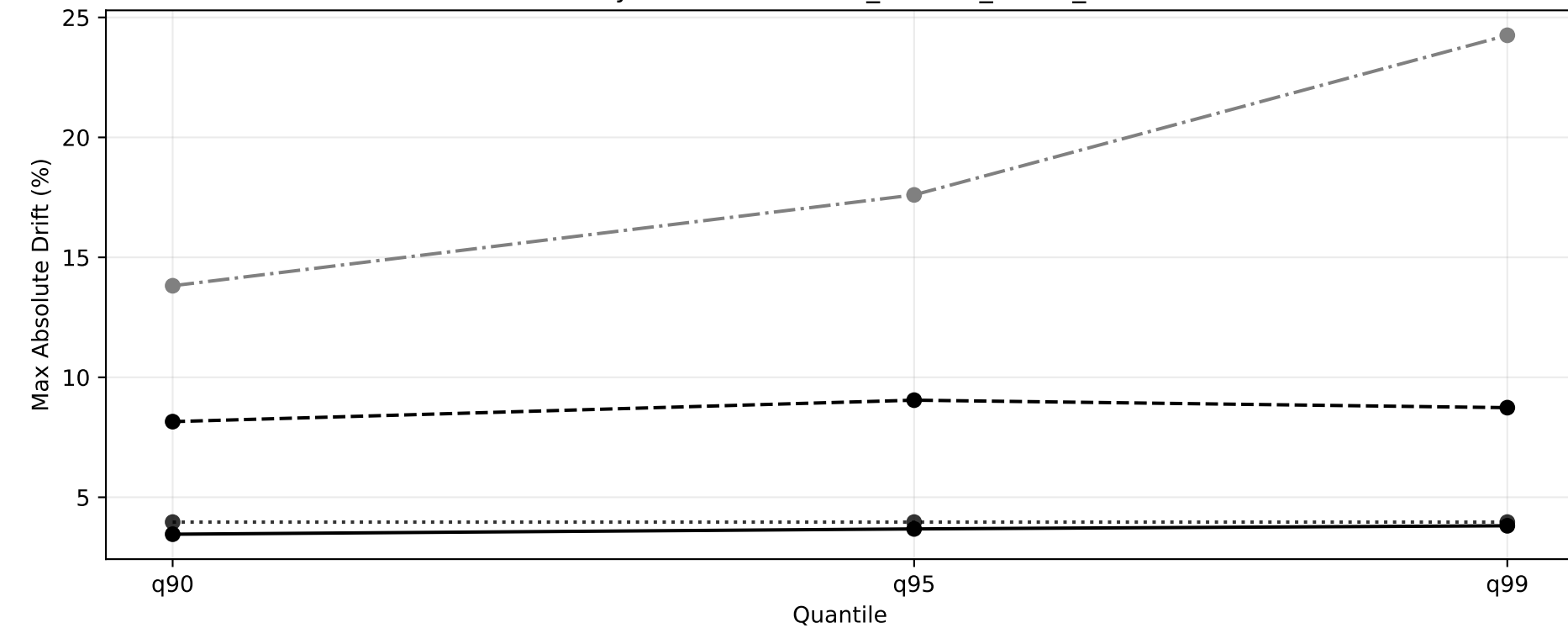
META - Daily Line View - trade_volume_below_mid - UP



META - Daily Line View - trade_volume_below_mid - DOWN



META - Daily Line View - trade_volume_below_mid - ABS



META - Weekly Summary Results

Weekly results are shown as combined Up / Down / Abs overview charts.

Post-abnormal future drift means next-week Up / Down / Abs excursion after the abnormal week has been identified.

Matched baseline future drift is the corresponding non-abnormal baseline reference.

Additional bars show global average next-week drift and realized drift already inside the abnormal week itself.

Within each pattern group, the weekly bars are ordered from left to right as q90, q95, q99.

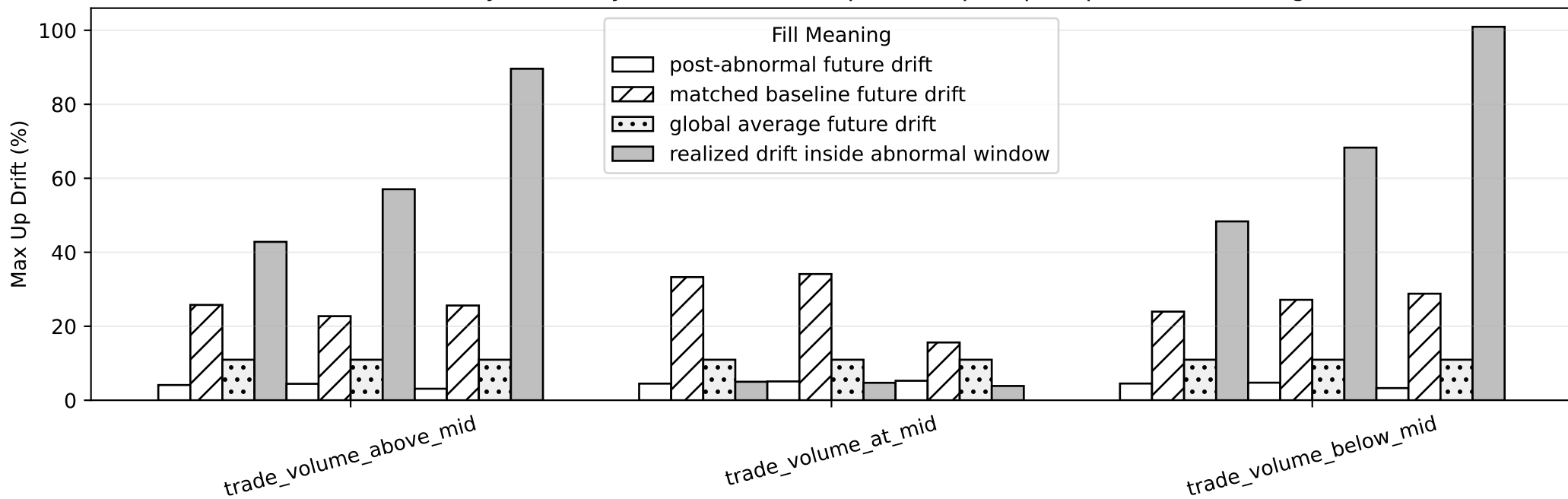
Legend fill and hatch styles explain which bars are post-abnormal future drift, matched baseline, global average, and realized drift.

Independent line-view charts are also included, using quantile on the x-axis.

Values are next-week excursions (%) relative to current week close.

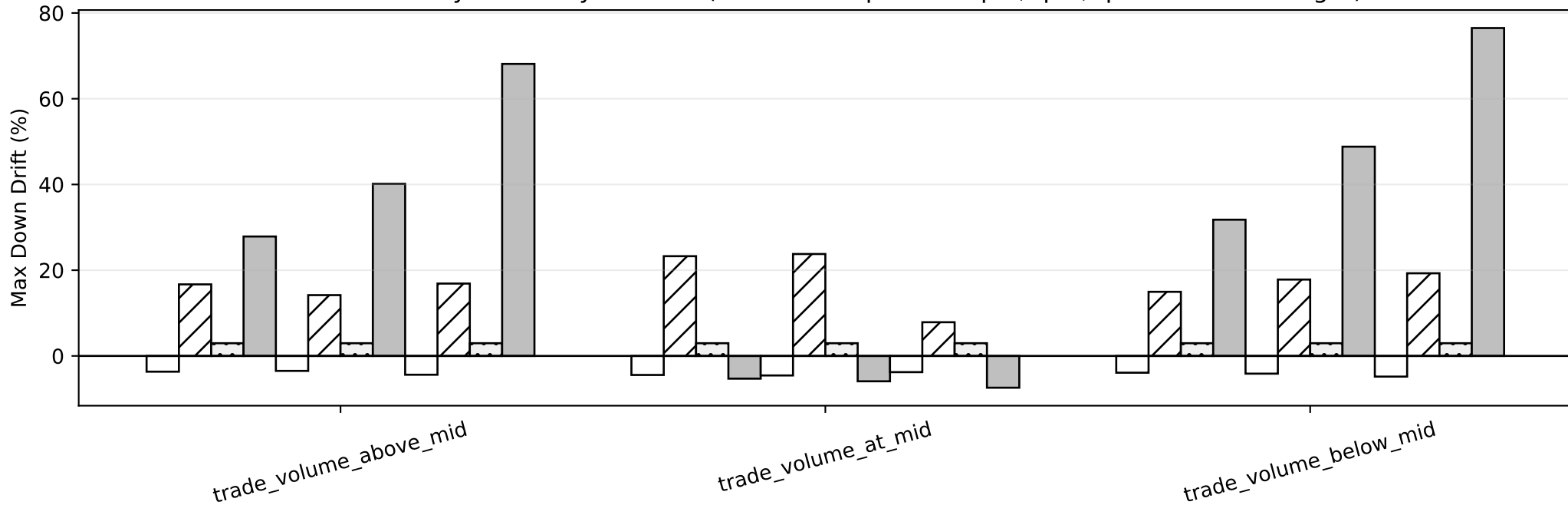
Down-side values are signed and are typically negative.

META - Weekly Summary - UP (within each pattern: q90, q95, q99 from left to right)



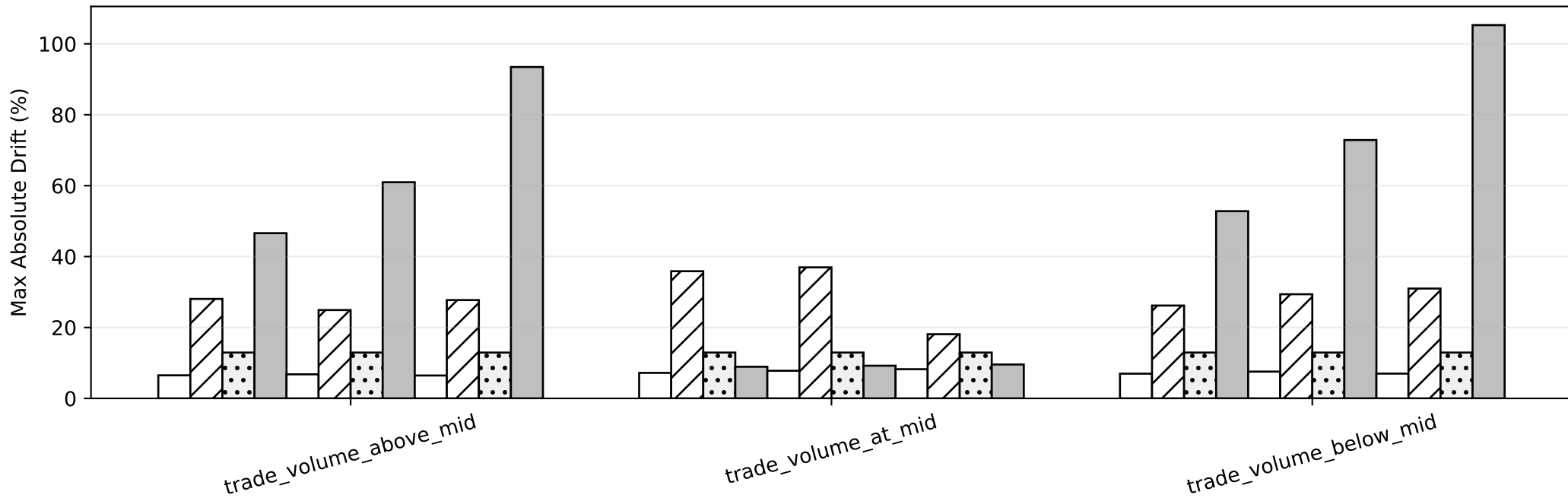
Pattern group (within each group: q90, q95, q99 from left to right)

META - Weekly Summary - DOWN (within each pattern: q90, q95, q99 from left to right)



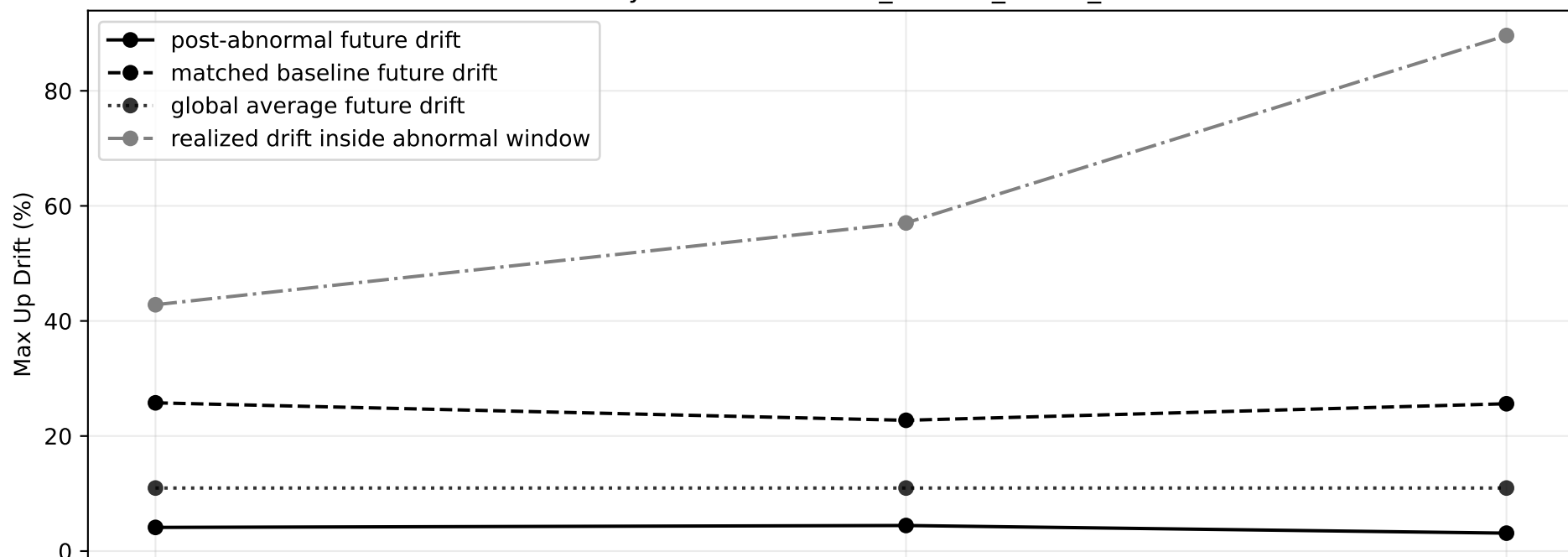
Pattern group (within each group: q90, q95, q99 from left to right)

META - Weekly Summary - ABS (within each pattern: q90, q95, q99 from left to right)

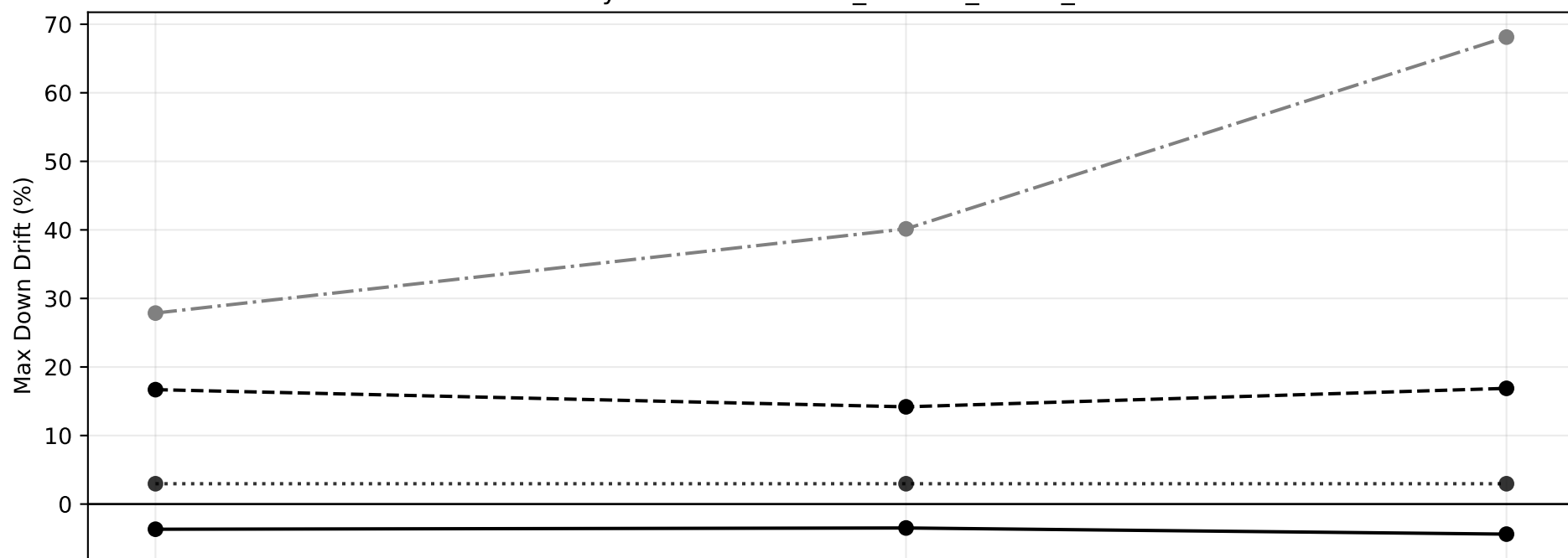


Pattern group (within each group: q90, q95, q99 from left to right)
For each pattern group, bars are ordered left to right as q90, q95, q99.

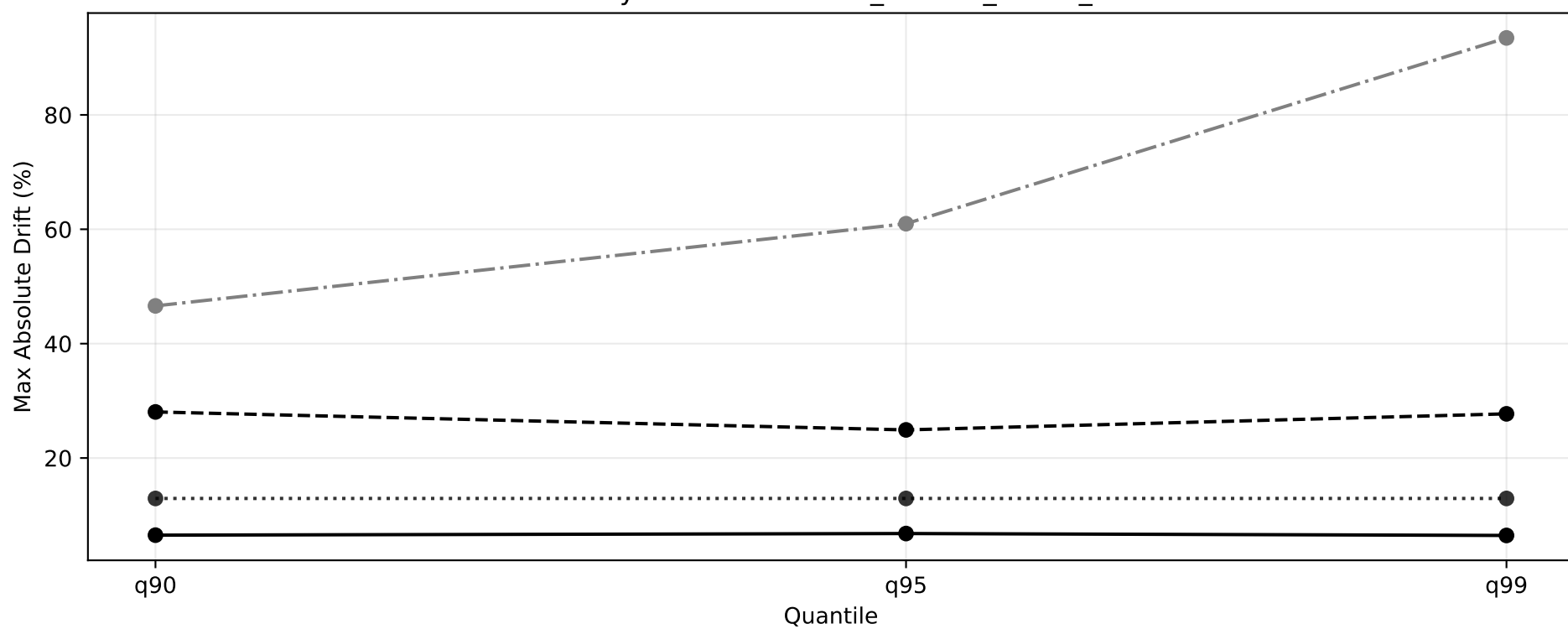
META - Weekly Line View - trade_volume_above_mid - UP

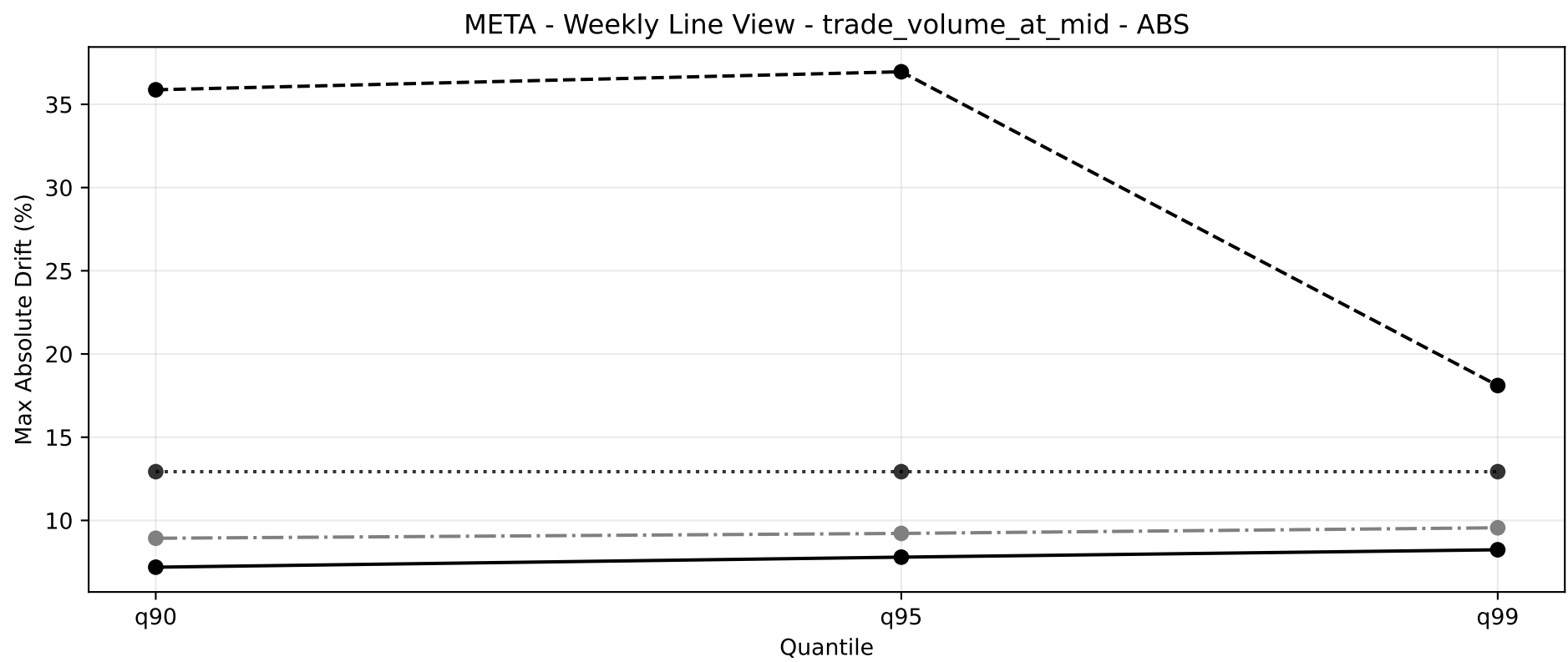
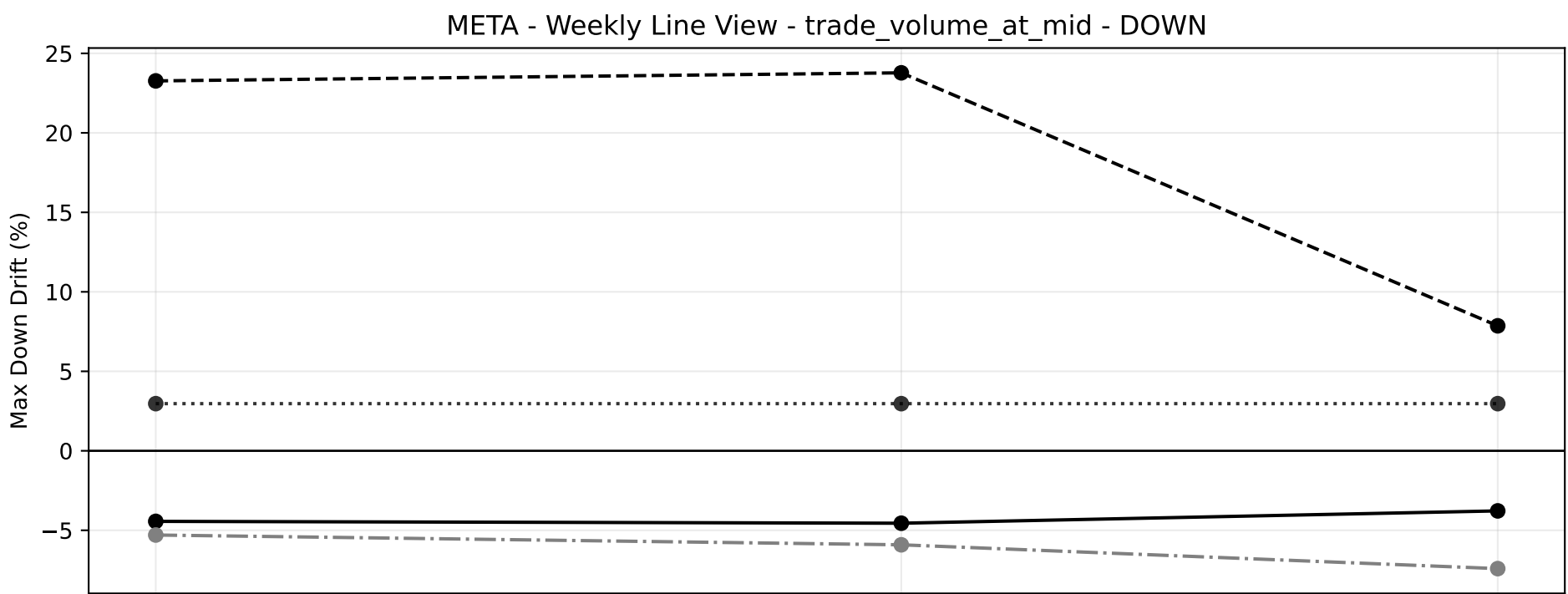
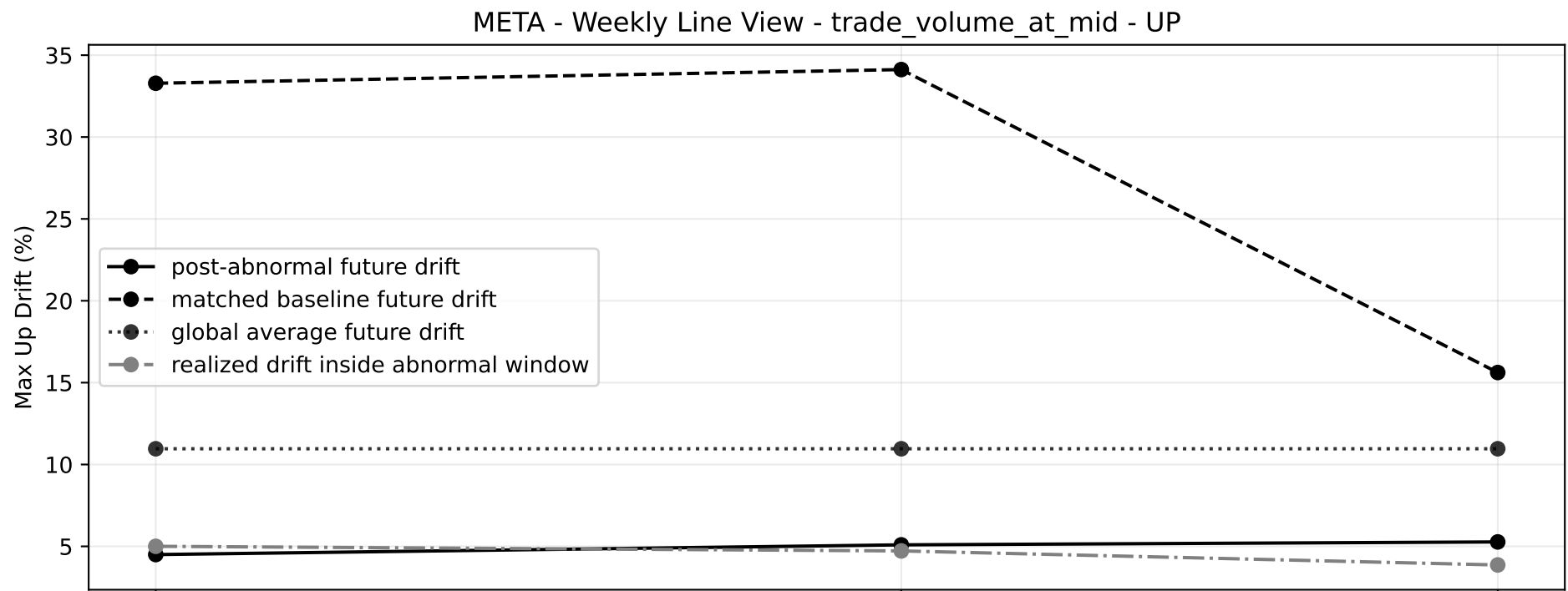


META - Weekly Line View - trade_volume_above_mid - DOWN

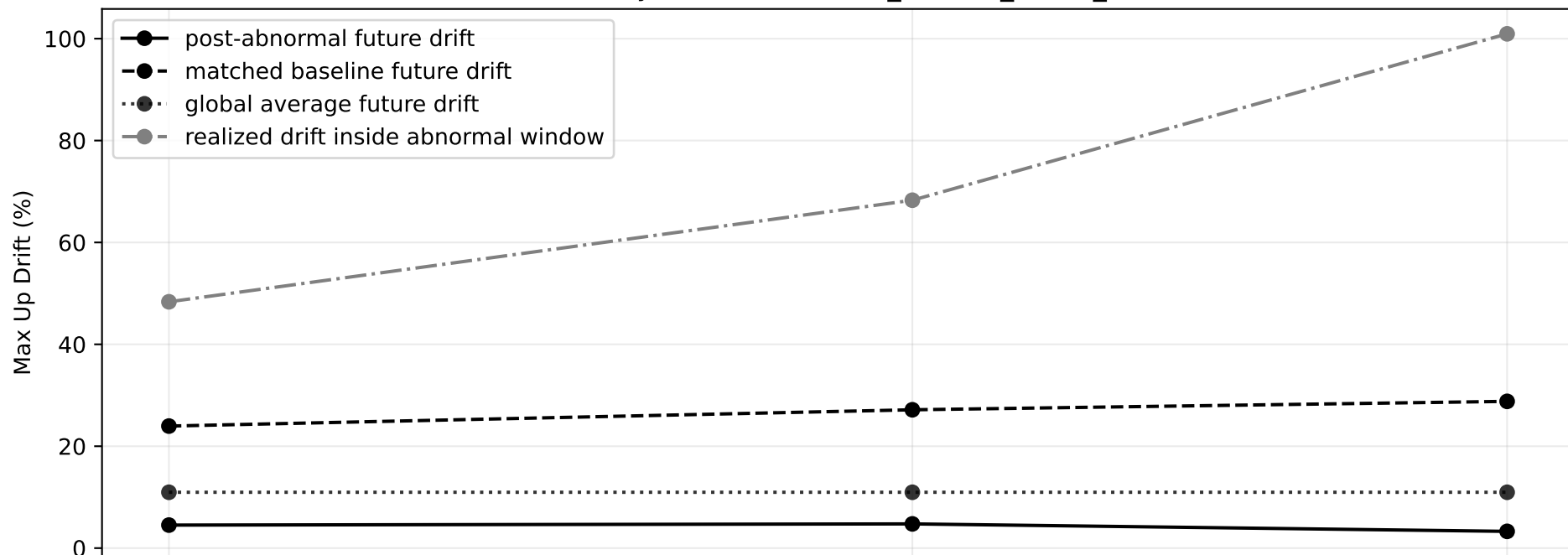


META - Weekly Line View - trade_volume_above_mid - ABS

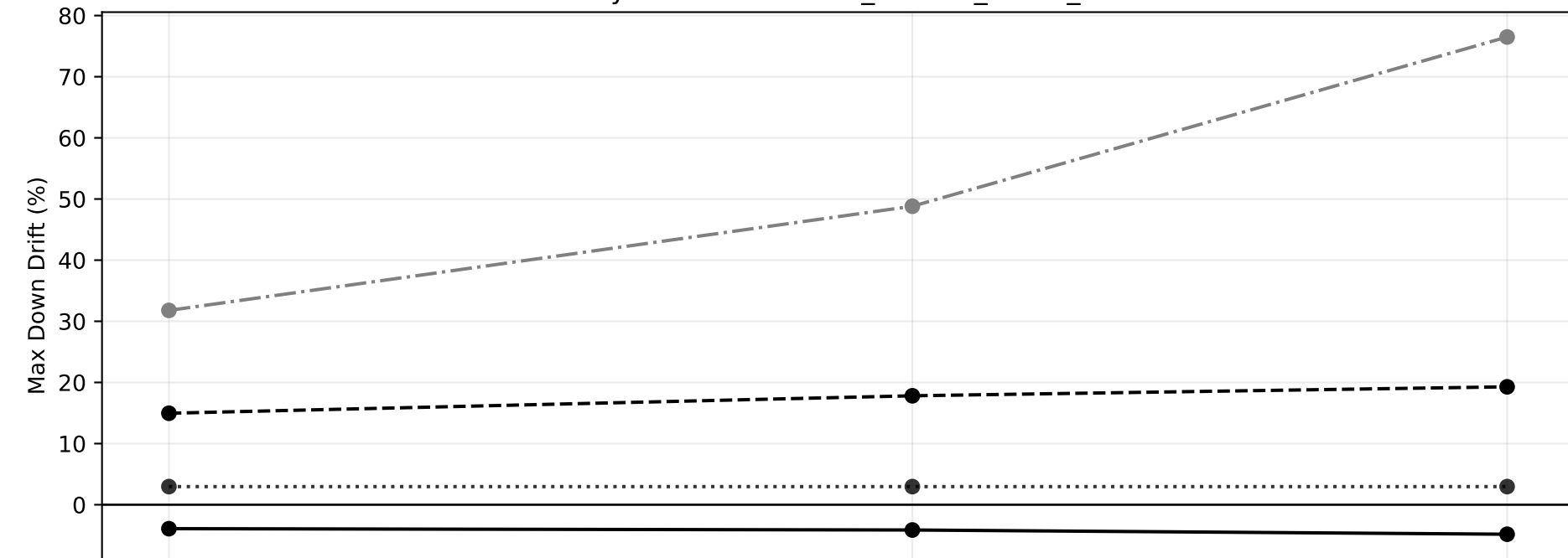




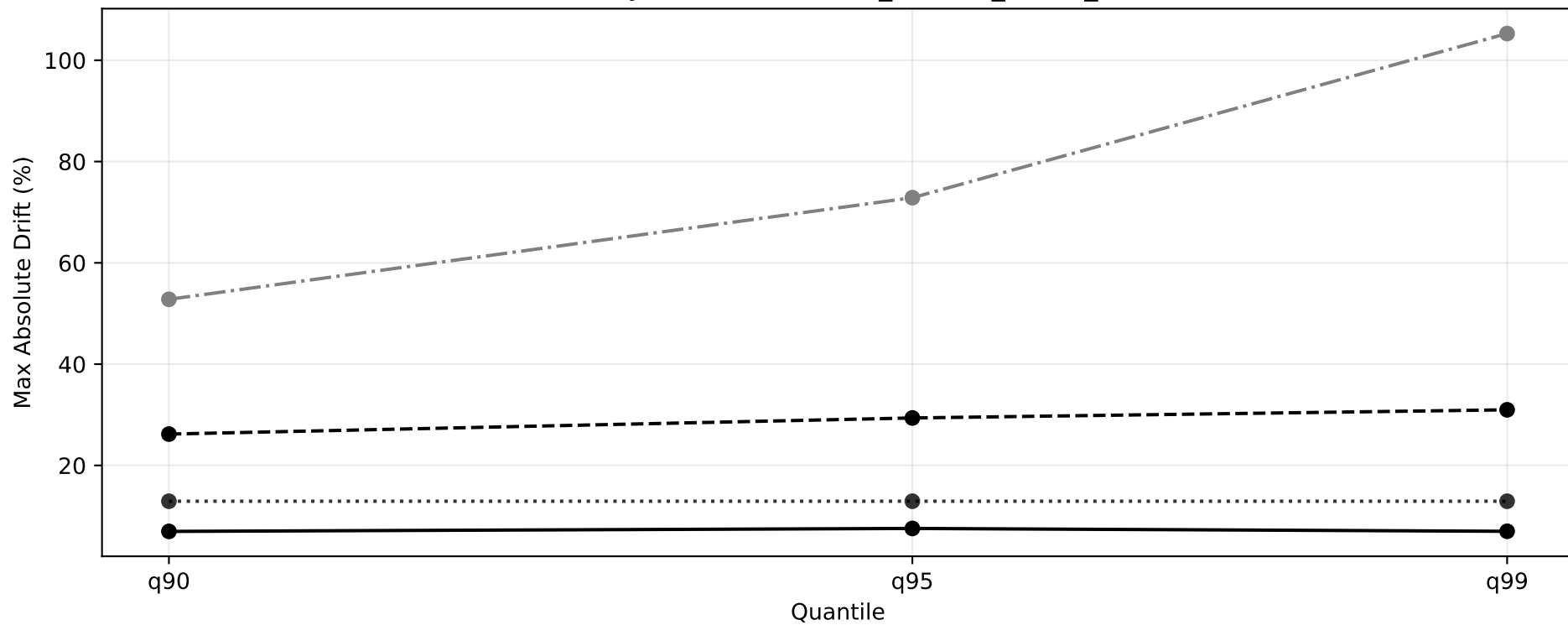
META - Weekly Line View - trade_volume_below_mid - UP



META - Weekly Line View - trade_volume_below_mid - DOWN



META - Weekly Line View - trade_volume_below_mid - ABS



META - Drift and Volume-Change Link

This section focuses on direct volume-drift links without repeating lower-value rolling-correlation charts. Scatter plots compare total / above-mid / at-mid / below-mid volume against max up/down drift. Correlation p-values for these plots are summarized in the P-Value Summary tables. Intraday volume-drift scatter plots are only produced for 30-minute and 60-minute windows.

META - P-Value Summary

This section reports correlation p-values for the scatter/regression plots.
Intraday rows are restricted to 30m and 60m windows for readability.

META - P-Value Summary - Correlation p-values

layer	window	volume	n_up	up_corr_p
intraday	30m	total	399535	0.00331102
intraday	30m	above_mid	399535	0.000134128
intraday	30m	at_mid	399535	0.000270033
intraday	30m	below_mid	399535	0.00382048
intraday	60m	total	399506	5.4056e-21
intraday	60m	above_mid	399506	1.47392e-26
intraday	60m	at_mid	399506	0.0392281
intraday	60m	below_mid	399506	4.27091e-24
daily	daily	total	1041	0.272001
daily	daily	above_mid	1041	0.295295
daily	daily	at_mid	1041	0.85249
daily	daily	below_mid	1041	0.156923

META - P-Value Summary - Correlation p-values

n_down	down_corr_p
399535	4.29008e-105
399535	2.34002e-98
399535	3.94863e-104
399535	7.98744e-95
399506	2.00834e-164
399506	1.43209e-152
399506	1.47745e-190
399506	1.37109e-151
1041	0.157463
1041	0.177843
1041	0.855538
1041	0.084391

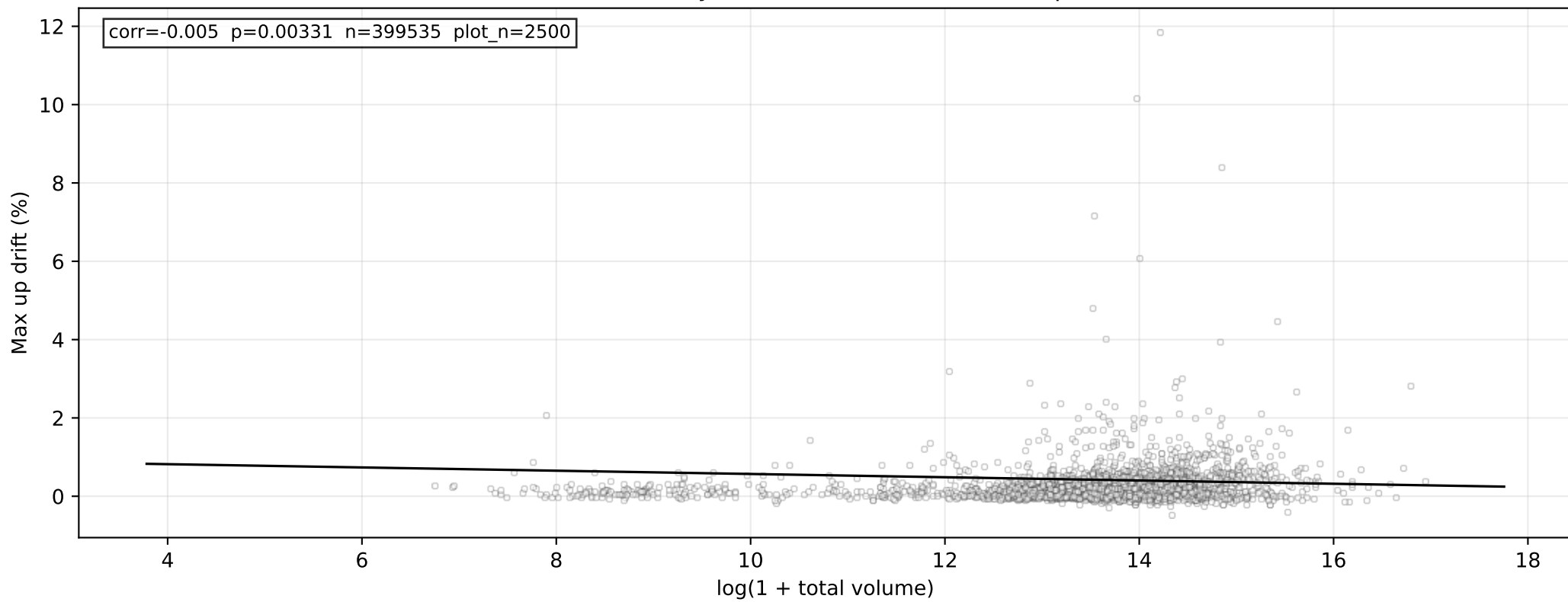
META - P-Value Summary - Correlation p-values

layer	window	volume	n_up	up_corr_p
weekly	weekly	total	217	0.350687
weekly	weekly	above_mid	217	0.263692
weekly	weekly	at_mid	217	0.64768
weekly	weekly	below_mid	217	0.311531

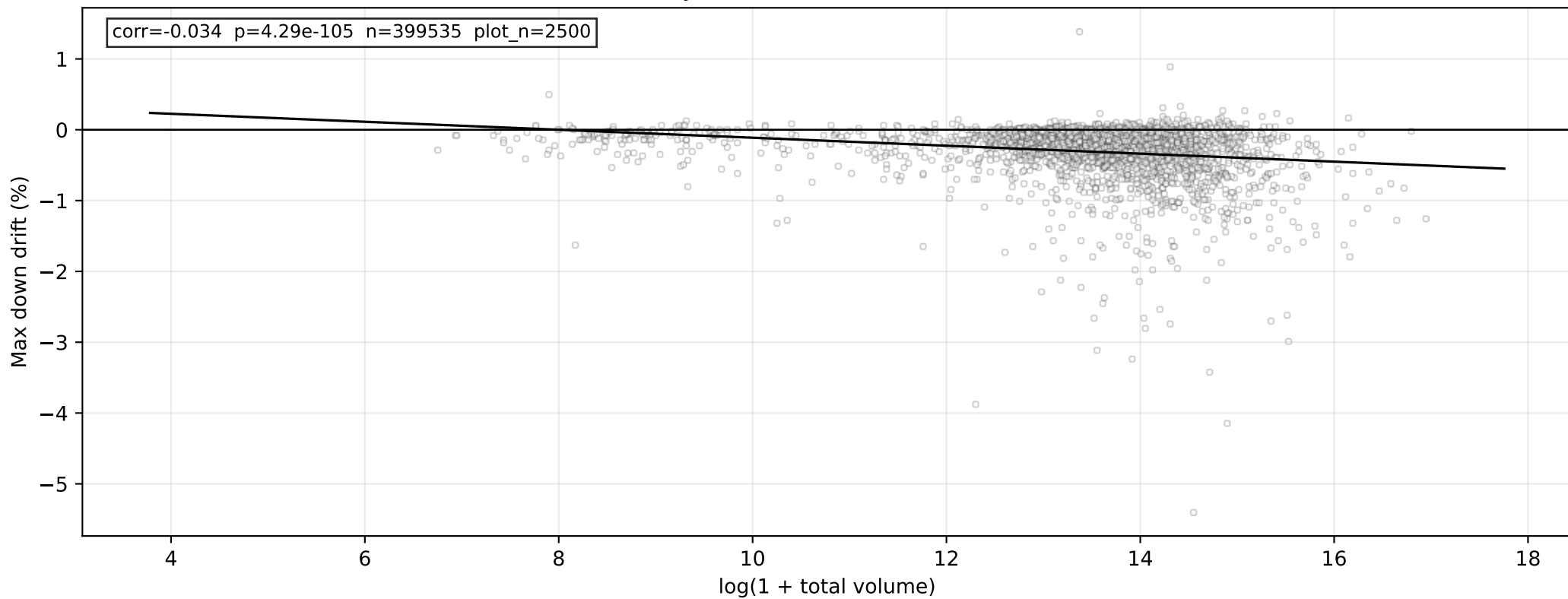
META - P-Value Summary - Correlation p-values

n_down	down_corr_p
217	0.260172
217	0.192294
217	0.82357
217	0.22974

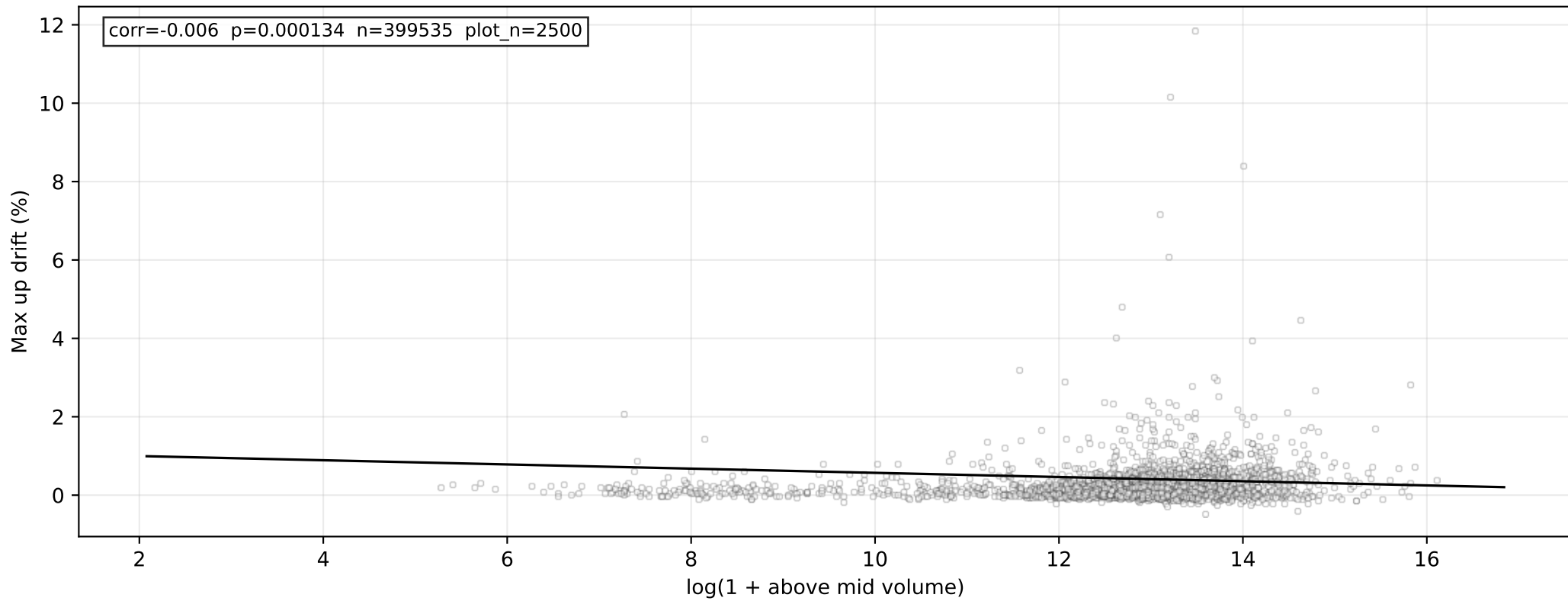
META - Intraday 30M Total Volume vs Max Up Drift



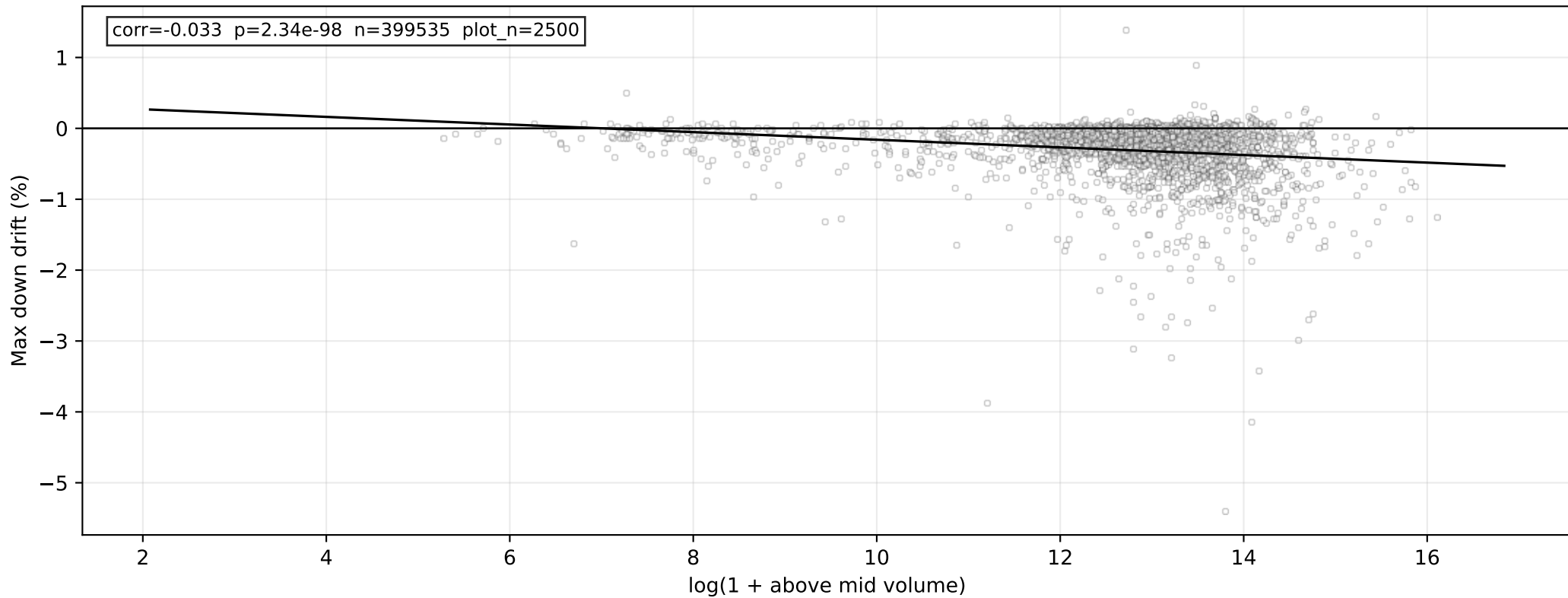
META - Intraday 30M Total Volume vs Max Down Drift



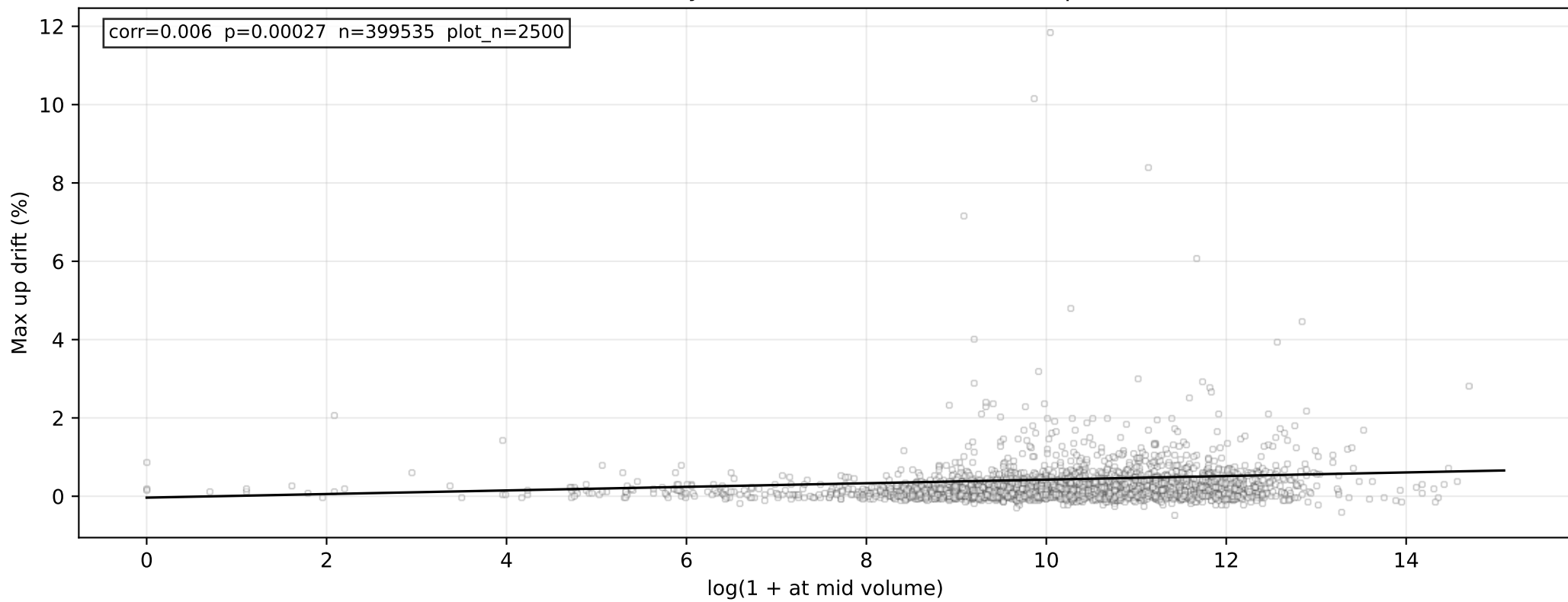
META - Intraday 30M Above Mid Volume vs Max Up Drift



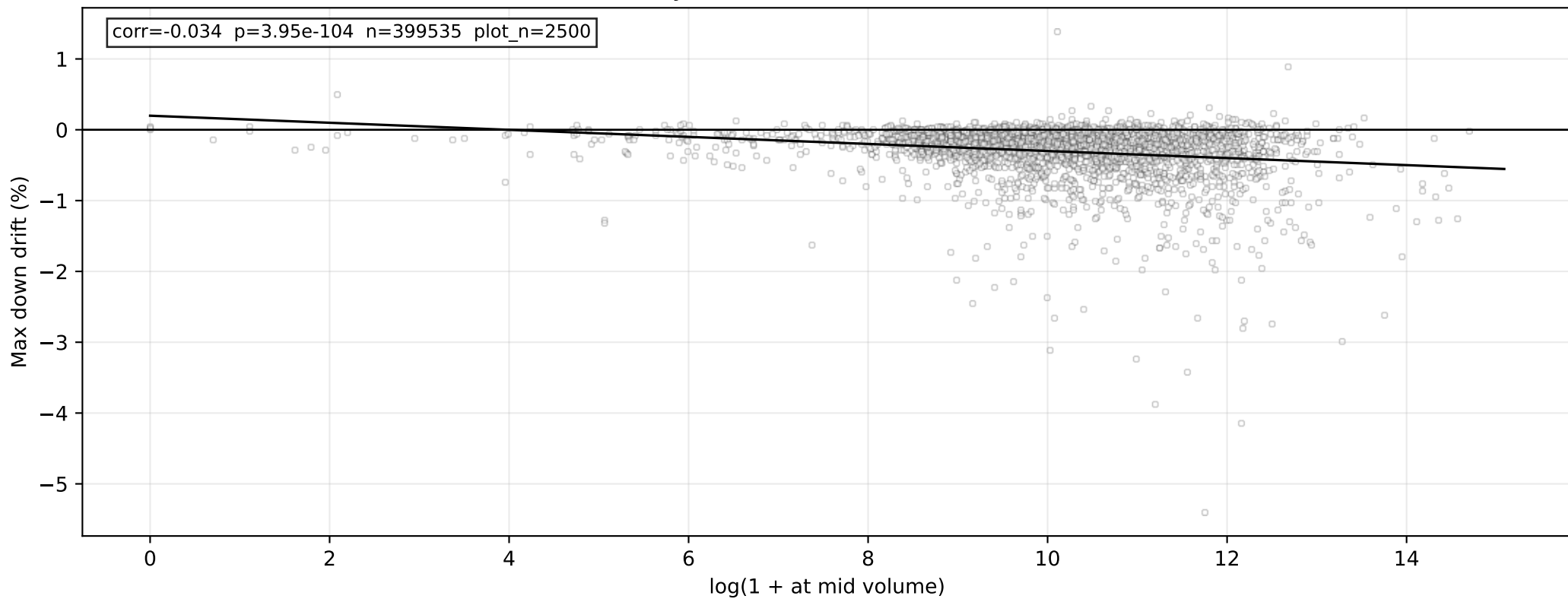
META - Intraday 30M Above Mid Volume vs Max Down Drift



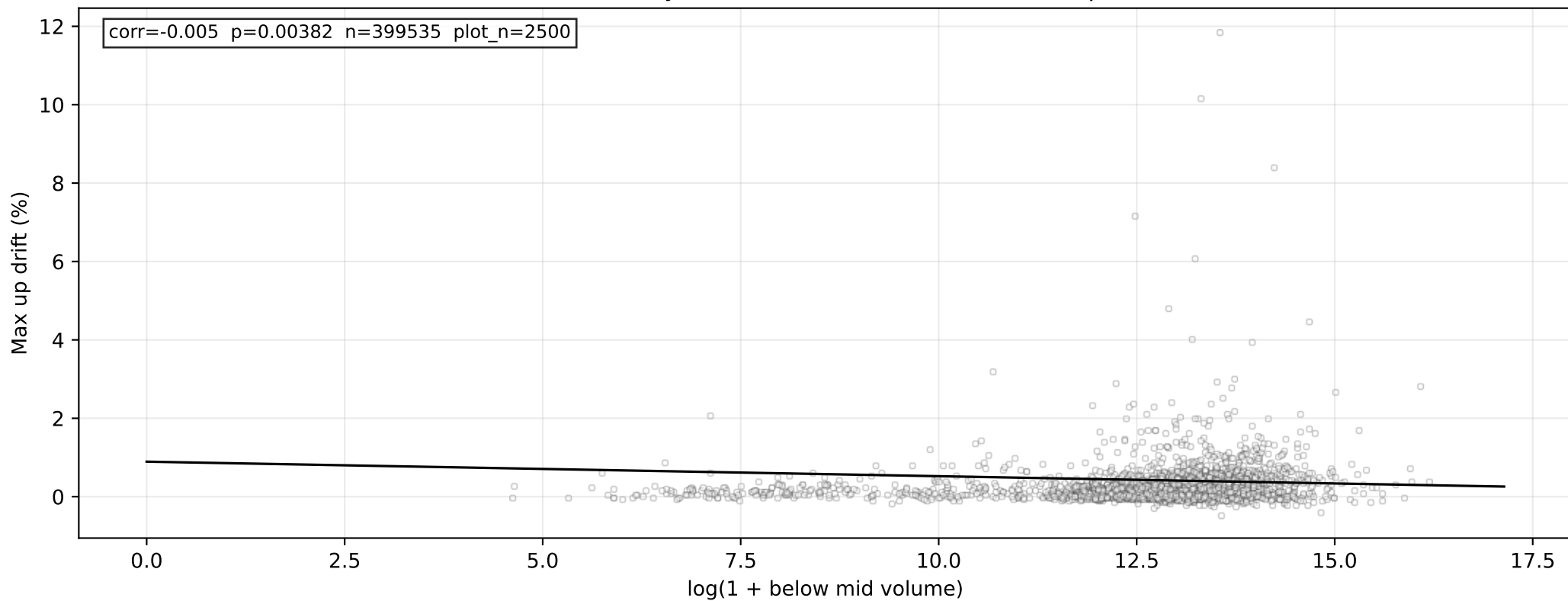
META - Intraday 30M At Mid Volume vs Max Up Drift



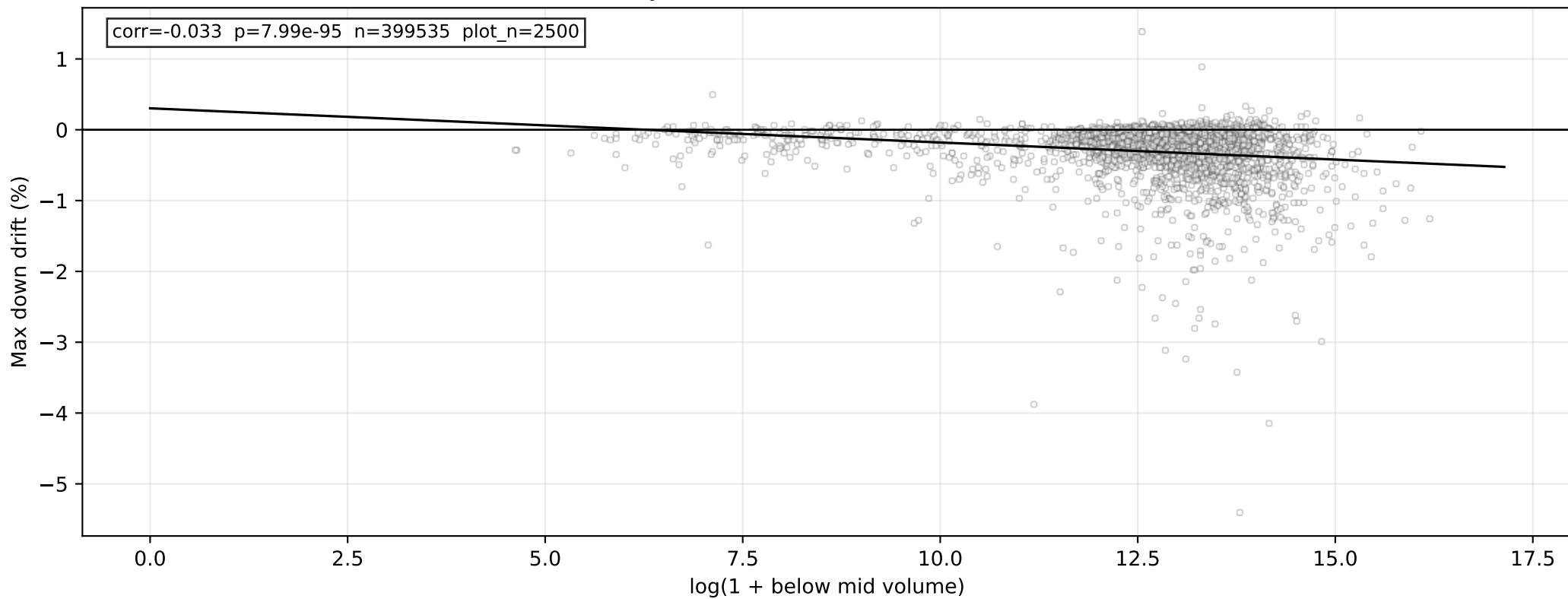
META - Intraday 30M At Mid Volume vs Max Down Drift



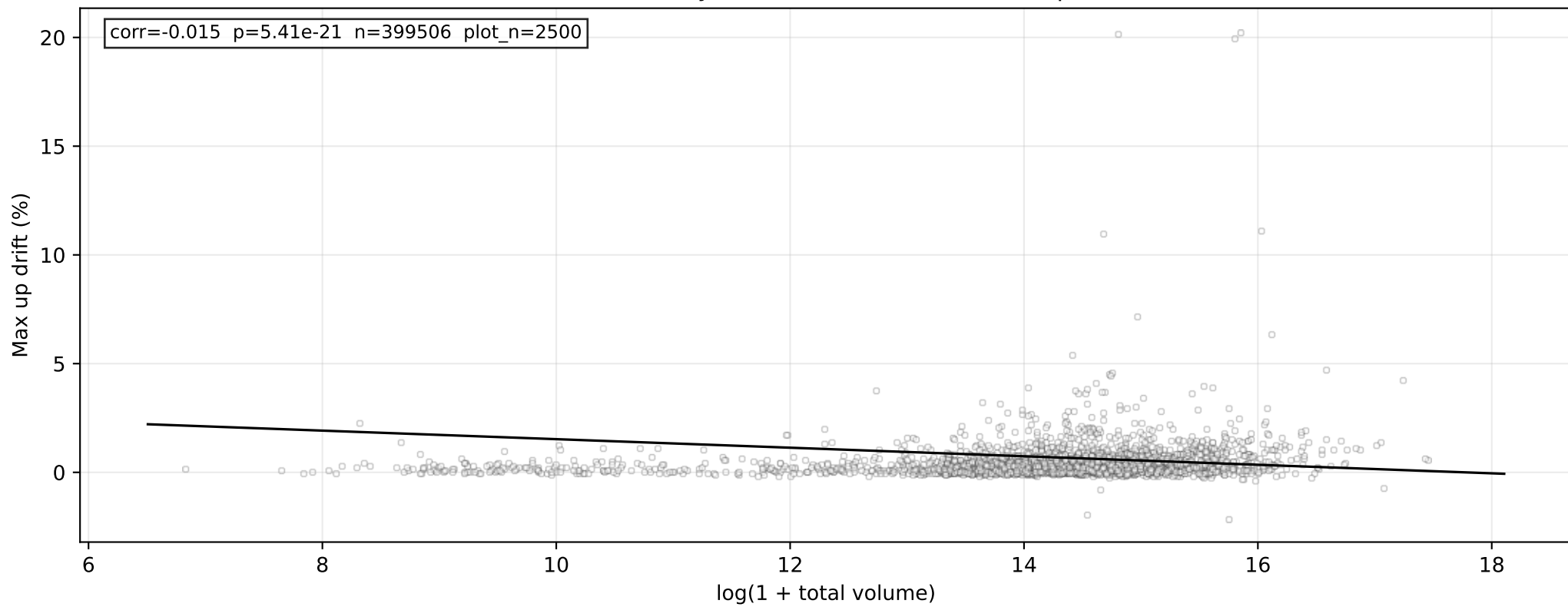
META - Intraday 30M Below Mid Volume vs Max Up Drift



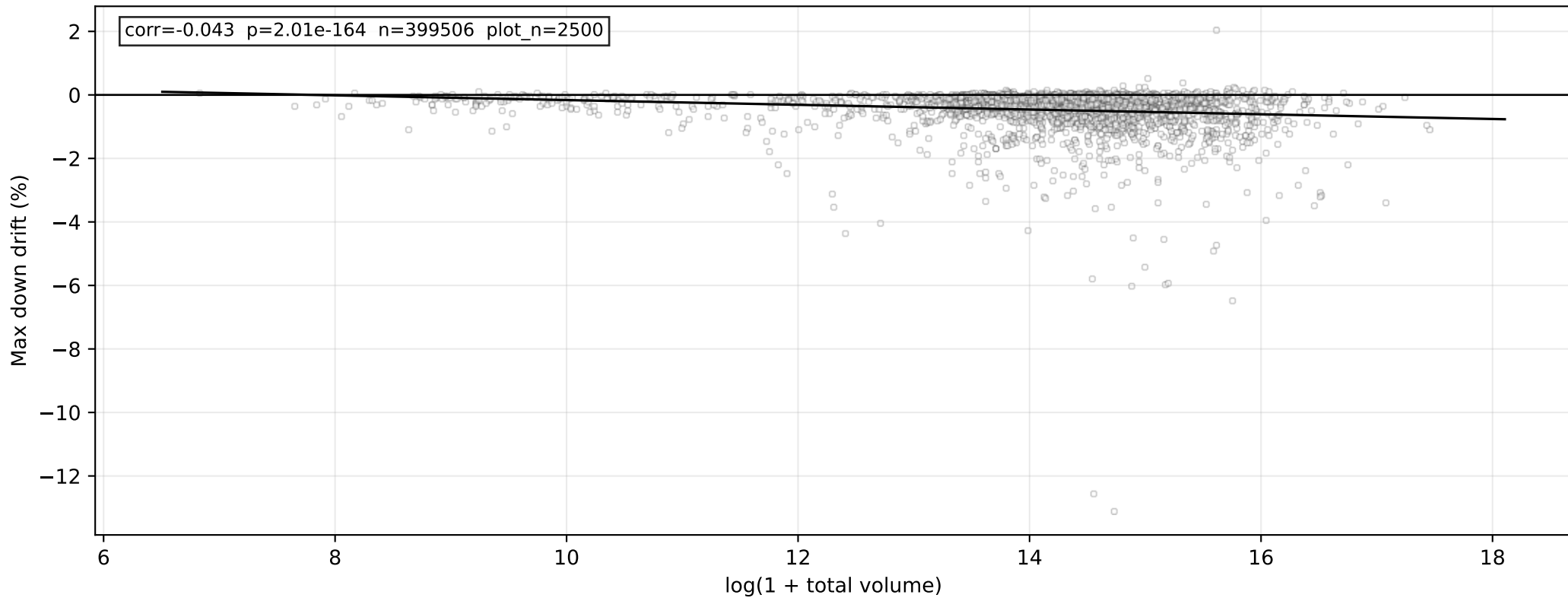
META - Intraday 30M Below Mid Volume vs Max Down Drift



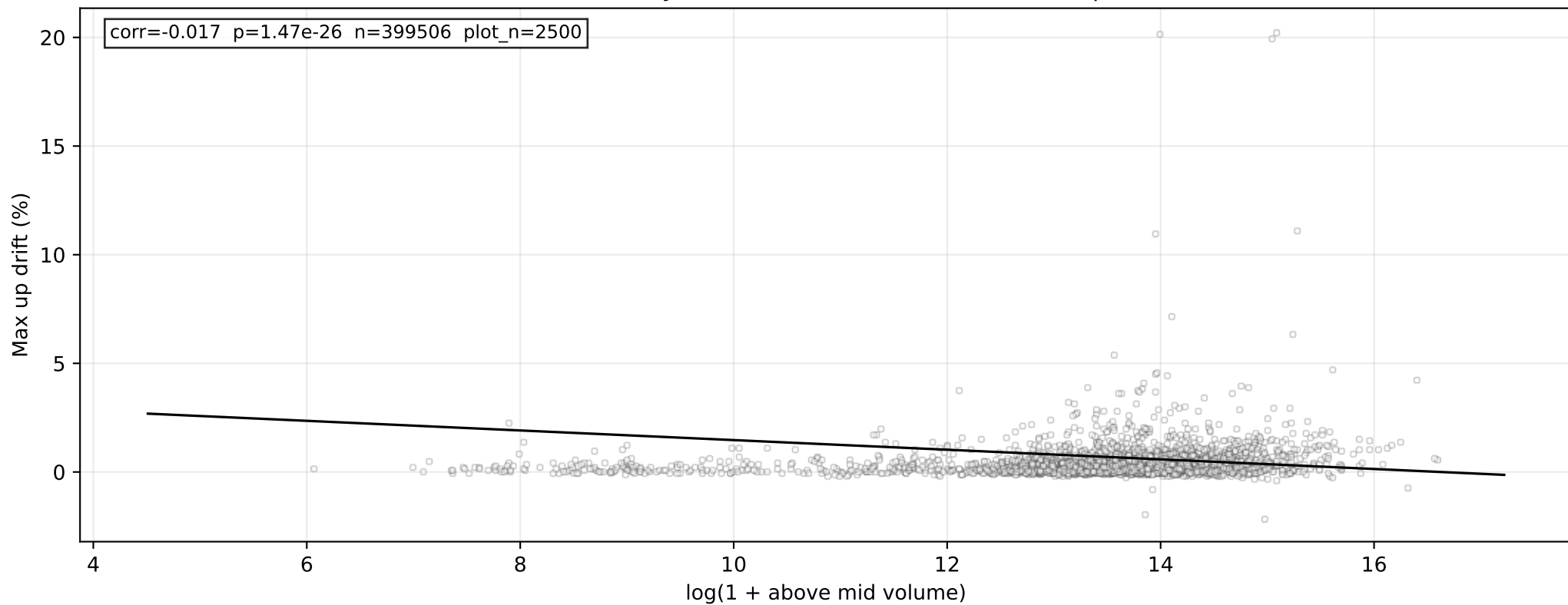
META - Intraday 60M Total Volume vs Max Up Drift



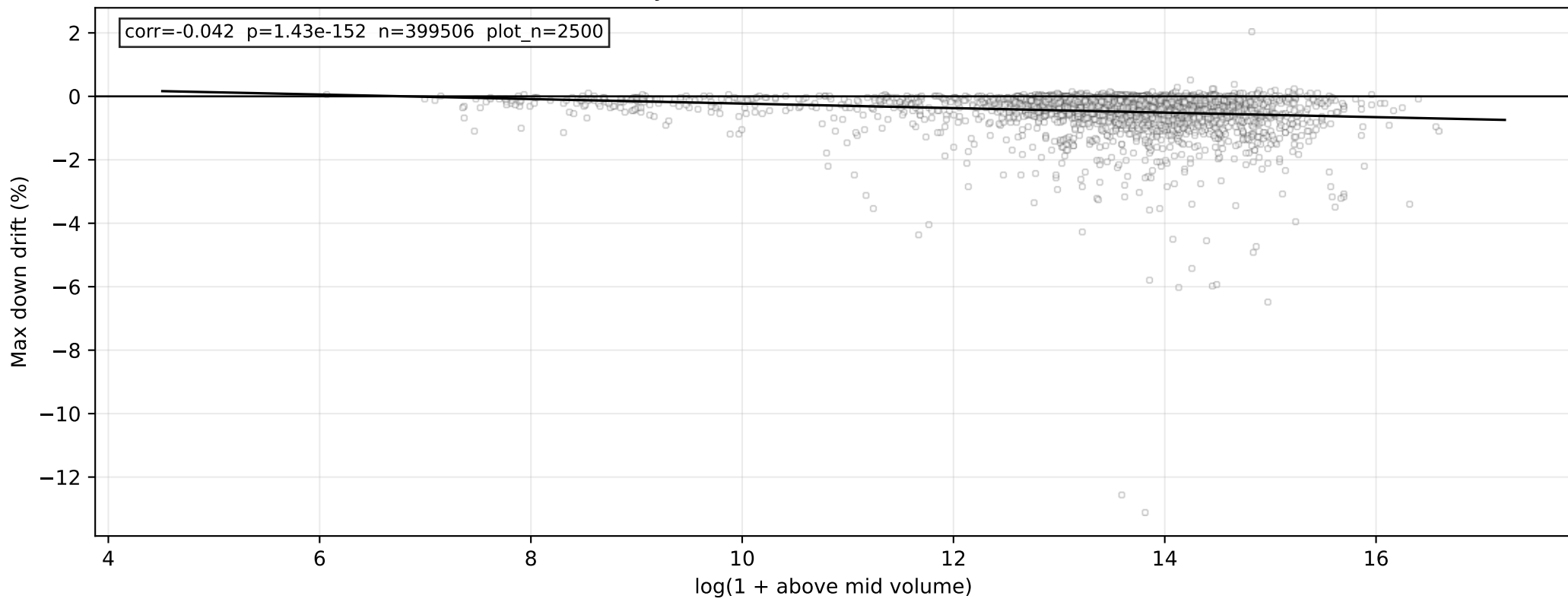
META - Intraday 60M Total Volume vs Max Down Drift



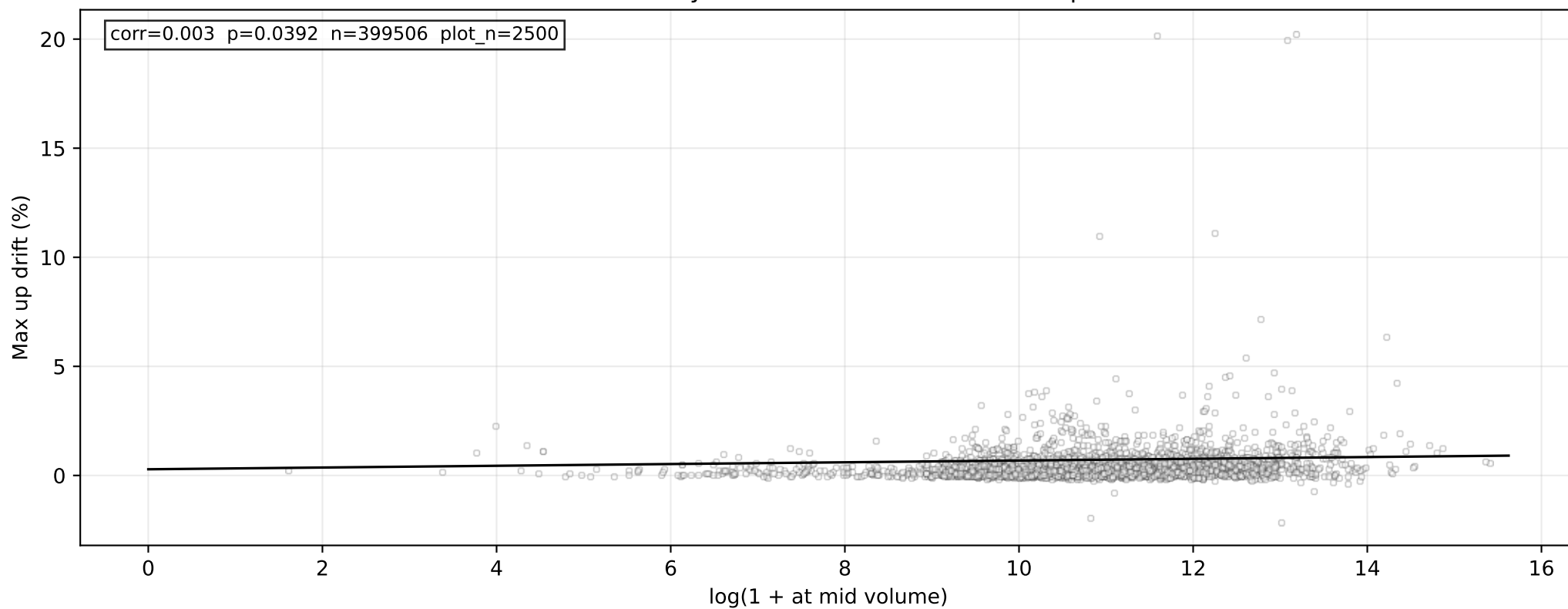
META - Intraday 60M Above Mid Volume vs Max Up Drift



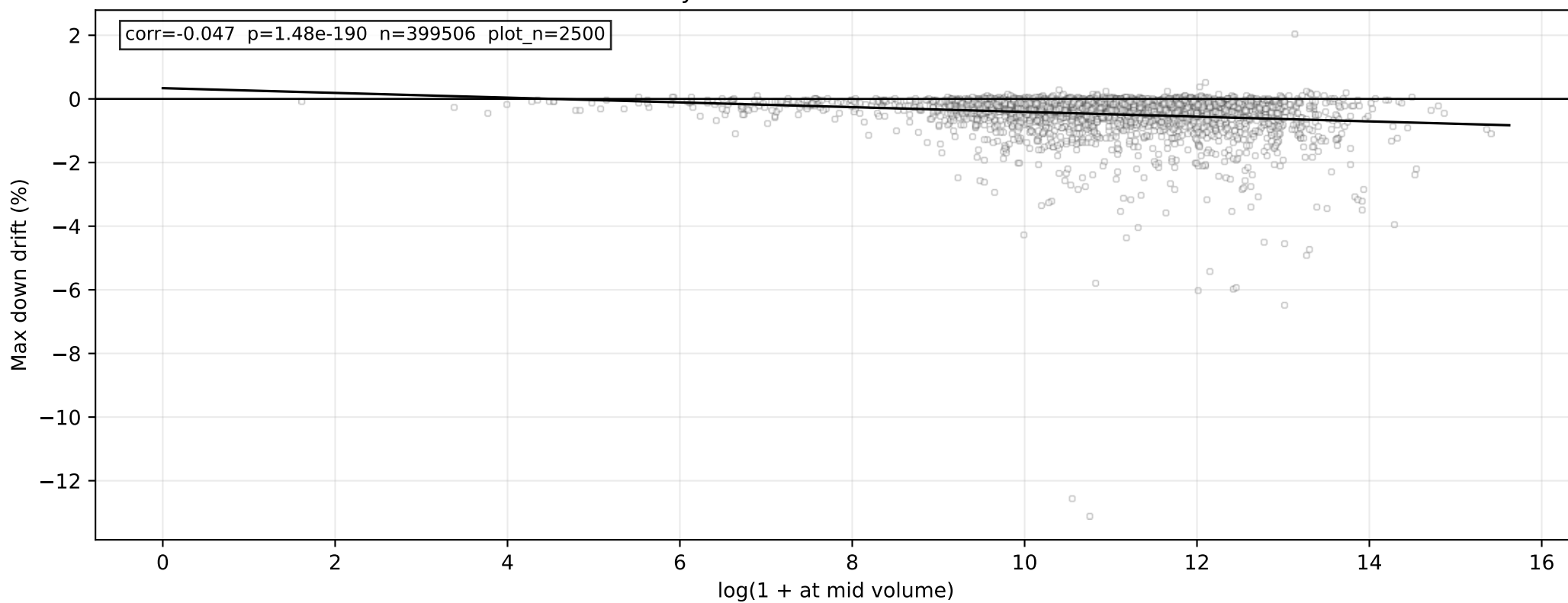
META - Intraday 60M Above Mid Volume vs Max Down Drift



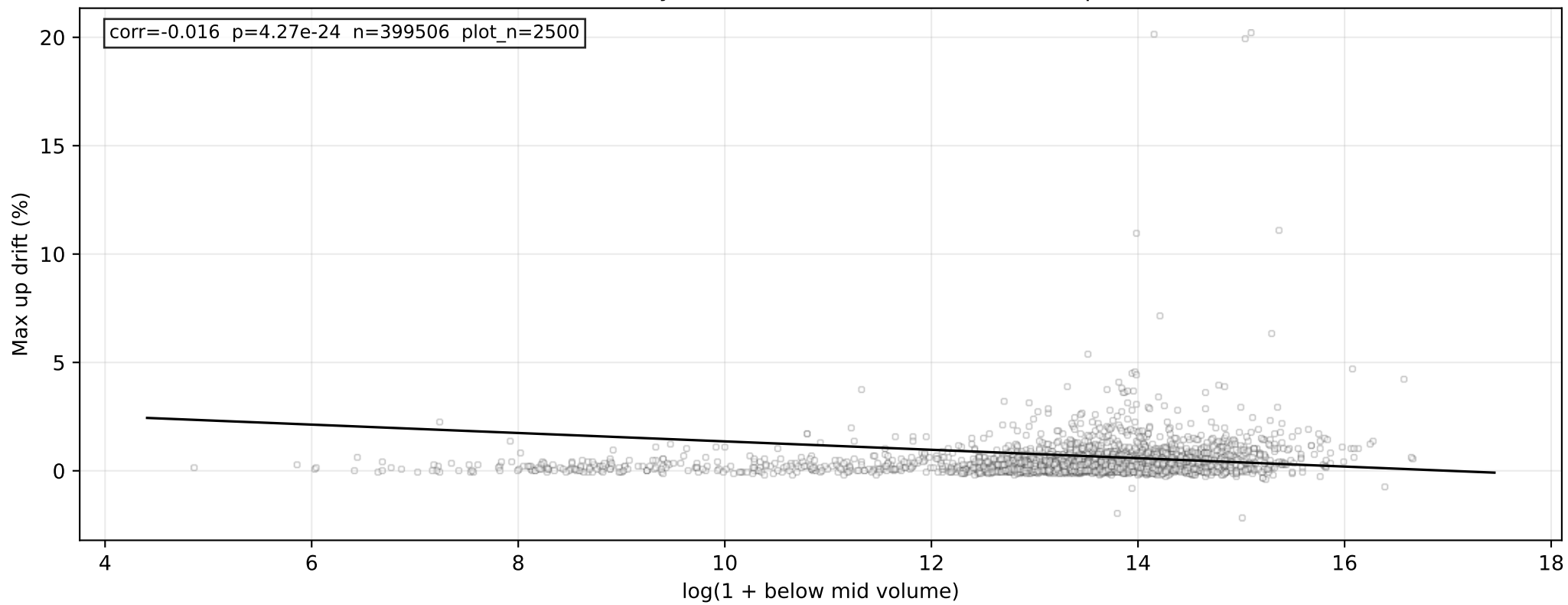
META - Intraday 60M At Mid Volume vs Max Up Drift



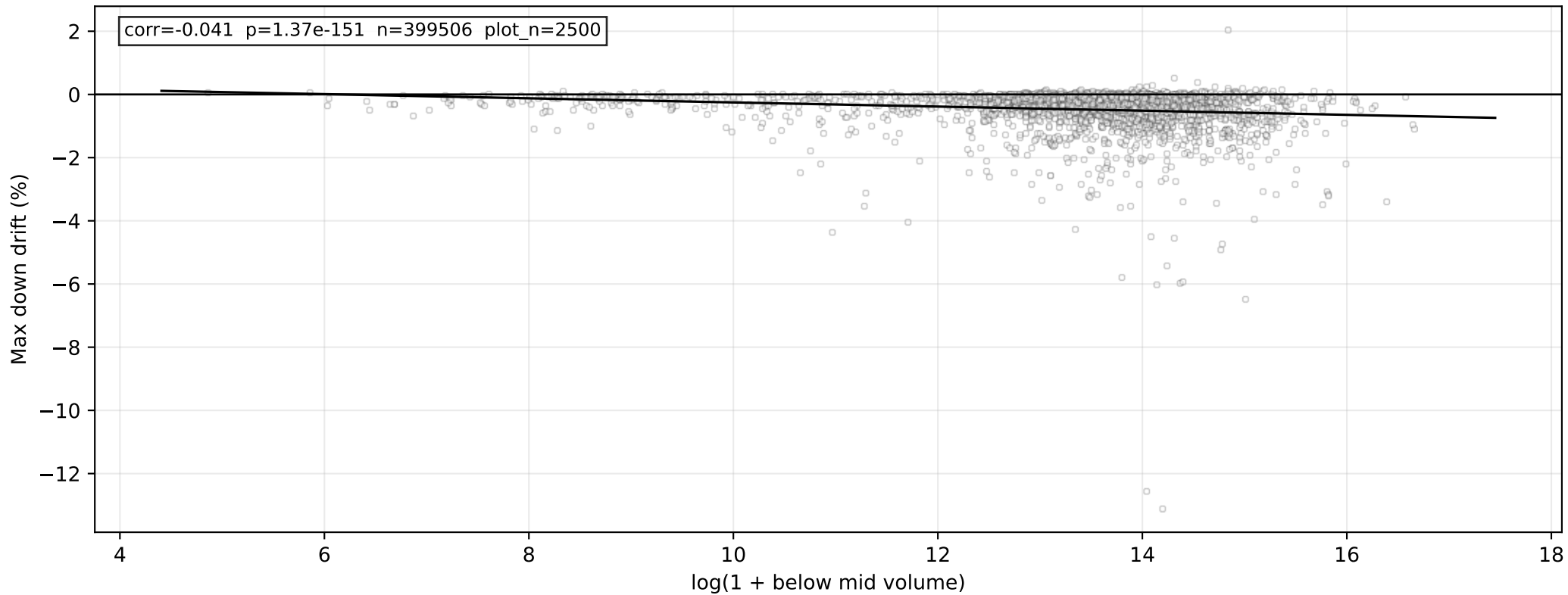
META - Intraday 60M At Mid Volume vs Max Down Drift



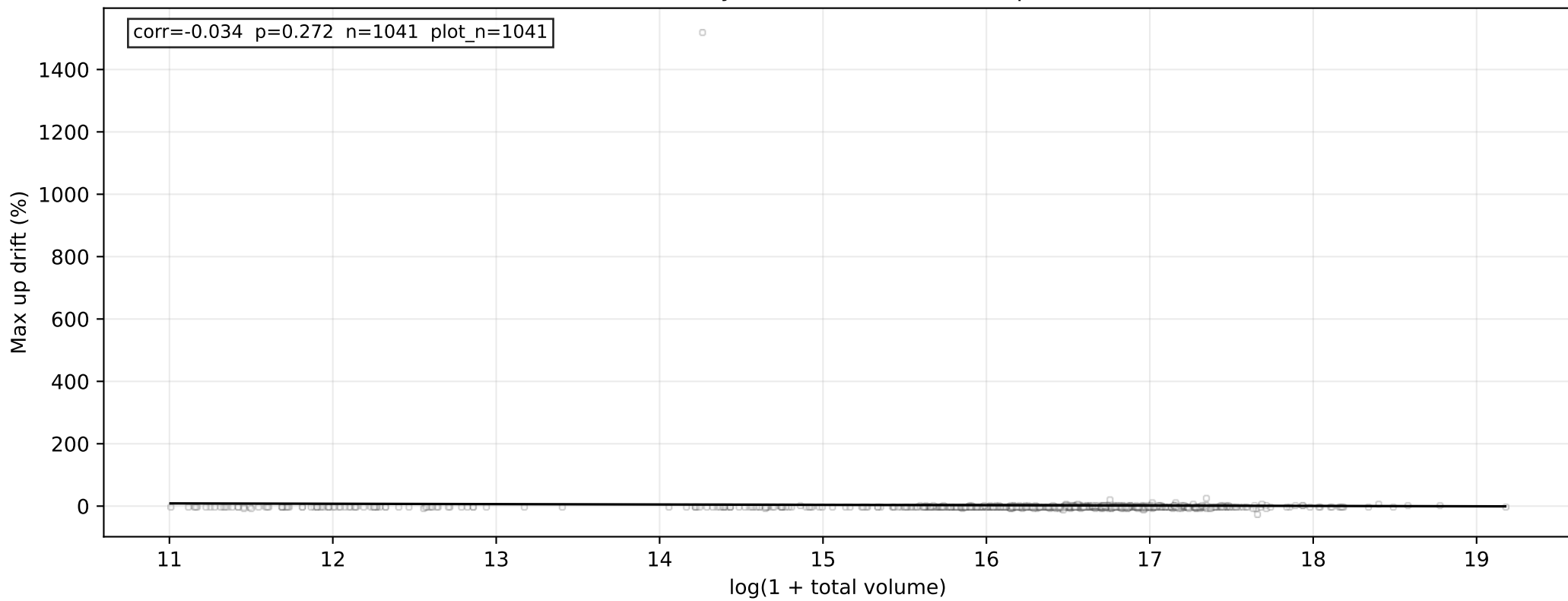
META - Intraday 60M Below Mid Volume vs Max Up Drift



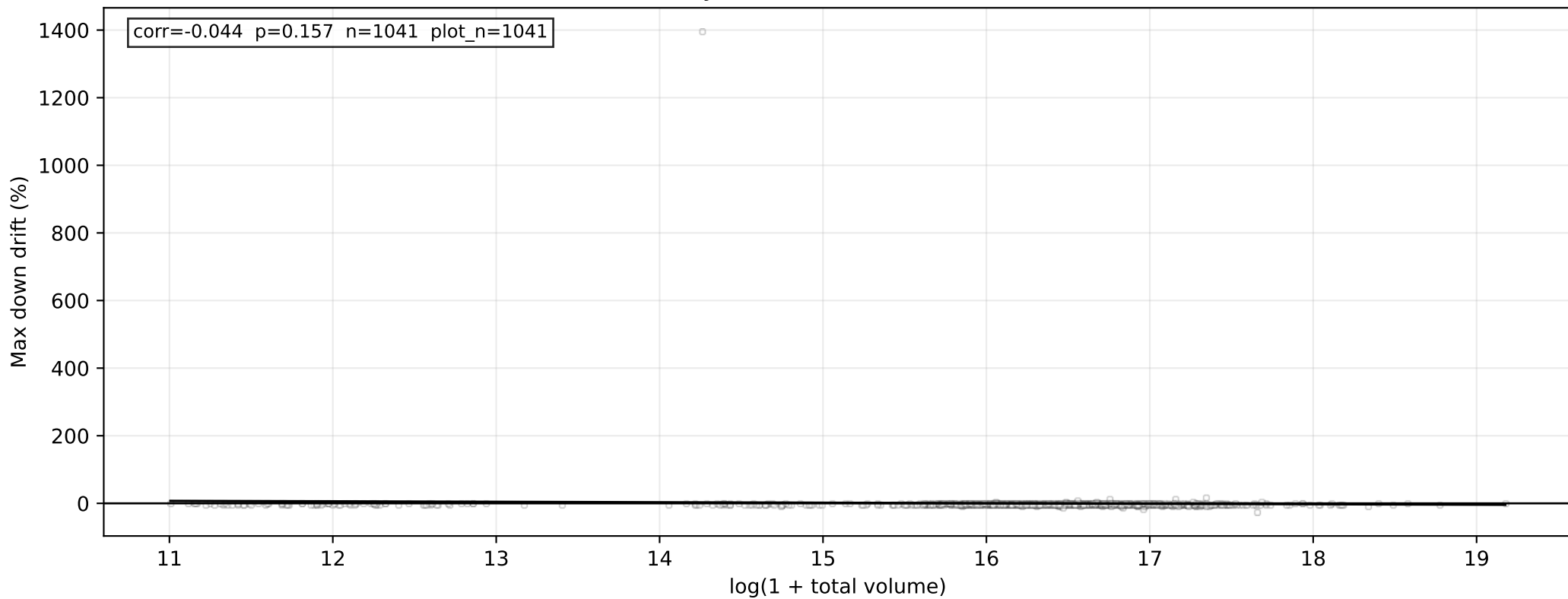
META - Intraday 60M Below Mid Volume vs Max Down Drift



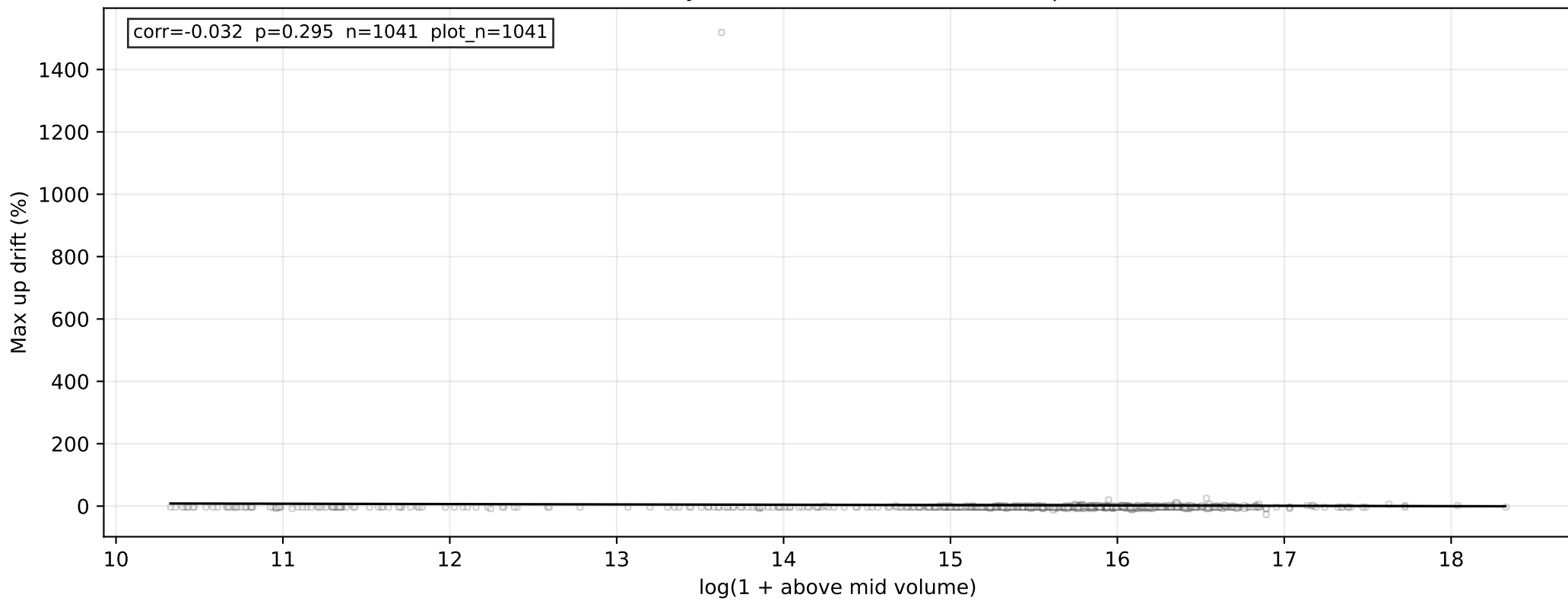
META - Daily Total Volume vs Max Up Drift



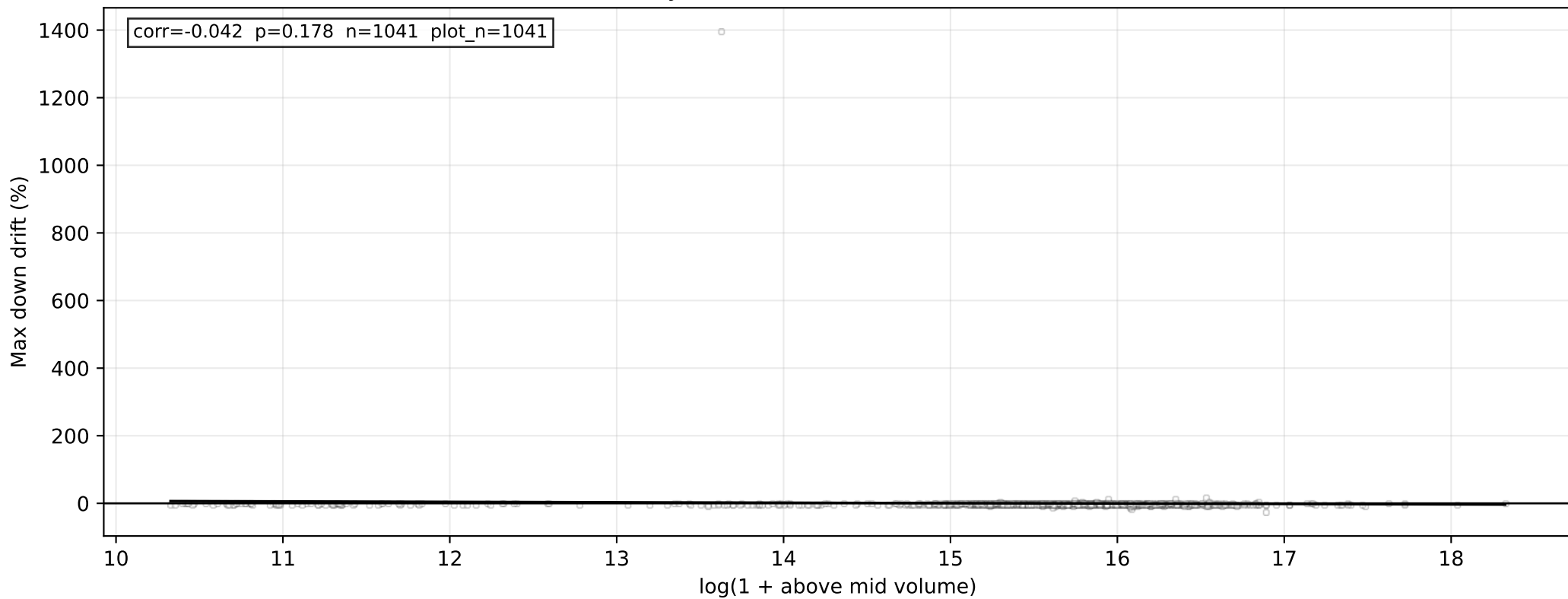
META - Daily Total Volume vs Max Down Drift



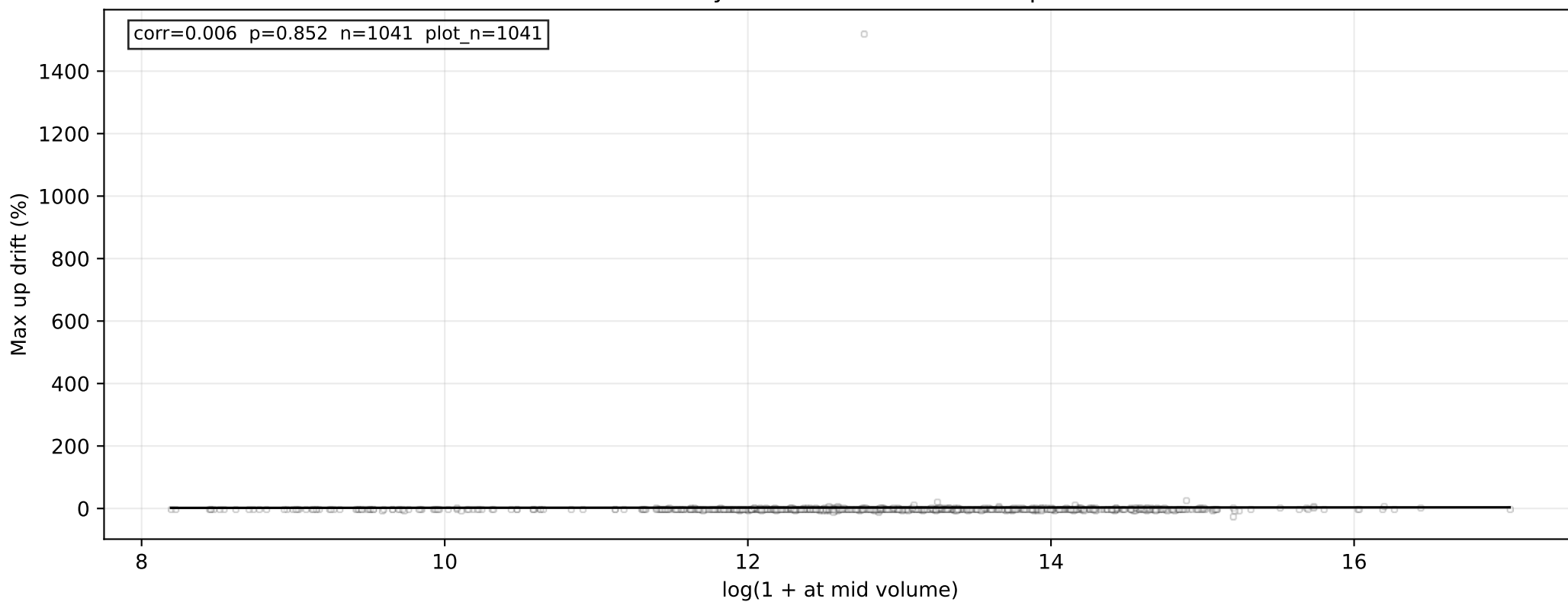
META - Daily Above Mid Volume vs Max Up Drift



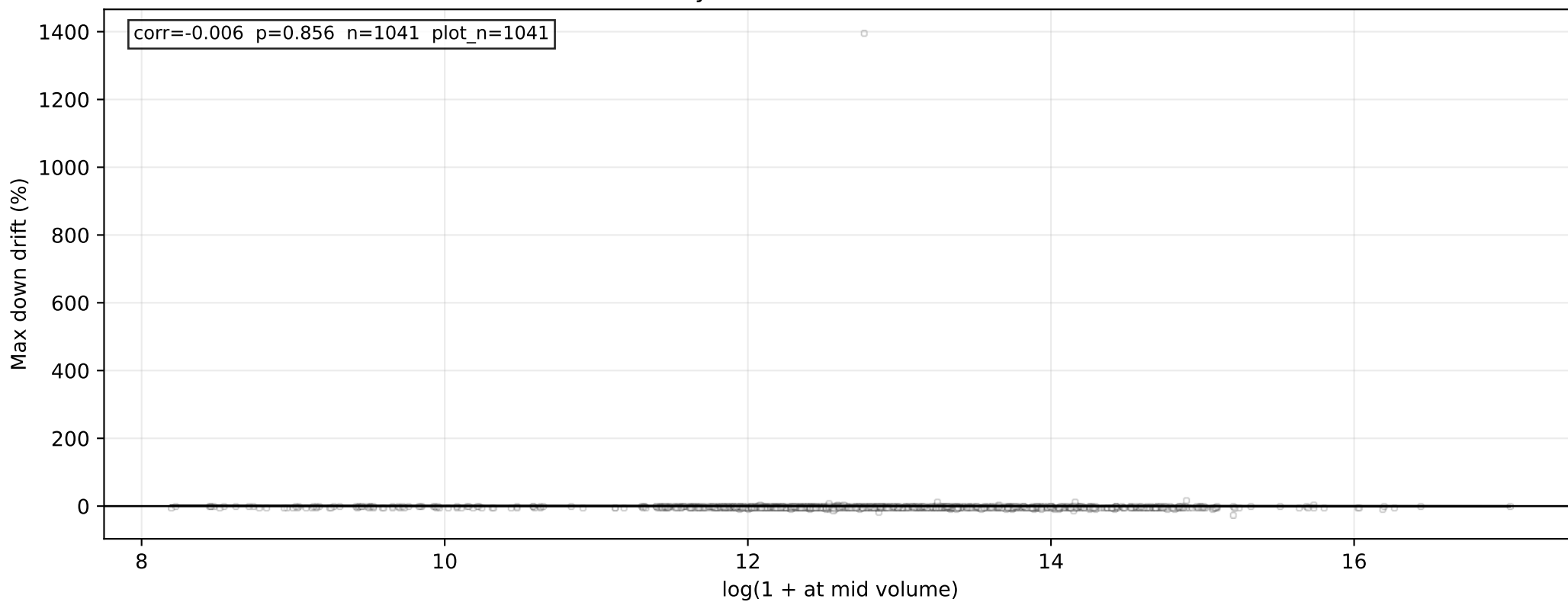
META - Daily Above Mid Volume vs Max Down Drift



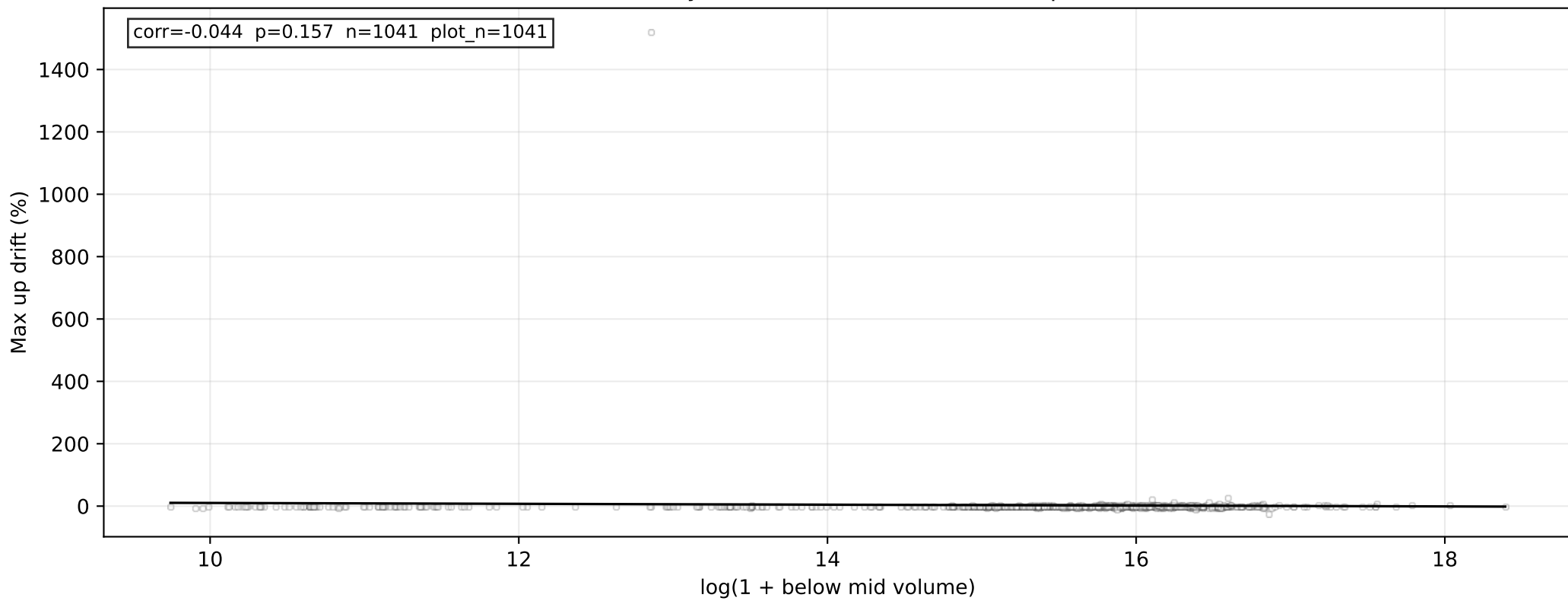
META - Daily At Mid Volume vs Max Up Drift



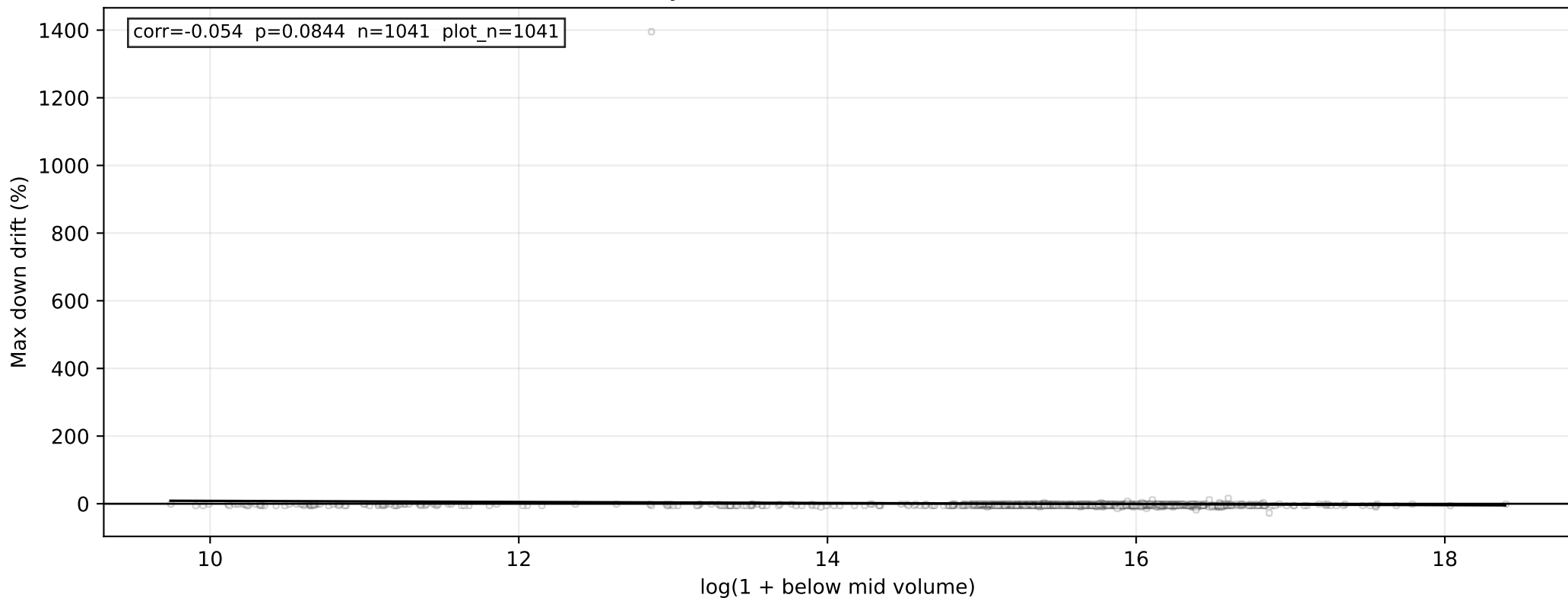
META - Daily At Mid Volume vs Max Down Drift



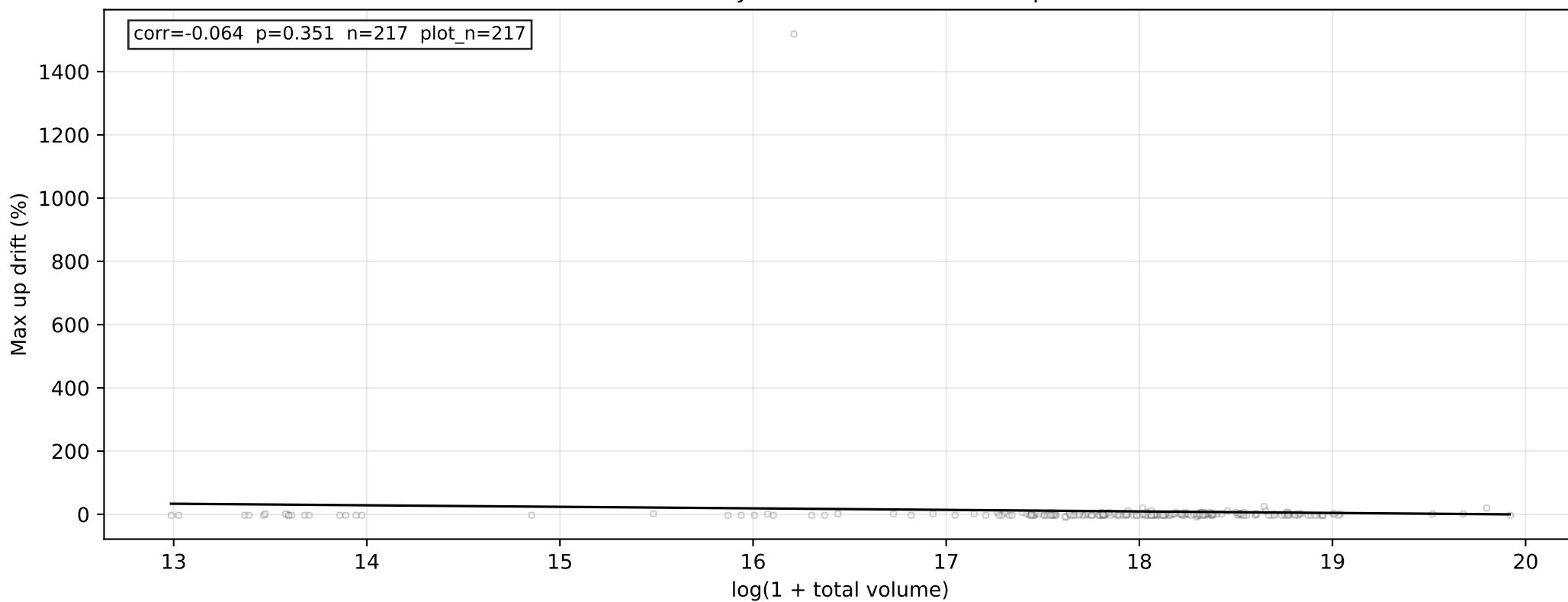
META - Daily Below Mid Volume vs Max Up Drift



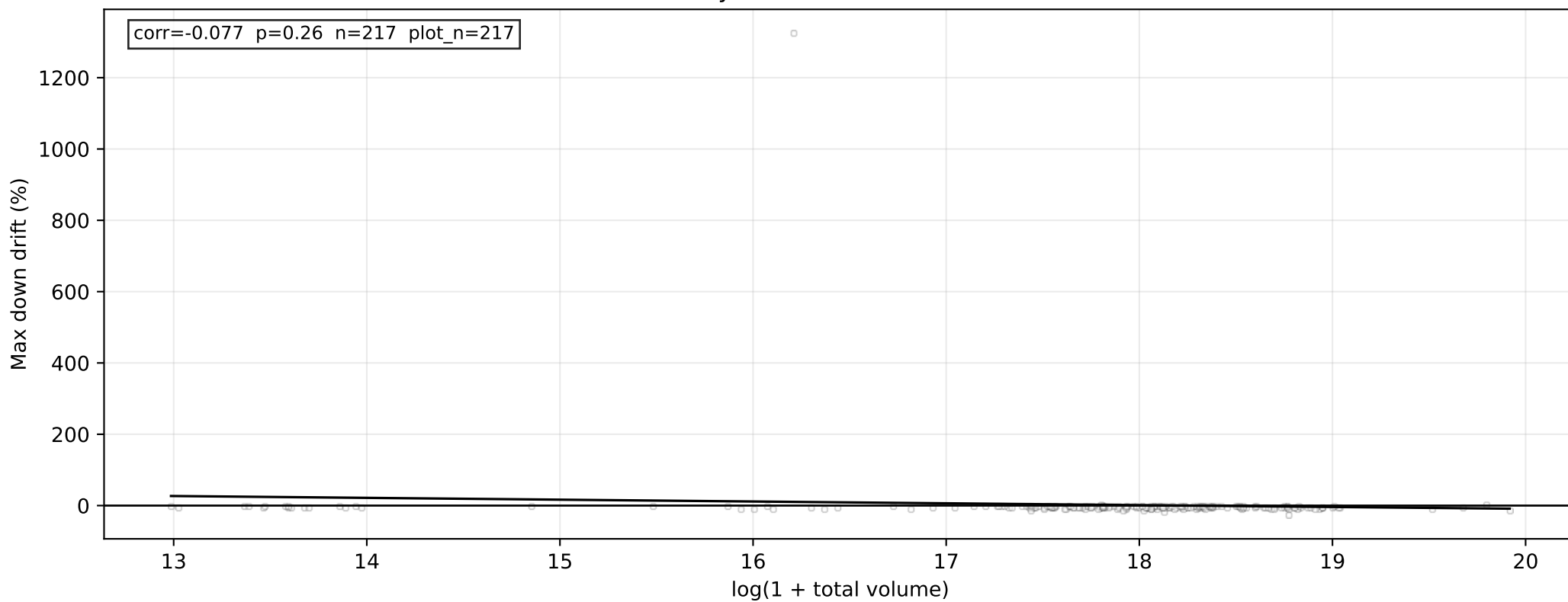
META - Daily Below Mid Volume vs Max Down Drift



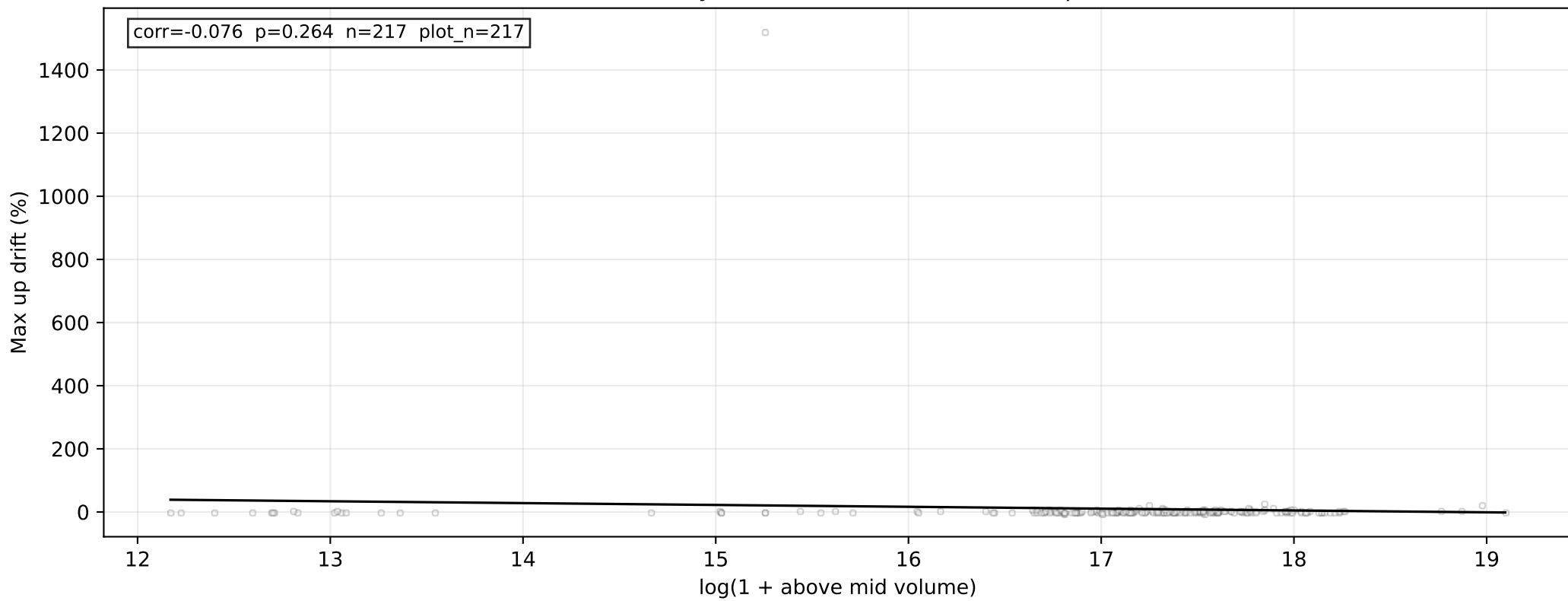
META - Weekly Total Volume vs Max Up Drift



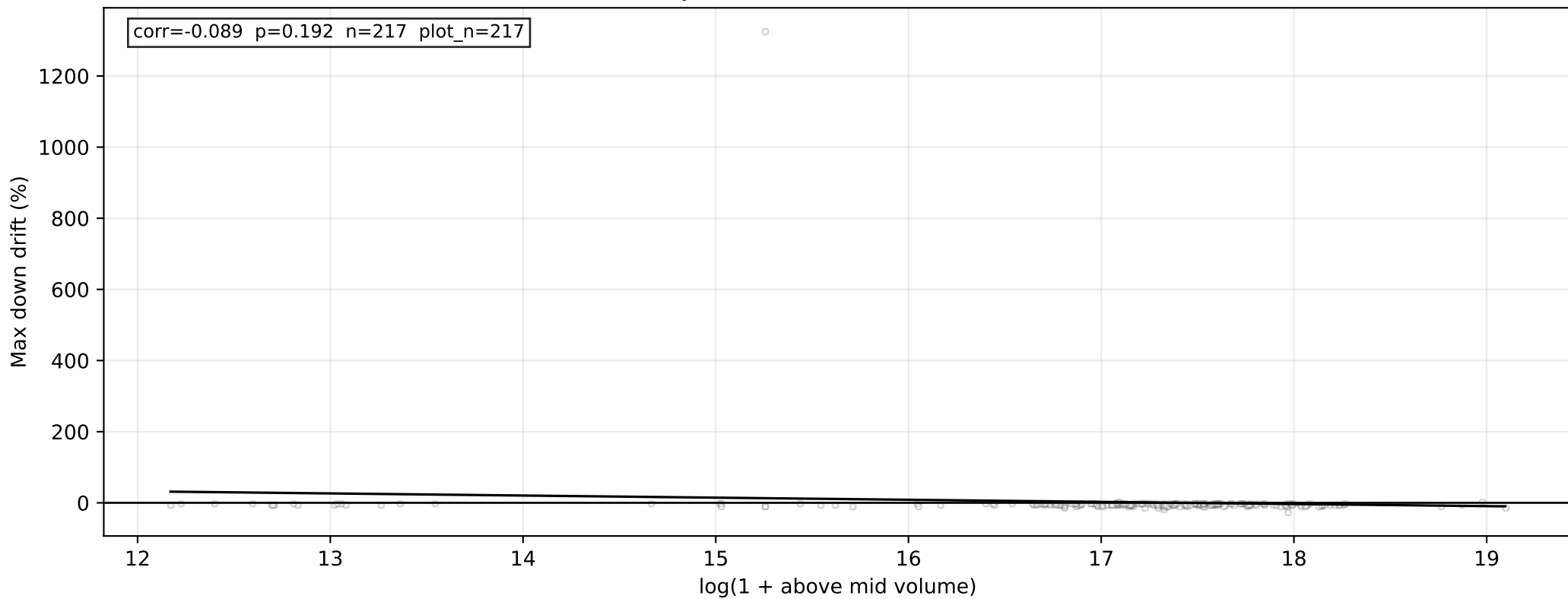
META - Weekly Total Volume vs Max Down Drift



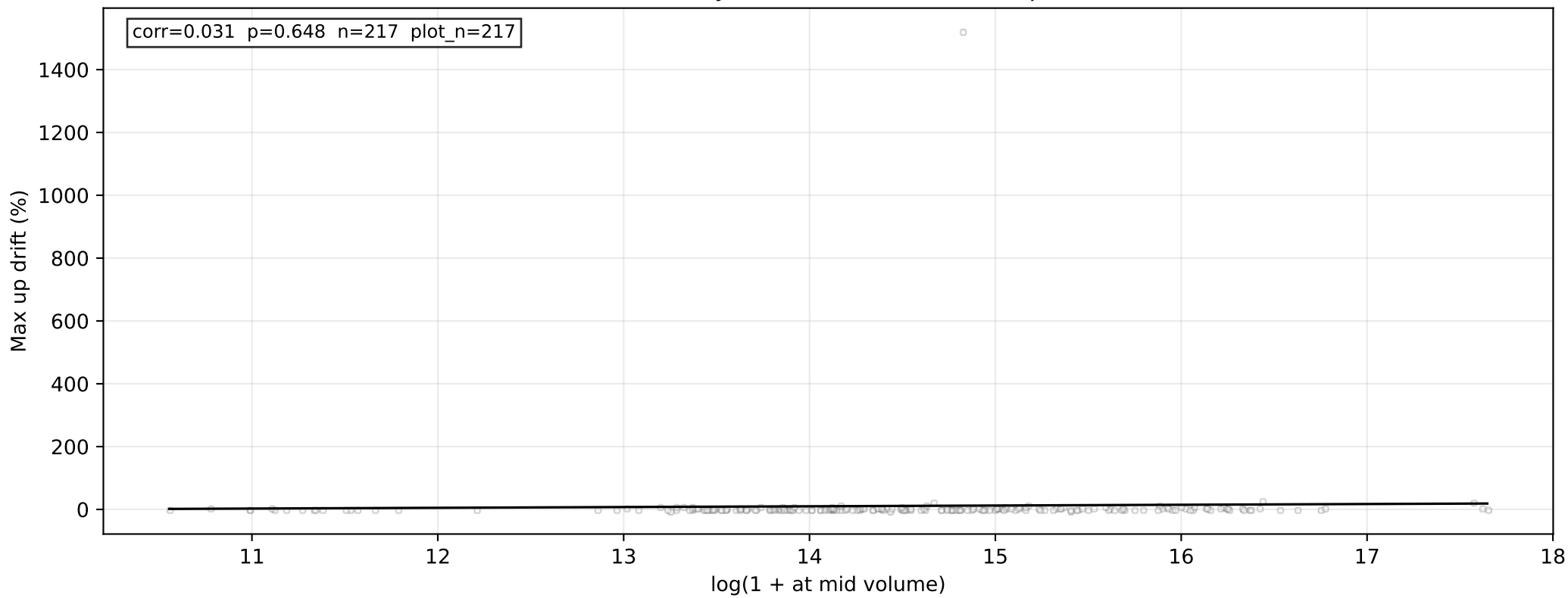
META - Weekly Above Mid Volume vs Max Up Drift



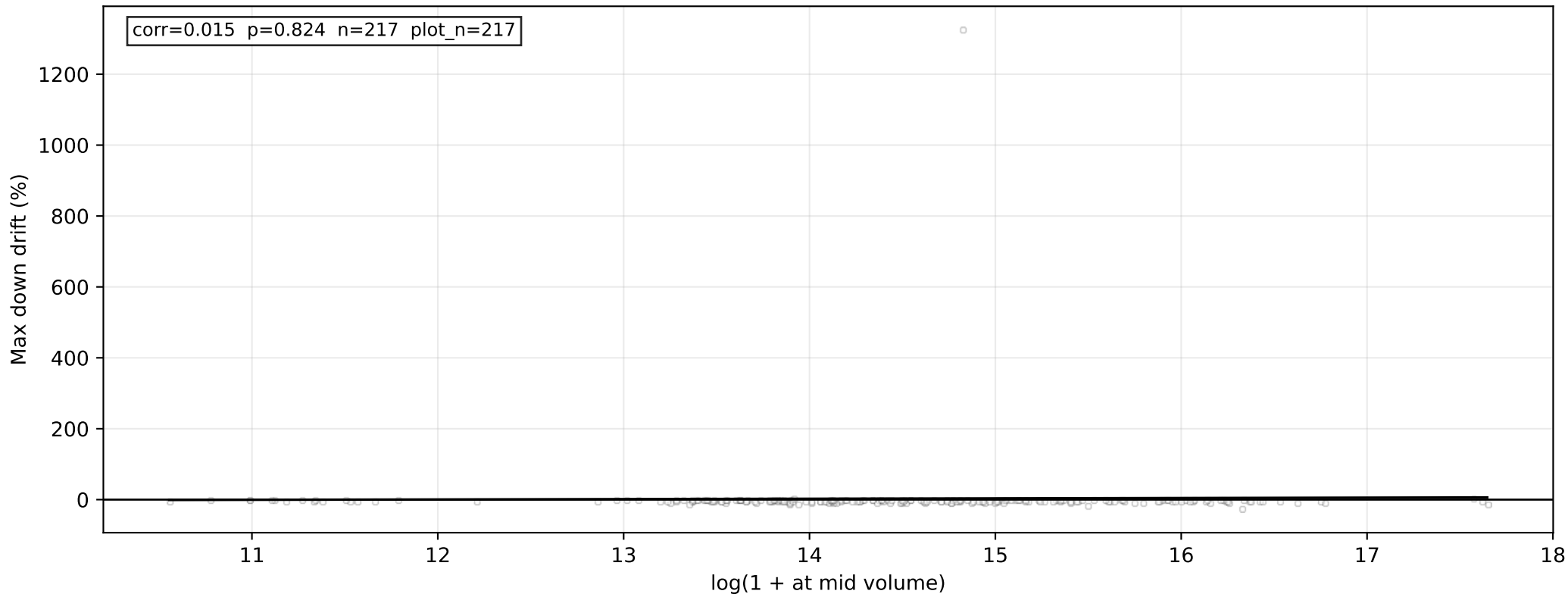
META - Weekly Above Mid Volume vs Max Down Drift



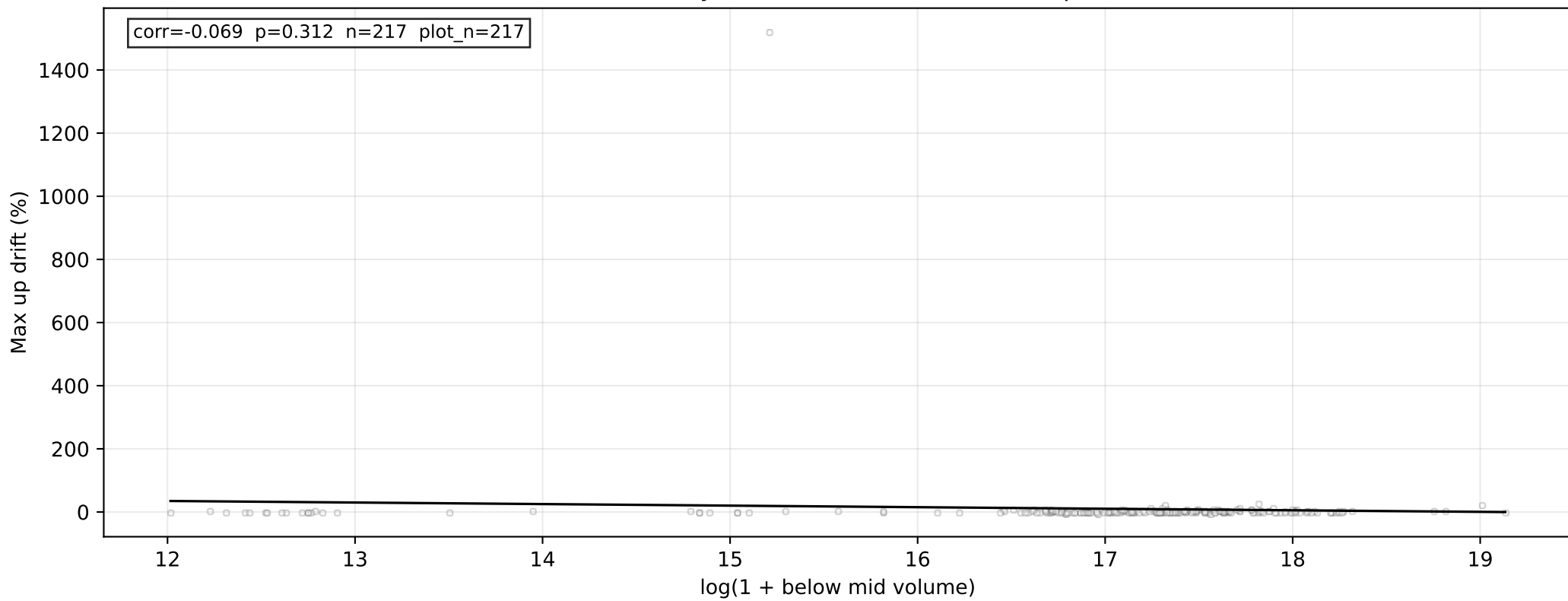
META - Weekly At Mid Volume vs Max Up Drift



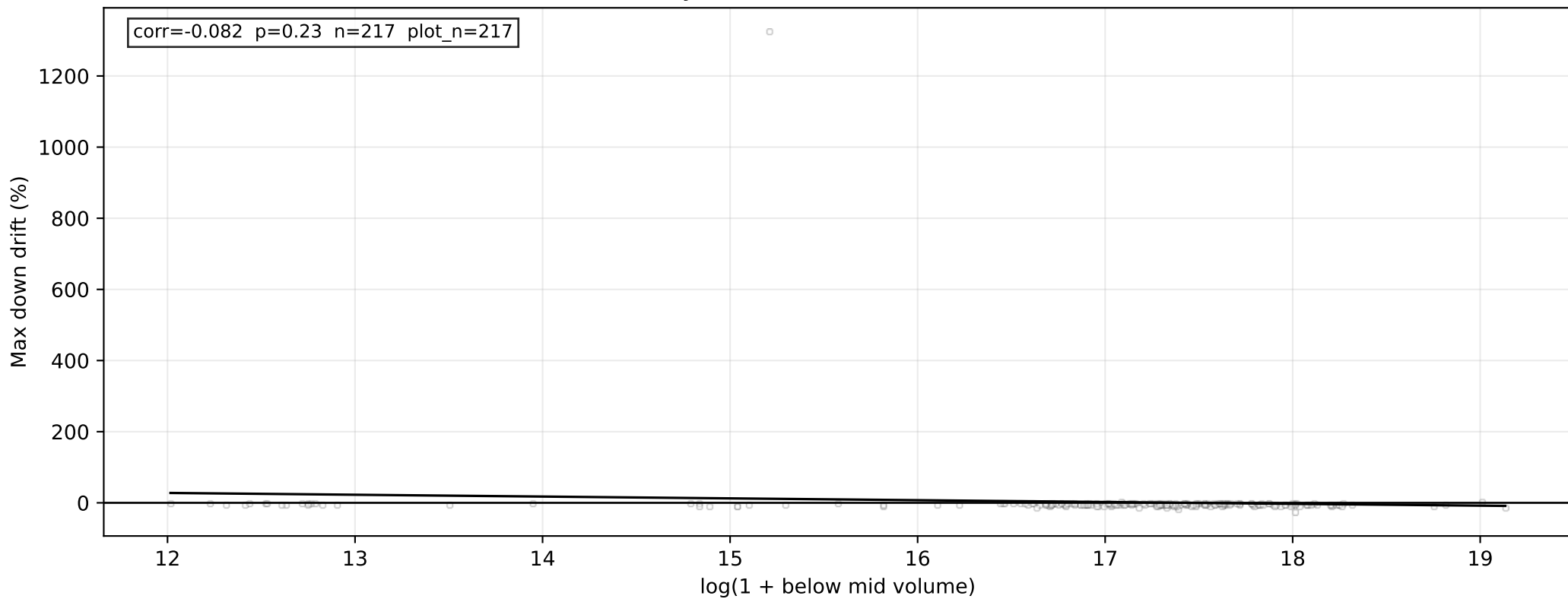
META - Weekly At Mid Volume vs Max Down Drift



META - Weekly Below Mid Volume vs Max Up Drift



META - Weekly Below Mid Volume vs Max Down Drift



META - Above/Below Mid Imbalance

This section isolates bid/ask-side imbalance from overall volume level.

The trigger statistic is the signed count-volume difference: `above_mid` minus `below_mid`.

Intraday uses rolling cumulative imbalance over each sequence window; daily and weekly use aggregated imbalance.

Upper-tail q90 / q95 / q99 events mark above-mid-dominant imbalance; lower-tail events mark below-mid-dominant imbalance.

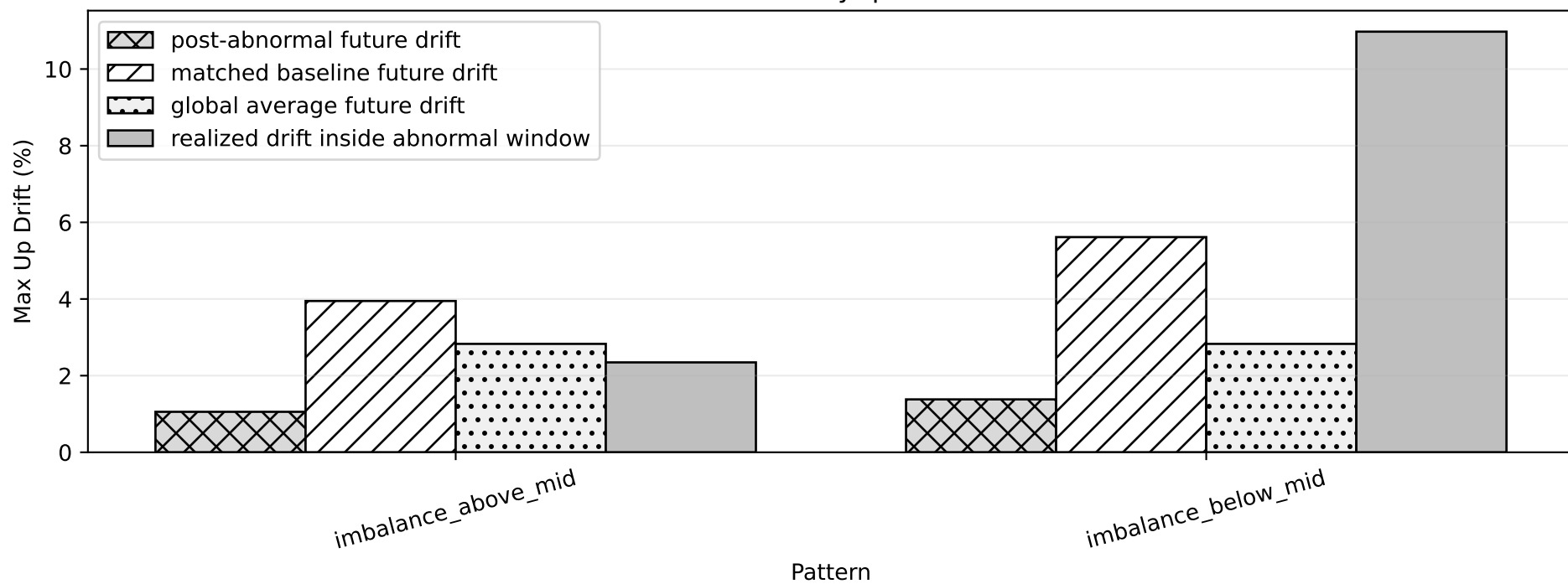
Baselines are rolling non-imbalanced observations at the same layer.

Post-abnormal future drift is distinct from realized drift already inside the abnormal window.

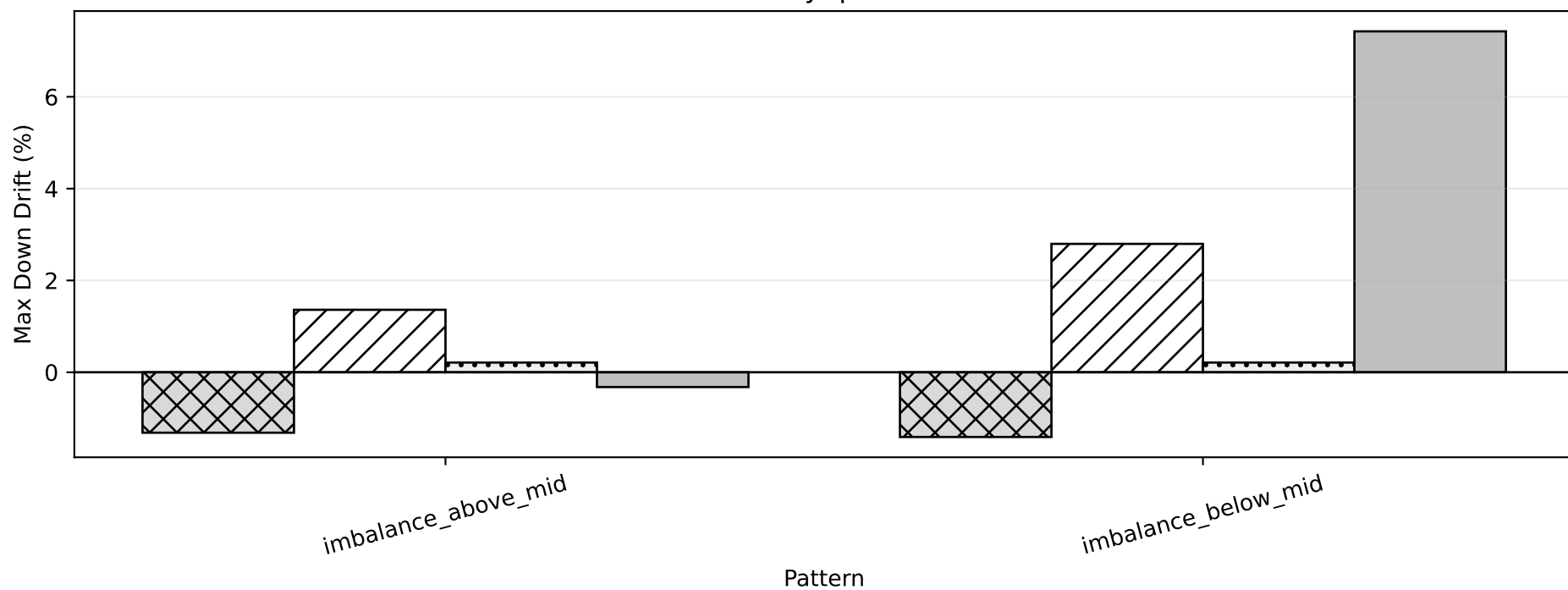
Global-drift and realized-abnormal-window references are also shown to check whether imbalance is just captured by the abnormal window.

Filled imbalance markers/bars indicate abnormal outcomes that are smaller than their matched baseline.

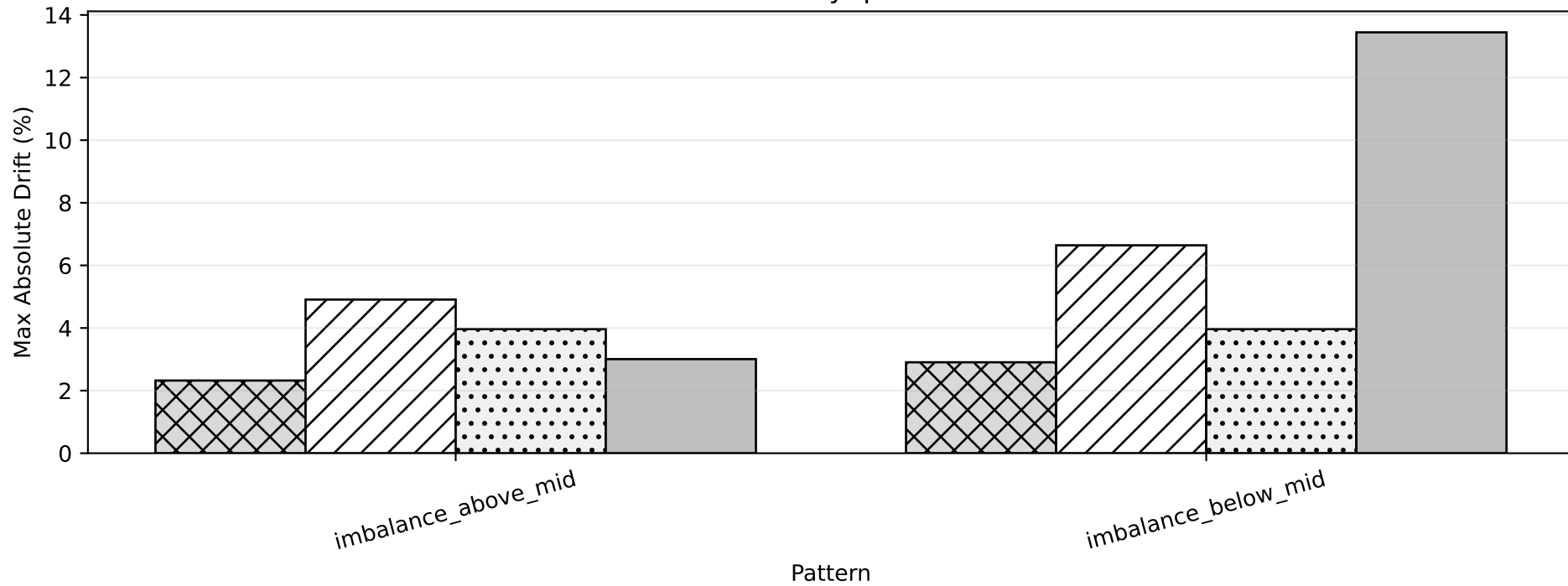
META - Daily q90 - UP



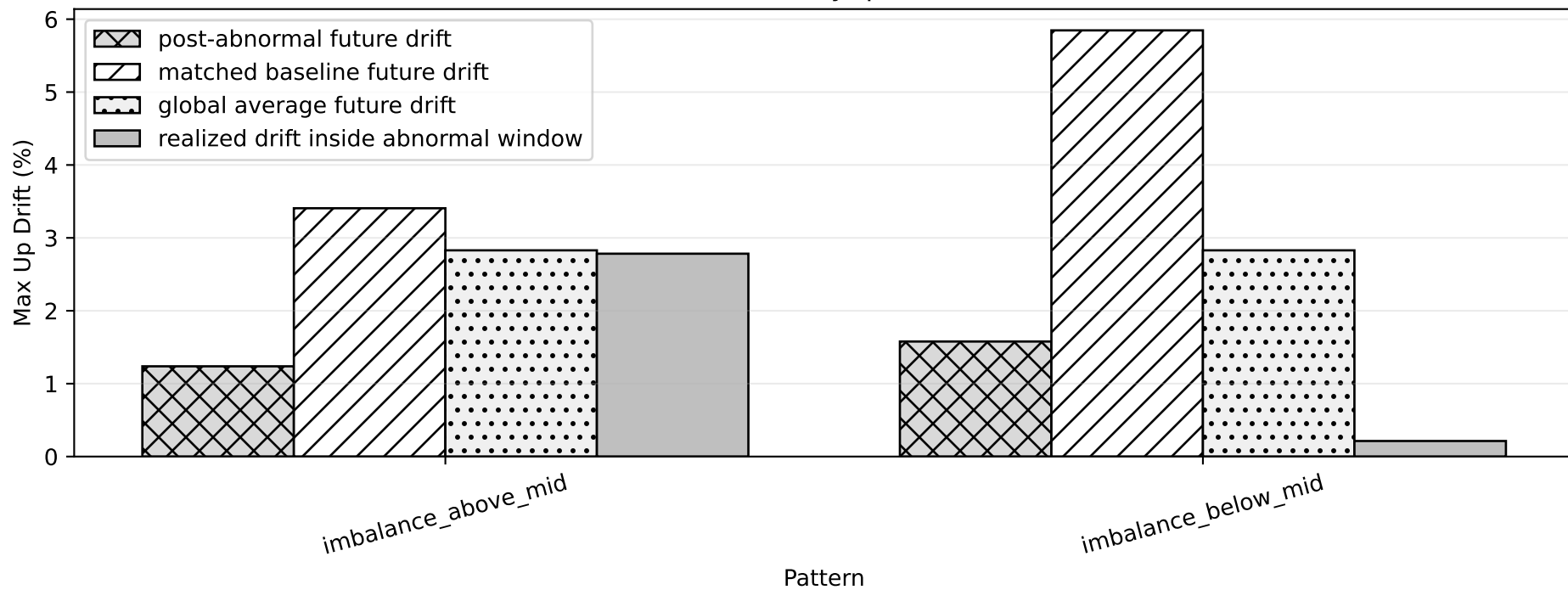
META - Daily q90 - DOWN



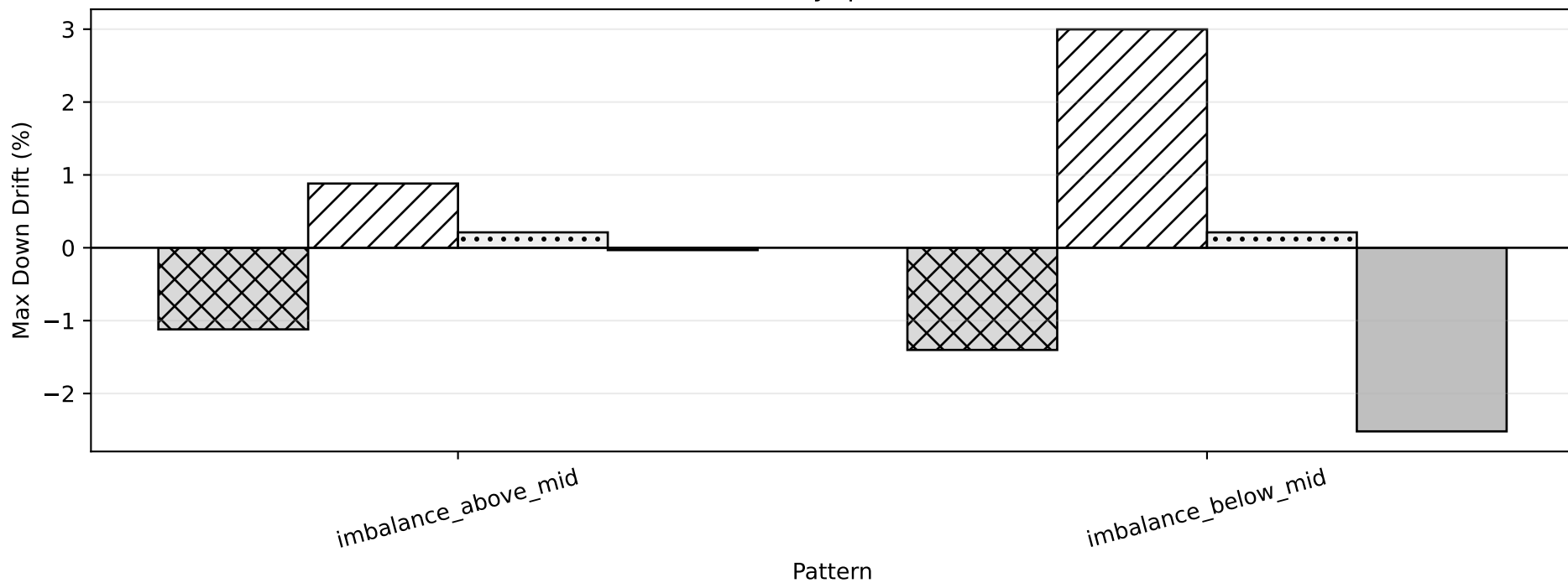
META - Daily q90 - ABS



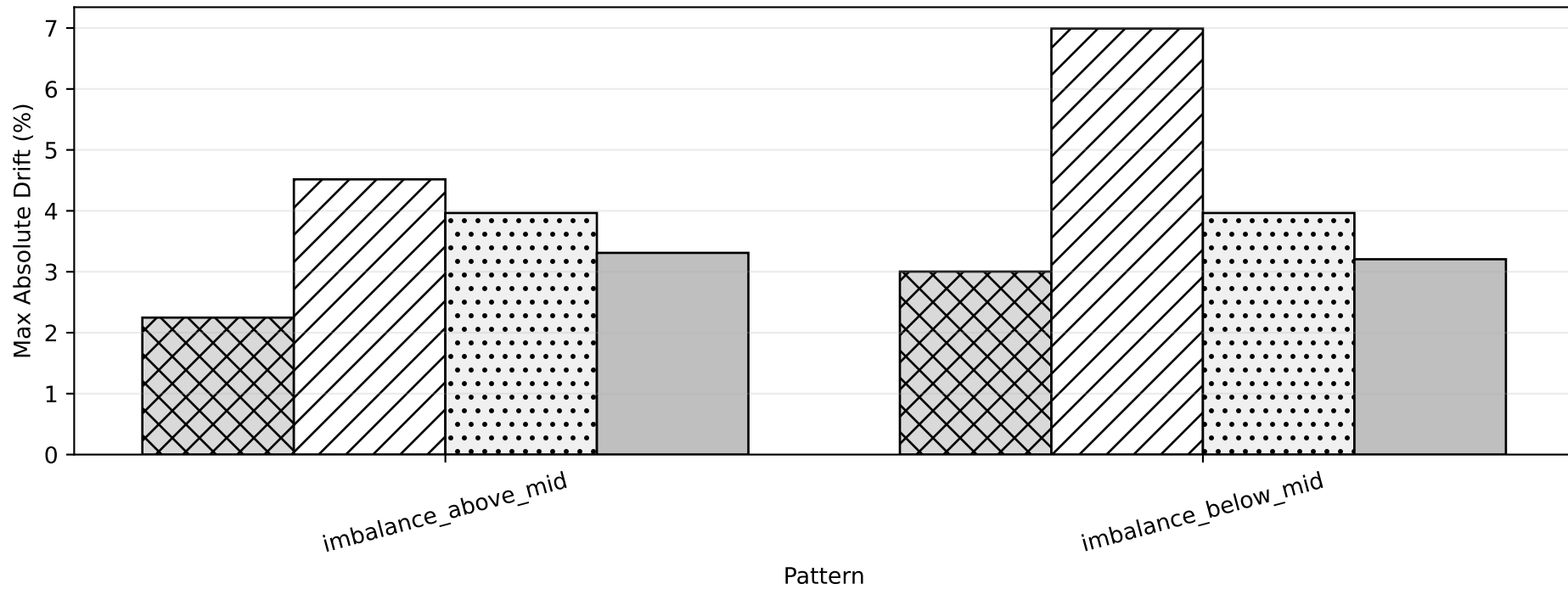
META - Daily q95 - UP



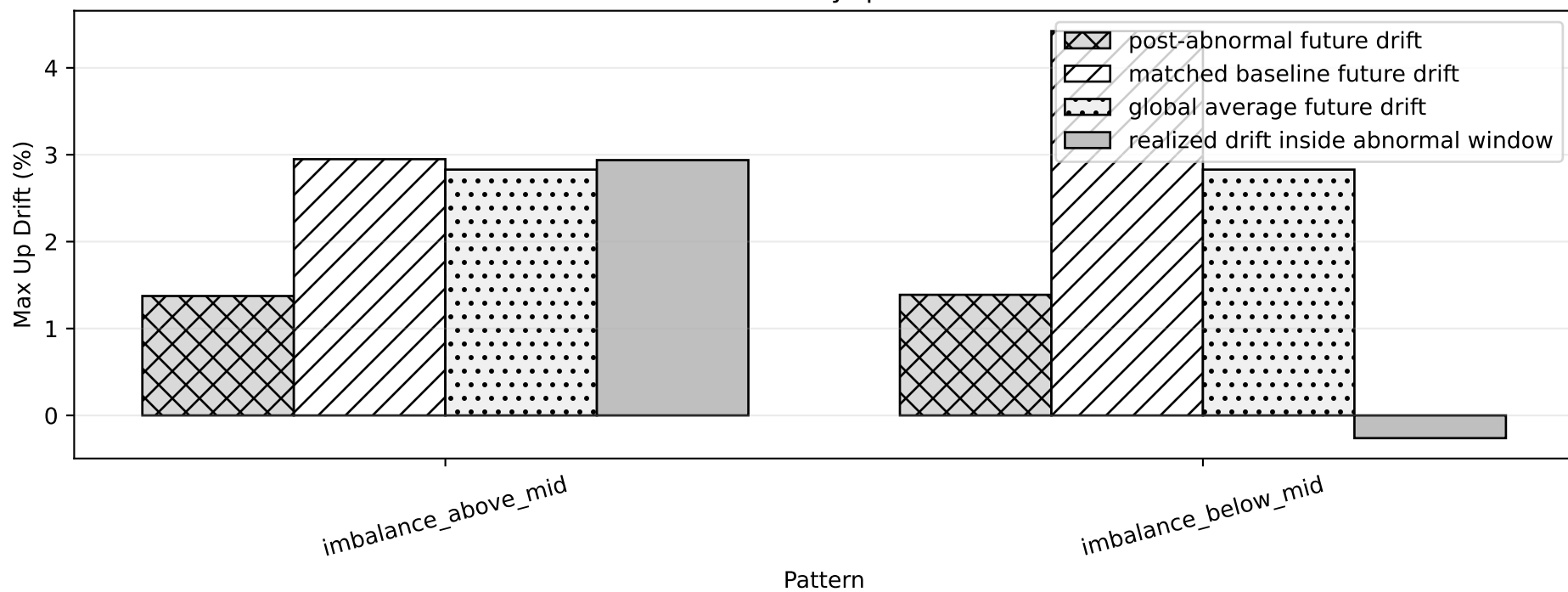
META - Daily q95 - DOWN



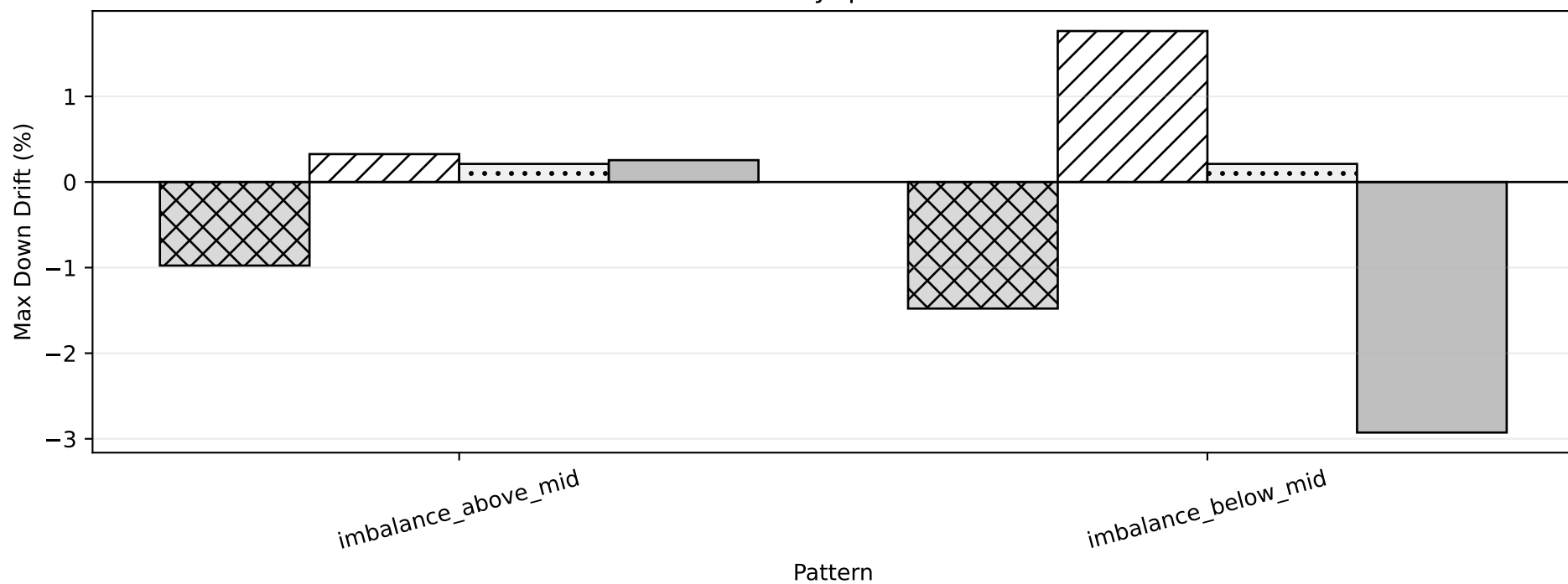
META - Daily q95 - ABS



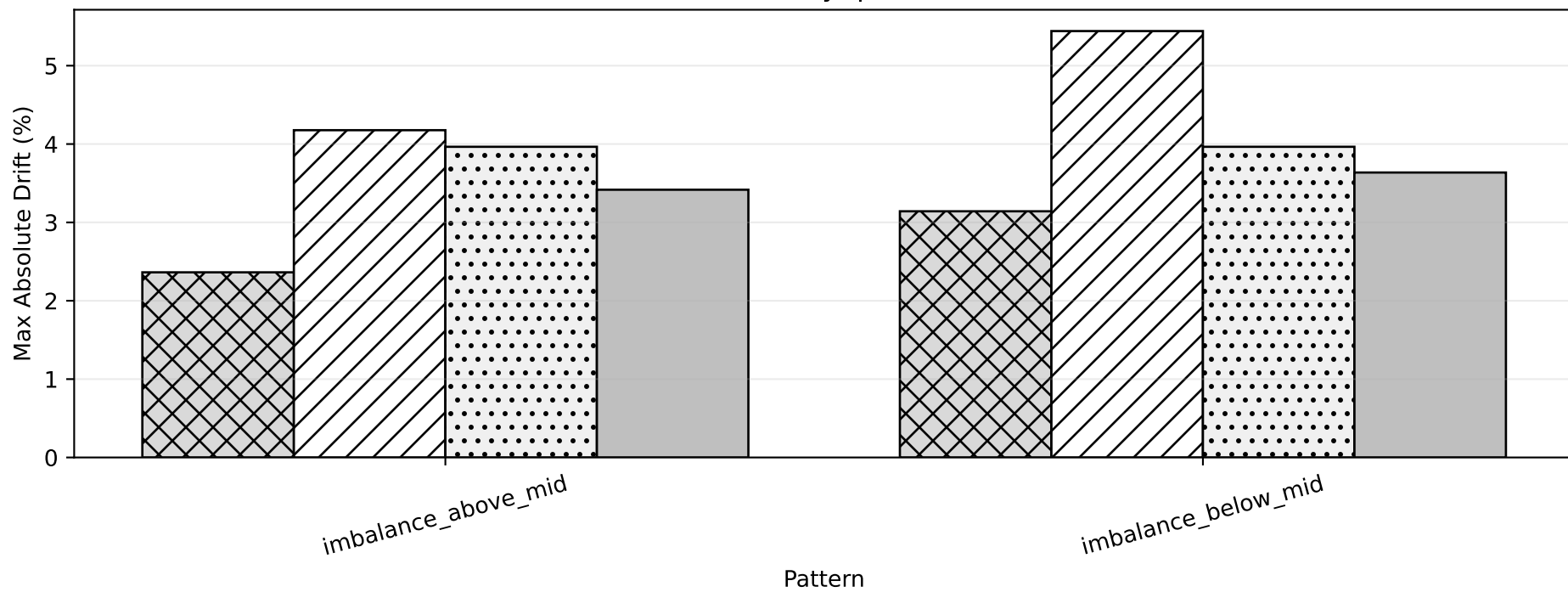
META - Daily q99 - UP



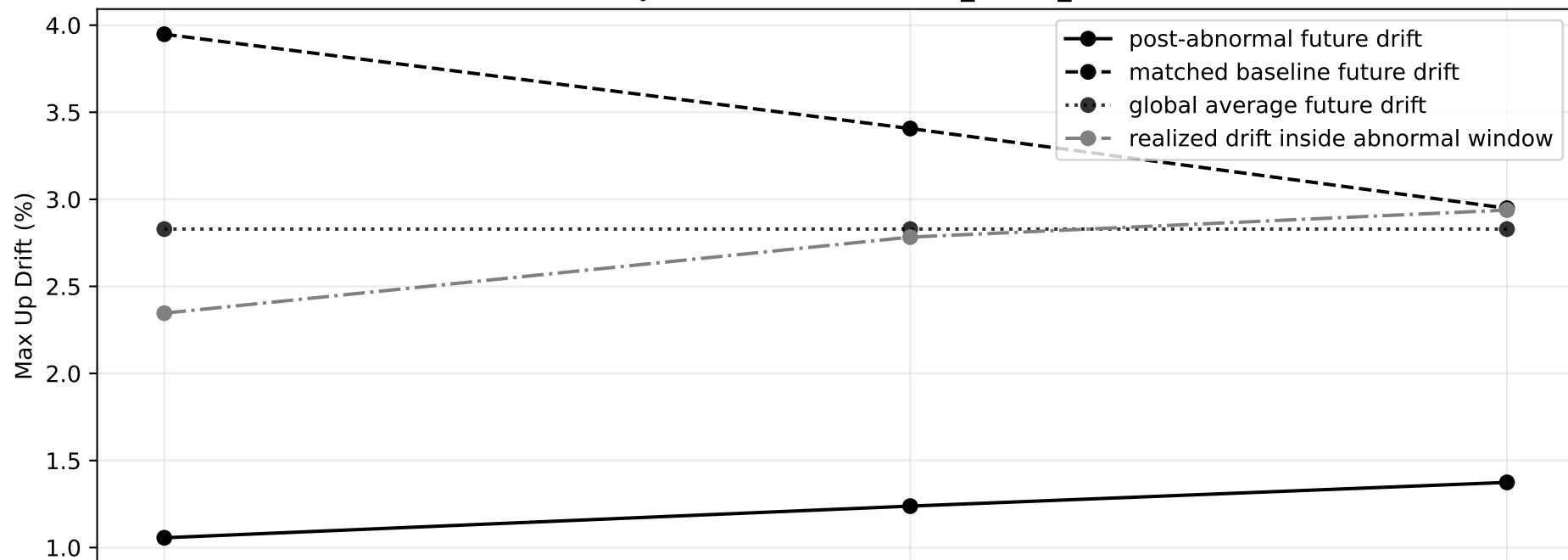
META - Daily q99 - DOWN



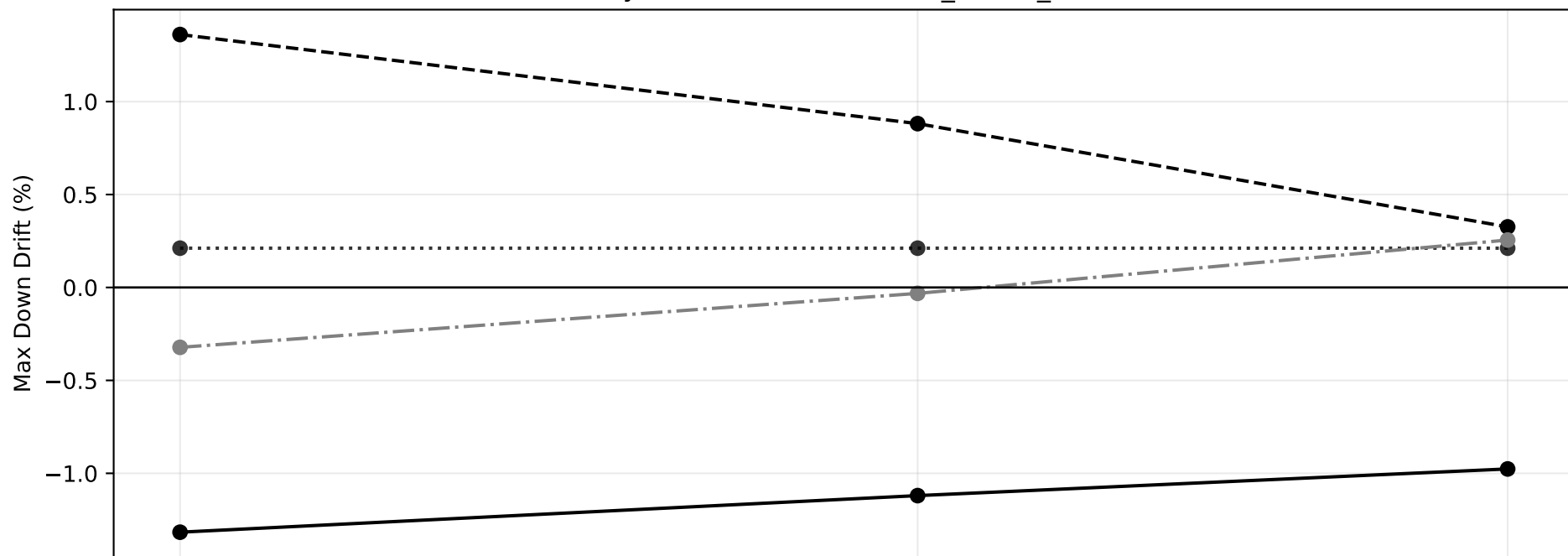
META - Daily q99 - ABS



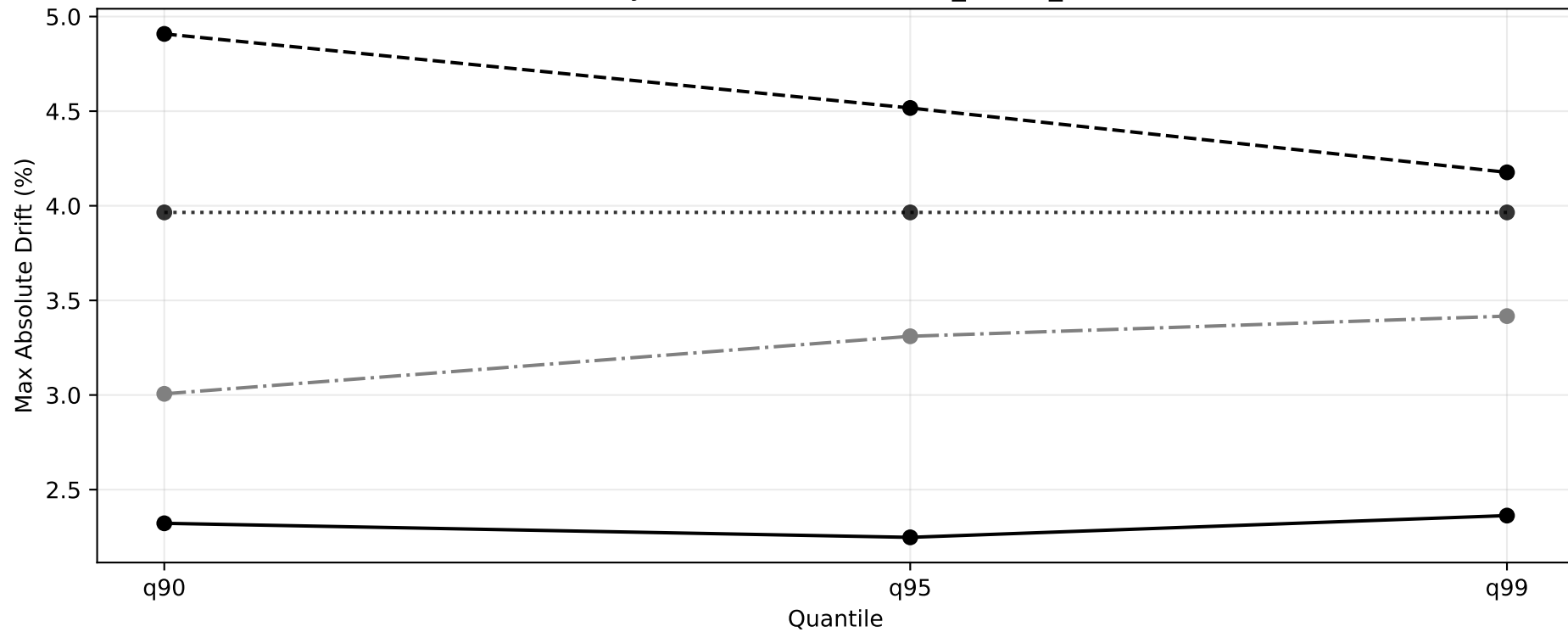
META - Daily Line View - imbalance_above_mid - UP



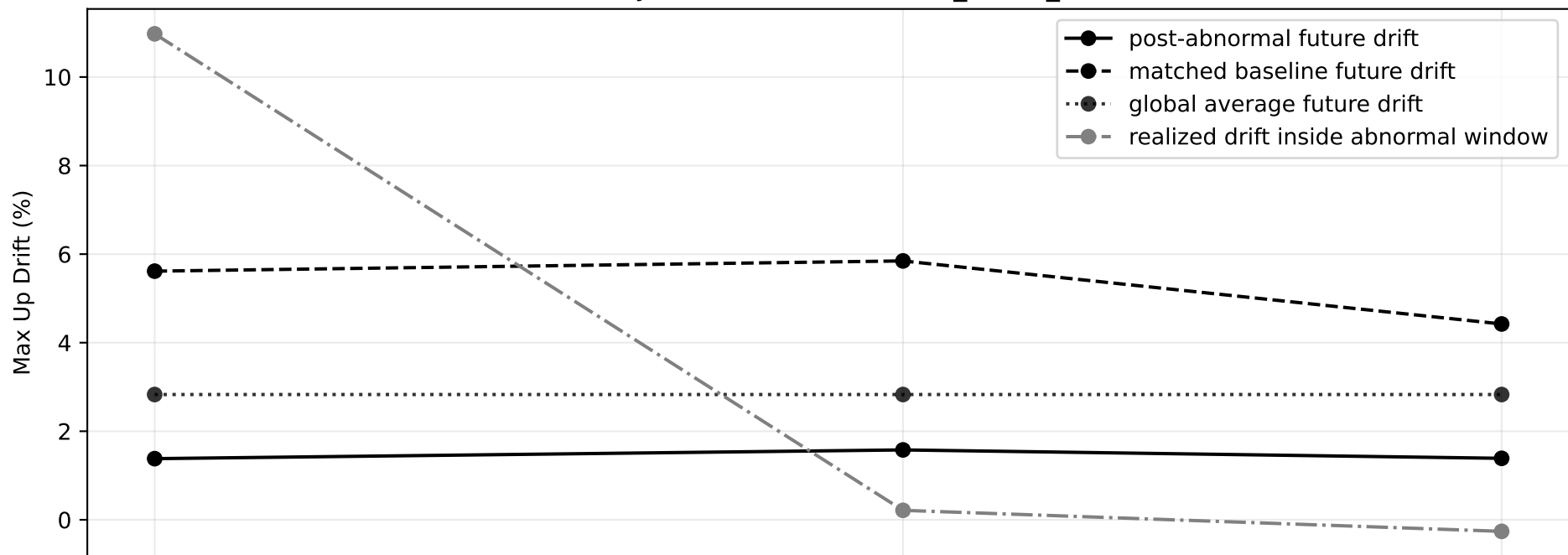
META - Daily Line View - imbalance_above_mid - DOWN



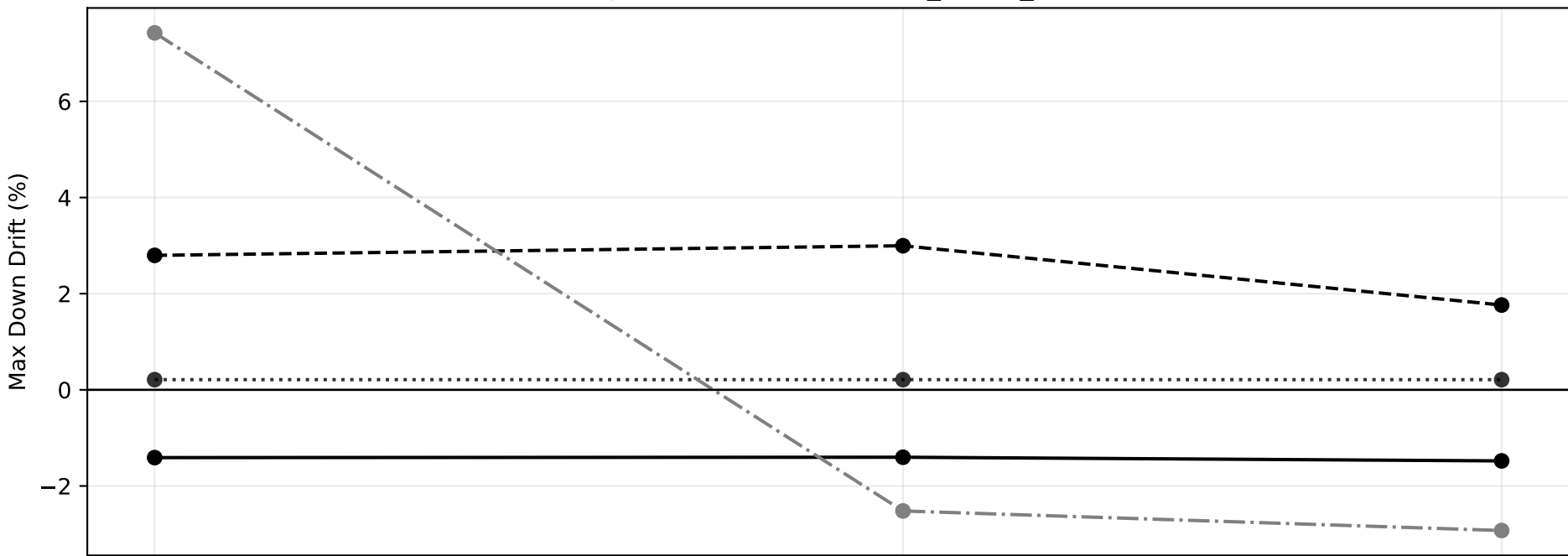
META - Daily Line View - imbalance_above_mid - ABS



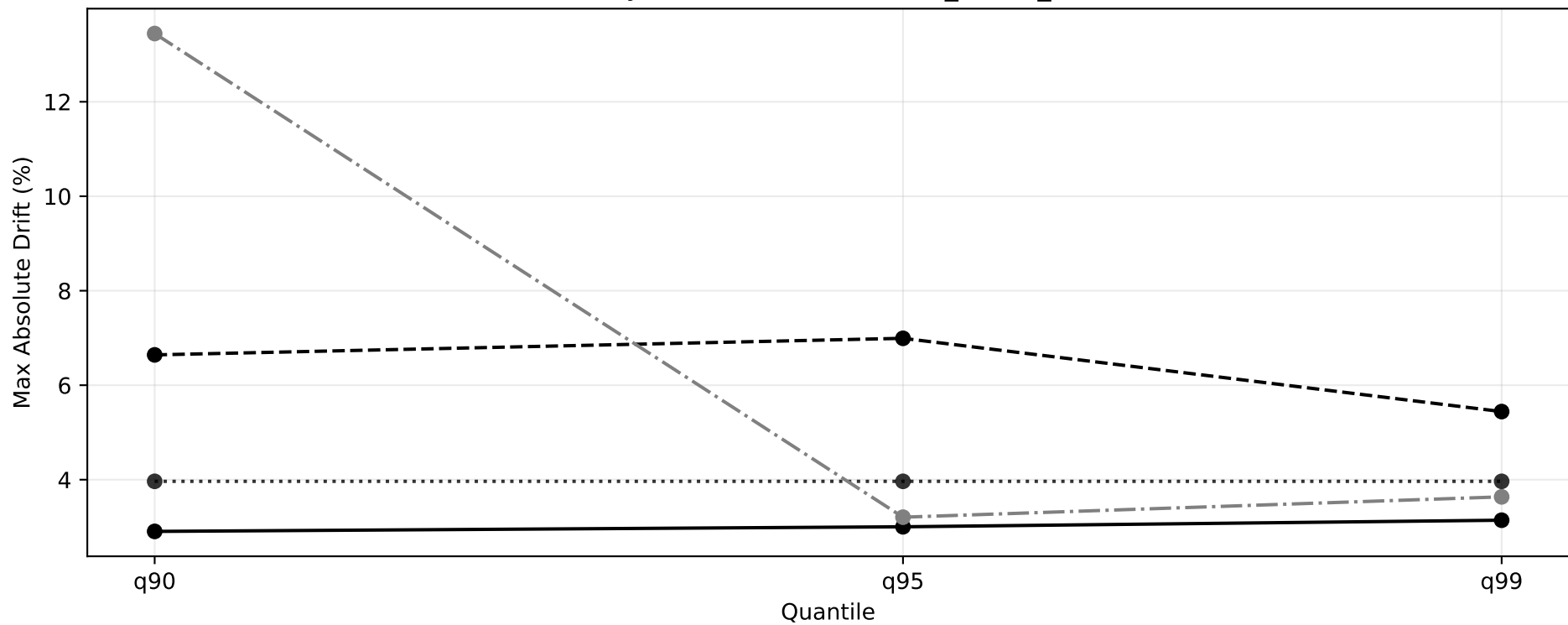
META - Daily Line View - imbalance_below_mid - UP

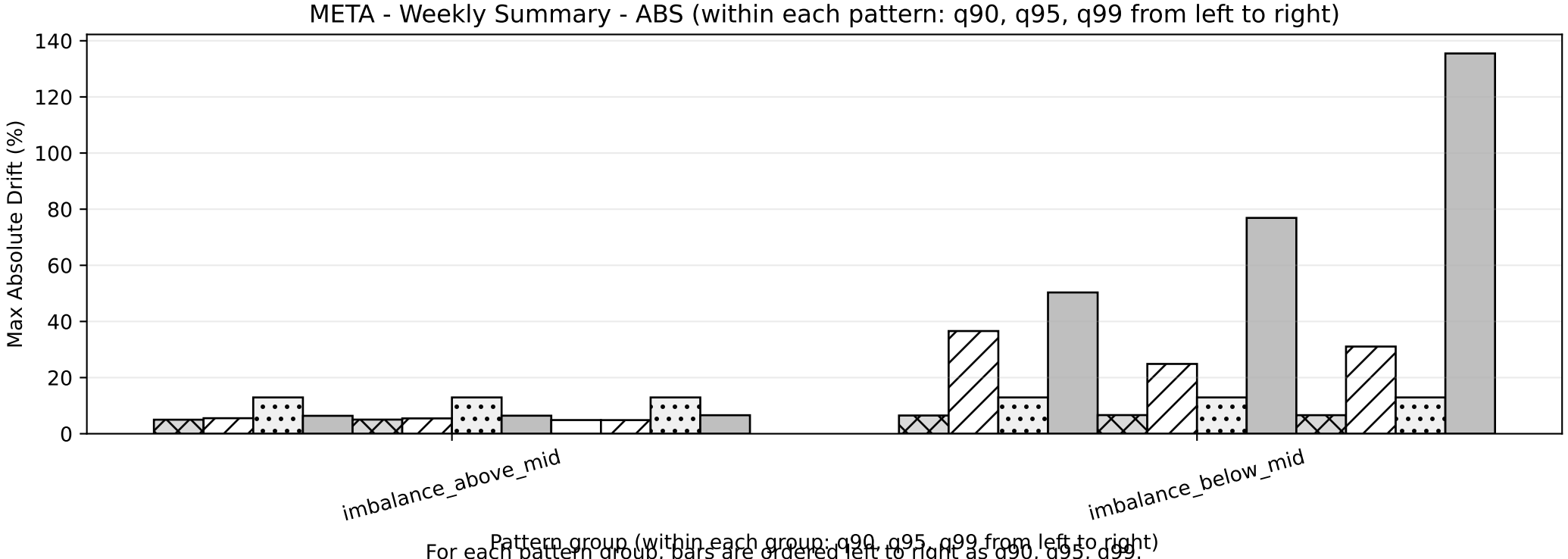
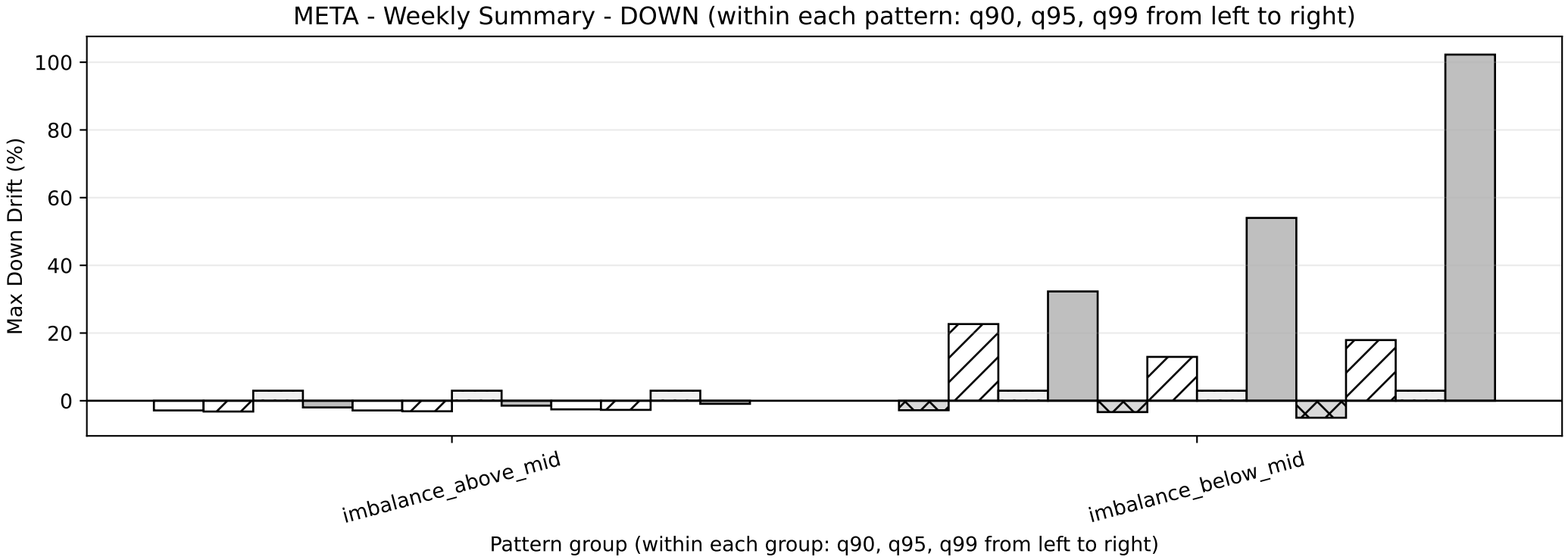
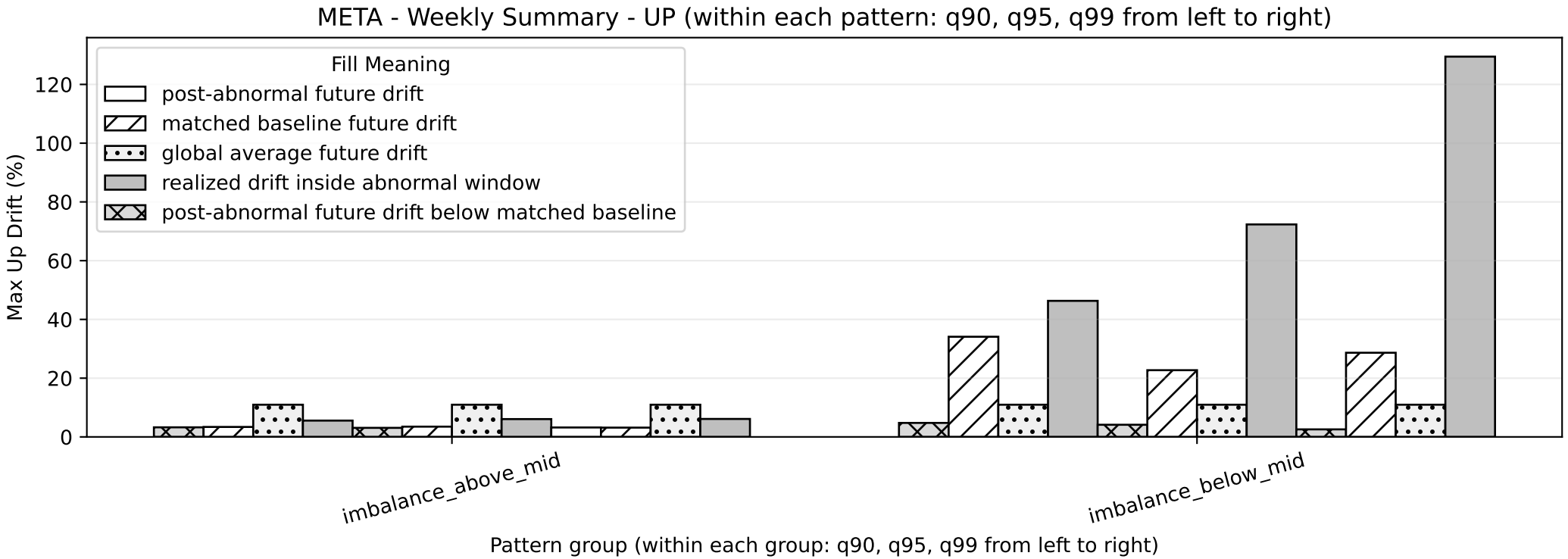


META - Daily Line View - imbalance_below_mid - DOWN

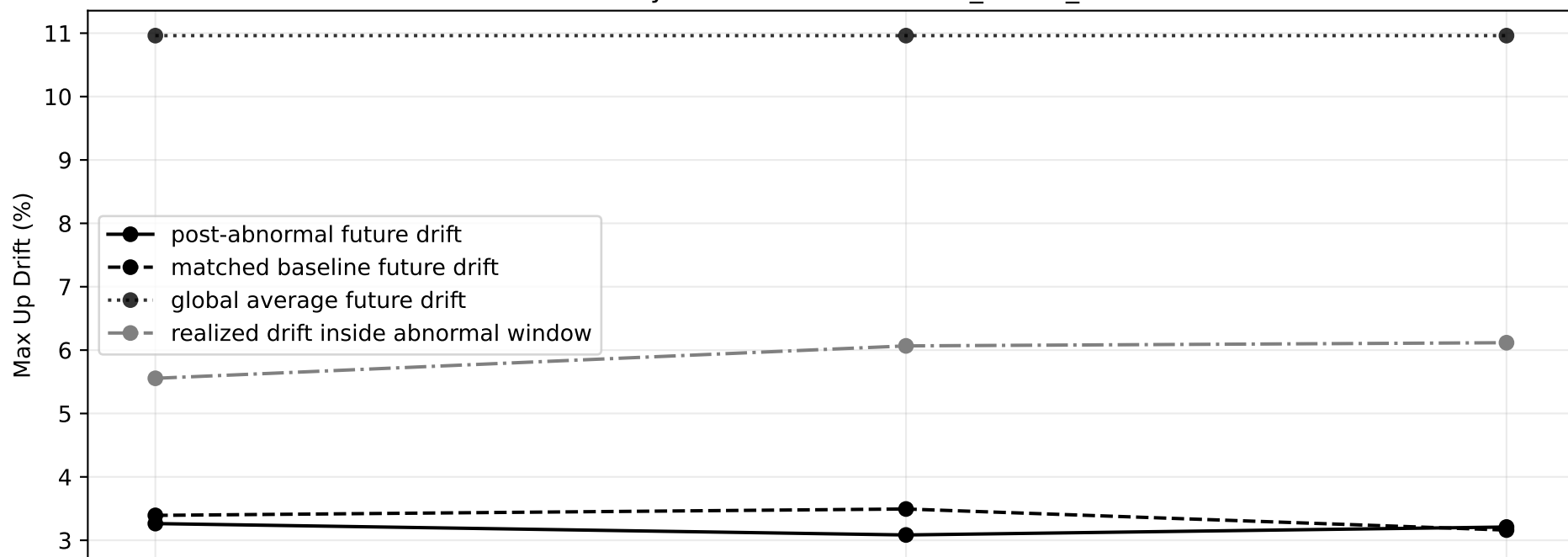


META - Daily Line View - imbalance_below_mid - ABS

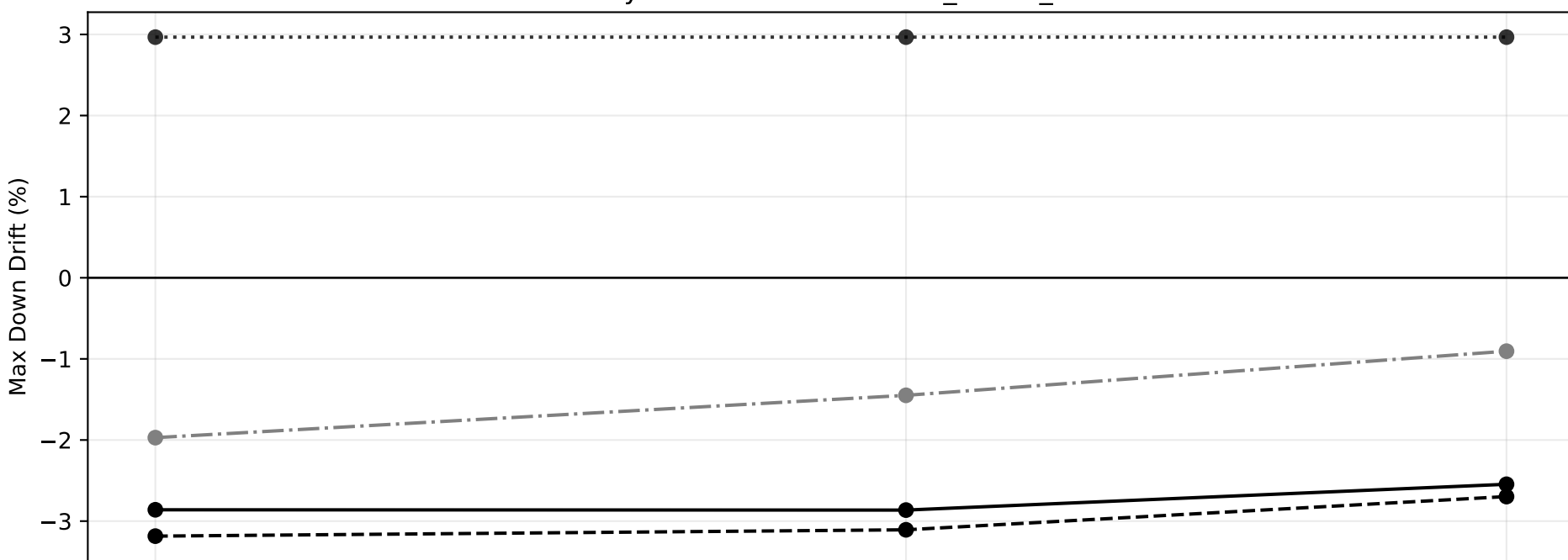




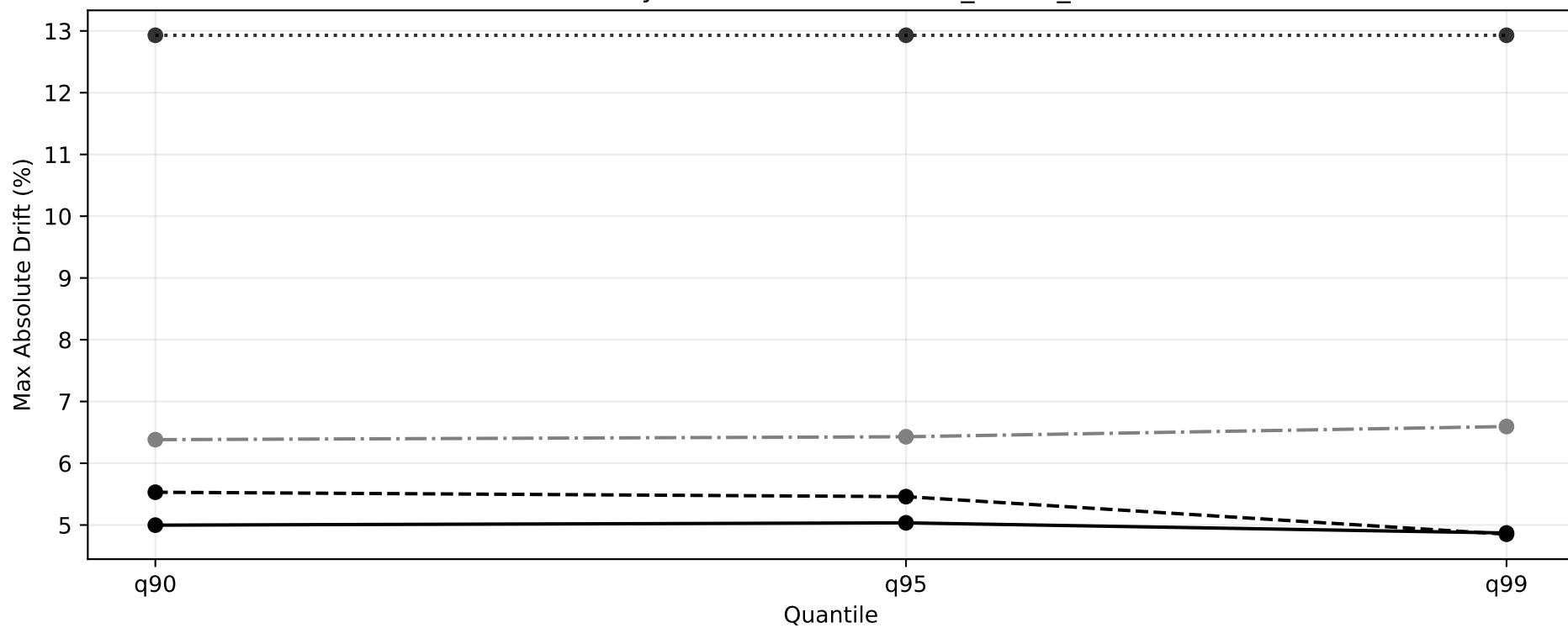
META - Weekly Line View - imbalance_above_mid - UP



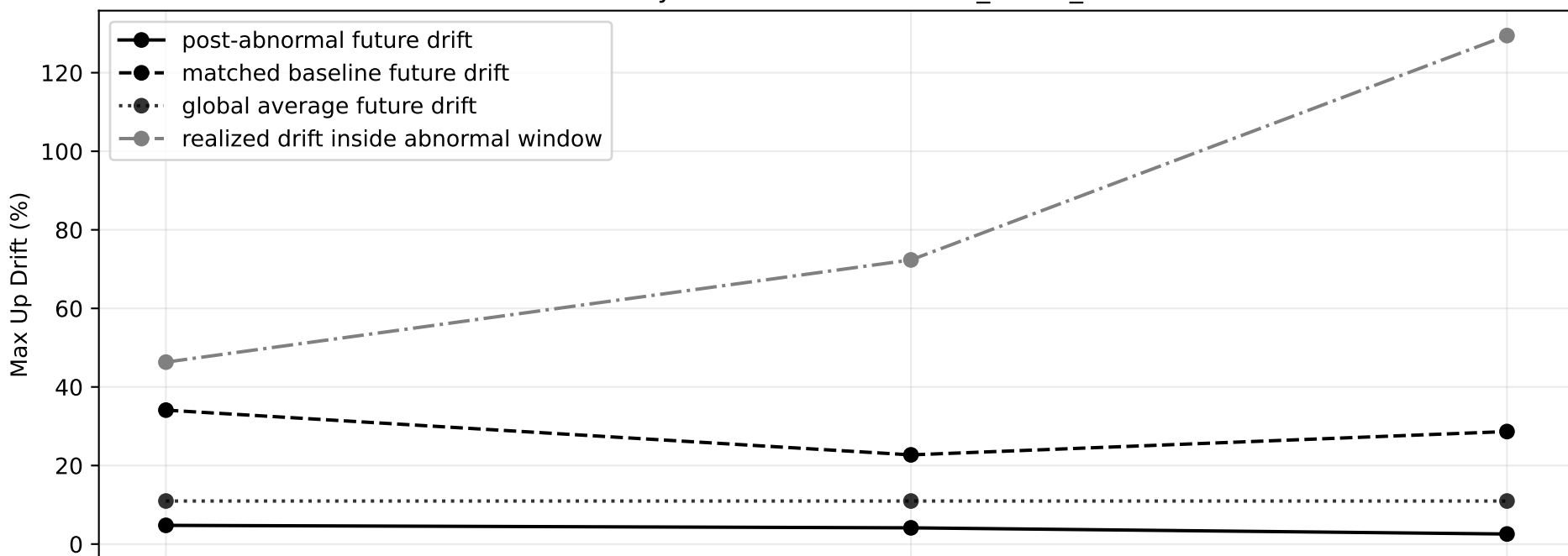
META - Weekly Line View - imbalance_above_mid - DOWN



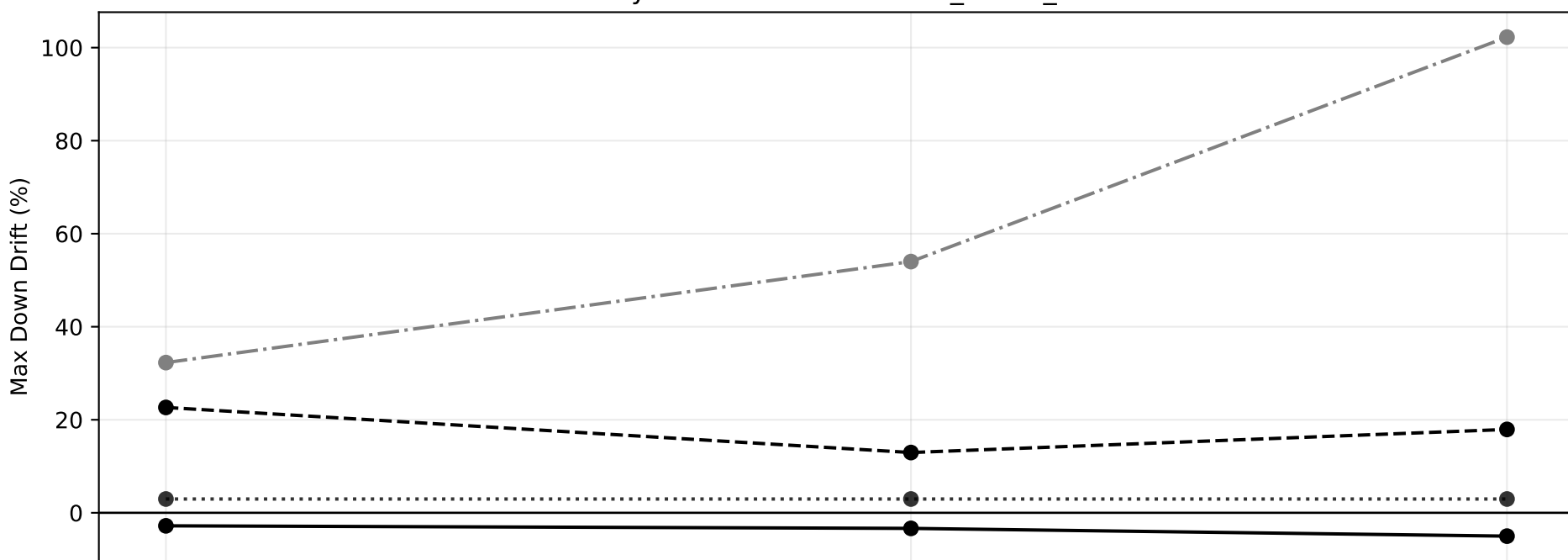
META - Weekly Line View - imbalance_above_mid - ABS



META - Weekly Line View - imbalance_below_mid - UP



META - Weekly Line View - imbalance_below_mid - DOWN



META - Weekly Line View - imbalance_below_mid - ABS

